

Commands - Creating a Privacy-aware Environment Map

```
labuser@lab-ubuntu-01:~$ roscore
... logging to
/home/labuser/.ros/log/b2a352c4-5dd3-11f1-9461-633a14440943/roslaunch-lab-ubuntu-01-2762
.log
Checking log directory for disk usage. This may take a while.
Press Ctrl-C to interrupt
Done checking log file disk usage. Usage is <1GB.

started roslaunch server http://lab-ubuntu-01:33595/
ros_comm version 1.17.4
```

SUMMARY

=====

PARAMETERS

- * /rostdistro: noetic
- * /rosversion: 1.17.4

NODES

```
auto-starting new master
process[master]: started with pid [2776]
ROS_MASTER_URI=http://lab-ubuntu-01:11311/

setting /run_id to b2a352c4-5dd3-11f1-9461-633a14440943
process[rosout-1]: started with pid [2786]
started core service [/rosout]
```

```
labuser@lab-ubuntu-01:~$ rospack find cob_simulation
[rospack] Error: package 'cob_simulation' not found
labuser@lab-ubuntu-01:~$
```

<http://lab-ubuntu-01:33595/>

Error response

Error code: 501

Message: Unsupported method ('GET').

Error code explanation: HTTPStatus.NOT_IMPLEMENTED - Server does not support this operation.

```
labuser@lab-ubuntu-01:~$ roscore
... logging to
/home/labuser/.ros/log/b2a352c4-5dd3-11f1-9461-633a14440943/roslaunch-lab-ubuntu-01-2762
.log
Checking log directory for disk usage. This may take a while.
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started roslaunch server http://lab-ubuntu-01:33595/
ros_comm version 1.17.4
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SUMMARY
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PARAMETERS
* /rostdistro: noetic
* /rosversion: 1.17.4

NODES

```
auto-starting new master
process[master]: started with pid [2776]
ROS_MASTER_URI=http://lab-ubuntu-01:11311/

setting /run_id to b2a352c4-5dd3-11f1-9461-633a14440943
process[rosout-1]: started with pid [2786]
```

```
started core service [/rosout]
^C[rosout-1] killing on exit
[master] killing on exit
shutting down processing monitor...
... shutting down processing monitor complete
done
labuser@lab-ubuntu-01:~$ rosversion -d
noetic
labuser@lab-ubuntu-01:~$ which roscore
/opt/ros/noetic/bin/roscore
labuser@lab-ubuntu-01:~$ echo $ROS_DISTRO
noetic
labuser@lab-ubuntu-01:~$ rospack find cob_simulation
[rospack] Error: package 'cob_simulation' not found
labuser@lab-ubuntu-01:~$ rospack list | grep cob
cob_actions /opt/ros/noetic/share/cob_actions
cob_android_msgs /opt/ros/noetic/share/cob_android_msgs
cob_android_script_server /opt/ros/noetic/share/cob_android_script_server
cob_base_controller_utils /opt/ros/noetic/share/cob_base_controller_utils
cob_base_velocity_smoother /opt/ros/noetic/share/cob_base_velocity_smoother
cob_bms_driver /opt/ros/noetic/share/cob_bms_driver
cob_bringup /opt/ros/noetic/share/cob_bringup
cob_calibration_data /opt/ros/noetic/share/cob_calibration_data
cob_cam3d_throttle /opt/ros/noetic/share/cob_cam3d_throttle
cob_collision_monitor /opt/ros/noetic/share/cob_collision_monitor
cob_collision_velocity_filter /opt/ros/noetic/share/cob_collision_velocity_filter
cob_command_gui /opt/ros/noetic/share/cob_command_gui
cob_control_mode_adapter /opt/ros/noetic/share/cob_control_mode_adapter
cob_control_msgs /opt/ros/noetic/share/cob_control_msgs
cob_dashboard /opt/ros/noetic/share/cob_dashboard
cob_default_env_config /opt/ros/noetic/share/cob_default_env_config
cob_default_robot_behavior /opt/ros/noetic/share/cob_default_robot_behavior
cob_default_robot_config /opt/ros/noetic/share/cob_default_robot_config
cob_description /opt/ros/noetic/share/cob_description
cob_docker_control /opt/ros/noetic/share/cob_docker_control
cob_footprint_observer /opt/ros/noetic/share/cob_footprint_observer
cob_frame_tracker /opt/ros/noetic/share/cob_frame_tracker
cob_gazebo /opt/ros/noetic/share/cob_gazebo
cob_gazebo_ros_control /opt/ros/noetic/share/cob_gazebo_ros_control
cob_hand_bridge /opt/ros/noetic/share/cob_hand_bridge
cob_hardware_config /opt/ros/noetic/share/cob_hardware_config
cob_helper_tools /opt/ros/noetic/share/cob_helper_tools
cob_image_flip /opt/ros/noetic/share/cob_image_flip
cob_light /opt/ros/noetic/share/cob_light
```

```
cob_linear_nav /opt/ros/noetic/share/cob_linear_nav
cob_mecanum_controller /opt/ros/noetic/share/cob_mecanum_controller
cob_mimic /opt/ros/noetic/share/cob_mimic
cob_monitoring /opt/ros/noetic/share/cob_monitoring
cob_moveit_config /opt/ros/noetic/share/cob_moveit_config
cob_msgs /opt/ros/noetic/share/cob_msgs
cob_obstacle_distance /opt/ros/noetic/share/cob_obstacle_distance
cob_omni_drive_controller /opt/ros/noetic/share/cob_omni_drive_controller
cob_perception_msgs /opt/ros/noetic/share/cob_perception_msgs
cob_phidget_em_state /opt/ros/noetic/share/cob_phidget_em_state
cob_phidget_power_state /opt/ros/noetic/share/cob_phidget_power_state
cob_phidgets /opt/ros/noetic/share/cob_phidgets
cob_reflector_referencing /opt/ros/noetic/share/cob_reflector_referencing
cob_safety_controller /opt/ros/noetic/share/cob_safety_controller
cob_scan_unifier /opt/ros/noetic/share/cob_scan_unifier
cob_script_server /opt/ros/noetic/share/cob_script_server
cob_sick_lms1xx /opt/ros/noetic/share/cob_sick_lms1xx
cob_sick_s300 /opt/ros/noetic/share/cob_sick_s300
cob_sound /opt/ros/noetic/share/cob_sound
cob_srvs /opt/ros/noetic/share/cob_srvs
cob_teleop /opt/ros/noetic/share/cob_teleop
cob_twist_controller /opt/ros/noetic/share/cob_twist_controller
cob_voltage_control /opt/ros/noetic/share/cob_voltage_control
labuser@lab-ubuntu-01:~$ gazebo --version
Gazebo multi-robot simulator, version 11.15.1
Copyright (C) 2012 Open Source Robotics Foundation.
Released under the Apache 2 License.
http://gazebo.org
```

```
labuser@lab-ubuntu-01:~$ which gazebo
/usr/bin/gazebo
labuser@lab-ubuntu-01:~$ which rviz
/opt/ros/noetic/bin/rviz
labuser@lab-ubuntu-01:~$ rospack list | grep slam
labuser@lab-ubuntu-01:~$ rospack find gmapping
[rospack] Error: package 'gmapping' not found
labuser@lab-ubuntu-01:~$ python3 --version
Python 3.8.10
labuser@lab-ubuntu-01:~$
```

```
labuser@lab-ubuntu-01:~$ rospack list | grep gazebo
```



```
cob_gazebo /opt/ros/noetic/share/cob_gazebo
cob_gazebo_ros_control /opt/ros/noetic/share/cob_gazebo_ros_control
gazebo_dev /opt/ros/noetic/share/gazebo_dev
gazebo_msgs /opt/ros/noetic/share/gazebo_msgs
gazebo_plugins /opt/ros/noetic/share/gazebo_plugins
gazebo_ros /opt/ros/noetic/share/gazebo_ros
gazebo_ros_control /opt/ros/noetic/share/gazebo_ros_control
labuser@lab-ubuntu-01:~$ echo $ROS_PACKAGE_PATH
/opt/ros/noetic/share
labuser@lab-ubuntu-01:~$
```

```
labuser@lab-ubuntu-01:~$ labuser@lab-ubuntu-01:~$ roscore
labuser@lab-ubuntu-01:~$: command not found
labuser@lab-ubuntu-01:~$ ... logging to
/home/labuser/.ros/log/b2a352c4-5dd3-11f1-9461-633a14440943/roslaunch-lab-ubuntu-01-2762
.log
...: command not found
labuser@lab-ubuntu-01:~$ Checking log directory for disk usage. This may take a while.
Checking: command not found
labuser@lab-ubuntu-01:~$ Press Ctrl-C to interrupt
Press: command not found
labuser@lab-ubuntu-01:~$ Done checking log file disk usage. Usage is <1GB.
bash: 1GB.: No such file or directory
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ started roslaunch server http://lab-ubuntu-01:33595/
started: command not found
labuser@lab-ubuntu-01:~$ ros_comm version 1.17.4
ros_comm: command not found
labuser@lab-ubuntu-01:~$
/labuser@lab-ubuntu-01:~$
glabuser@lab-ubuntu-01:~$ SUMMARY
SUMMARY: command not found
labuser@lab-ubuntu-01:~$ =====
=====: command not found
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ PARAMETERS
PARAMETERS: command not found
labuser@lab-ubuntu-01:~$ */rostdistro: noetic
catkin_ws: command not found
labuser@lab-ubuntu-01:~$ */rosversion: 1.17.4
catkin_ws: command not found
labuser@lab-ubuntu-01:~$
```

```

labuser@lab-ubuntu-01:~$ NODES
NODES: command not found
labuser@lab-ubuntu-01:~$
ulabuser@lab-ubuntu-01:~$ auto-starting new master
auto-starting: command not found
labuser@lab-ubuntu-01:~$ process[master]: started with pid [2776]
process[master]:: command not found
labuser@lab-ubuntu-01:~$ ROS_MASTER_URI=http://lab-ubuntu-01:11311/
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ setting /run_id to b2a352c4-5dd3-11f1-9461-633a14440943
setting: command not found
labuser@lab-ubuntu-01:~$ process[rosout-1]: started with pid [2786]
process[rosout-1]:: command not found
labuser@lab-ubuntu-01:~$ started core service [/rosout]
started: command not found
labuser@lab-ubuntu-01:~$ ^C[rosout-1] killing on exit
bash: :s^C[rosout-1] killing on exit: substitution failed
labuser@lab-ubuntu-01:~$ [master] killing on exit
[master]: command not found
labuser@lab-ubuntu-01:~$ shutting down processing monitor...
shutting: command not found
labuser@lab-ubuntu-01:~$ ... shutting down processing monitor complete
...: command not found
labuser@lab-ubuntu-01:~$ done
bash: syntax error near unexpected token `done'
labuser@lab-ubuntu-01:~$ labuser@lab-ubuntu-01:~$ rosversion -d
labuser@lab-ubuntu-01:~$: command not found
labuser@lab-ubuntu-01:~$ noetic
noetic: command not found
labuser@lab-ubuntu-01:~$ labuser@lab-ubuntu-01:~$ which roscore
labuser@lab-ubuntu-01:~$: command not found
labuser@lab-ubuntu-01:~$ /opt/ros/noetic/bin/roscore
... logging to
/home/labuser/.ros/log/2f4a19f6-5dee-11f1-9461-633a14440943/roslaunch-lab-ubuntu-01-7445.l
og
Checking log directory for disk usage. This may take a while.
Press Ctrl-C to interrupt
Done checking log file disk usage. Usage is <1GB.

started roslaunch server http://lab-ubuntu-01:43933/
ros_comm version 1.17.4

```

SUMMARY

=====

PARAMETERS

* /roscdistro: noetic
* /rosversion: 1.17.4

NODES

auto-starting new master
process[master]: started with pid [7453]
ROS_MASTER_URI=http://lab-ubuntu-01:11311/

setting /run_id to 2f4a19f6-5dee-11f1-9461-633a14440943
process[rosout-1]: started with pid [7463]
started core service [/rosout]
^C[rosout-1] killing on exit
[master] killing on exit
shutting down processing monitor...
... shutting down processing monitor complete
done
labuser@lab-ubuntu-01:~\$

```
labuser@lab-ubuntu-01:~$ find /opt/ros/noetic/share/cob_* -name "*.launch" | less
/opt/ros/noetic/share/cob_android_script_server/launch/script_server.launch
/opt/ros/noetic/share/cob_base_controller_utils/launch/stop_detector.launch
/opt/ros/noetic/share/cob_base_controller_utils/launch/halt_detector.launch
/opt/ros/noetic/share/cob_base_controller_utils/launch/spin_detector.launch
/opt/ros/noetic/share/cob_base_controller_utils/launch/stuck_detector.launch
/opt/ros/noetic/share/cob_bringup/components/cam3d_d435_rbgd.launch
/opt/ros/noetic/share/cob_bringup/components/cam3d_openni2.launch
/opt/ros/noetic/share/cob_bringup/components/raw_mini_base.launch
/opt/ros/noetic/share/cob_bringup/components/sdhx_remote.launch
/opt/ros/noetic/share/cob_bringup/components/cob4_base.launch
/opt/ros/noetic/share/cob_bringup/components/cam3d_visionary_t.launch
/opt/ros/noetic/share/cob_bringup/components/cam3d_zr300_rbgd.launch
/opt/ros/noetic/share/cob_bringup/components/canopen_generic.launch
/opt/ros/noetic/share/cob_bringup/components/sdhx_legacy.launch
/opt/ros/noetic/share/cob_bringup/components/raw3_base.launch
/opt/ros/noetic/share/cob_bringup/components/ros_control_base.launch
/opt/ros/noetic/share/cob_bringup/components/cob4_head_camera.launch
/opt/ros/noetic/share/cob_bringup/components/cam3d_r200_rbgd.launch
/opt/ros/noetic/share/cob_bringup/tools/base_collision_observer.launch
```

/opt/ros/noetic/share/cob_bringup/tools/diagnostics_aggregator.launch
/opt/ros/noetic/share/cob_bringup/tools/autoinit.launch
/opt/ros/noetic/share/cob_bringup/tools/behavior.launch
/opt/ros/noetic/share/cob_bringup/tools/teleop.launch
/opt/ros/noetic/share/cob_bringup/tools/android.launch
/opt/ros/noetic/share/cob_bringup/tools/emergency_stop_monitor.launch
/opt/ros/noetic/share/cob_bringup/tools/fake_diagnostics.launch
/opt/ros/noetic/share/cob_bringup/tools/dashboard.launch
/opt/ros/noetic/share/cob_bringup/tools/twist_mux.launch
/opt/ros/noetic/share/cob_bringup/tools/wifi_monitor.launch
/opt/ros/noetic/share/cob_bringup/tools/pc_monitor.launch
/opt/ros/noetic/share/cob_bringup/tools/velocity_smoother.launch
/opt/ros/noetic/share/cob_bringup/tools/rviz.launch
/opt/ros/noetic/share/cob_bringup/tools/battery_monitor.launch
/opt/ros/noetic/share/cob_bringup/tools/hz_monitor.launch
/opt/ros/noetic/share/cob_bringup/tools/cob_obstacle_distance.launch
/opt/ros/noetic/share/cob_bringup/tools/docker_control.launch
/opt/ros/noetic/share/cob_bringup/tools/teleop_spacenv_twist.launch
/opt/ros/noetic/share/cob_bringup/tools/collision_monitor.launch
/opt/ros/noetic/share/cob_bringup/tools/robot_state_publisher.launch
/opt/ros/noetic/share/cob_bringup/controllers/generic_cartesian_controller.launch
/opt/ros/noetic/share/cob_bringup/controllers/generic_controller.launch
/opt/ros/noetic/share/cob_bringup/robot.launch
/opt/ros/noetic/share/cob_bringup/robots/cob4-13.launch
/opt/ros/noetic/share/cob_bringup/robots/cob4-9.launch
/opt/ros/noetic/share/cob_bringup/robots/component_solo_canopen.launch
/opt/ros/noetic/share/cob_bringup/robots/cob4-29.launch
/opt/ros/noetic/share/cob_bringup/robots/cob4-23.launch
/opt/ros/noetic/share/cob_bringup/robots/raw3-3.launch
/opt/ros/noetic/share/cob_bringup/robots/cob4-4.launch
/opt/ros/noetic/share/cob_bringup/robots/cob4-18.launch
/opt/ros/noetic/share/cob_bringup/robots/cob4-8.launch
/opt/ros/noetic/share/cob_bringup/robots/cob4-25.launch
/opt/ros/noetic/share/cob_bringup/robots/cob4-5.launch
/opt/ros/noetic/share/cob_bringup/robots/raw-mini.launch
/opt/ros/noetic/share/cob_bringup/robots/cob4-24.launch
/opt/ros/noetic/share/cob_bringup/robots/cob4-20.launch
/opt/ros/noetic/share/cob_bringup/robots/cob4-30.launch
/opt/ros/noetic/share/cob_bringup/robots/cob4-16.launch
/opt/ros/noetic/share/cob_bringup/robots/cob4-11.launch
/opt/ros/noetic/share/cob_bringup/robots/cob4-6.launch
/opt/ros/noetic/share/cob_bringup/robots/raw3-5.launch
/opt/ros/noetic/share/cob_bringup/robots/cob4-10.launch
/opt/ros/noetic/share/cob_bringup/robots/base_solo.launch

/opt/ros/noetic/share/cob_bringup/robots/cob4-3.launch
/opt/ros/noetic/share/cob_bringup/robots/cob4-7.launch
/opt/ros/noetic/share/cob_bringup/robots/raw3-1.launch
/opt/ros/noetic/share/cob_bringup/drivers/sick_visionary_t.launch
/opt/ros/noetic/share/cob_bringup/drivers/scan_unifier.launch
/opt/ros/noetic/share/cob_bringup/drivers/joy.launch
/opt/ros/noetic/share/cob_bringup/drivers/bms.launch
/opt/ros/noetic/share/cob_bringup/drivers/openni.launch
/opt/ros/noetic/share/cob_bringup/drivers/schunk_sdhx.launch
/opt/ros/noetic/share/cob_bringup/drivers/phidgets.launch
/opt/ros/noetic/share/cob_bringup/drivers/openni2.launch
/opt/ros/noetic/share/cob_bringup/drivers/battery_voltage_filter.launch
/opt/ros/noetic/share/cob_bringup/drivers/sound.launch
/opt/ros/noetic/share/cob_bringup/drivers/schunk_sdhx_local.launch
/opt/ros/noetic/share/cob_bringup/drivers/emstate_phidget.launch
/opt/ros/noetic/share/cob_bringup/drivers/battery_voltage.launch
/opt/ros/noetic/share/cob_bringup/drivers/usb_camera_node.launch
/opt/ros/noetic/share/cob_bringup/drivers/laser_scan_filter.launch
/opt/ros/noetic/share/cob_bringup/drivers/image_flip.launch
/opt/ros/noetic/share/cob_bringup/drivers/sick_s300.launch
/opt/ros/noetic/share/cob_bringup/drivers/sick_lms1xx.launch
/opt/ros/noetic/share/cob_bringup/drivers/light.launch
/opt/ros/noetic/share/cob_bringup/drivers/sick_flexisoft.launch
/opt/ros/noetic/share/cob_bringup/drivers/canopen_402.launch
/opt/ros/noetic/share/cob_bringup/drivers/image_flip_nodelet.launch
/opt/ros/noetic/share/cob_bringup/drivers/powerstate_phidget.launch
/opt/ros/noetic/share/cob_bringup/drivers/ur.launch
/opt/ros/noetic/share/cob_bringup/drivers/mimic.launch
/opt/ros/noetic/share/cob_bringup/drivers/realsense_d400_rgbd.launch
/opt/ros/noetic/share/cob_cam3d_throttle/launch/cam3d_throttle.launch
/opt/ros/noetic/share/cob_default_env_config/test_env_config.launch
/opt/ros/noetic/share/cob_default_env_config/upload_object_locations.launch
/opt/ros/noetic/share/cob_default_env_config/upload_navigation_goals.launch
/opt/ros/noetic/share/cob_default_robot_config/upload_param.launch
/opt/ros/noetic/share/cob_gazebo/launch/robot.launch
/opt/ros/noetic/share/cob_hardware_config/upload_robot.launch
/opt/ros/noetic/share/cob_hardware_config/display_robot.launch
/opt/ros/noetic/share/cob_image_flip/launch/image_flip.launch
/opt/ros/noetic/share/cob_image_flip/launch/image_flip_nodelet.launch
/opt/ros/noetic/share/cob_moveit_config/launch/upload_config.launch
/opt/ros/noetic/share/cob_obstacle_distance/launch/example.launch
/opt/ros/noetic/share/cob_script_server/launch/script_server.launch
/opt/ros/noetic/share/cob_teleop/launch/teleop_keyboard.launch
/opt/ros/noetic/share/cob_twist_controller/launch/debug_fk_vel_recursive.launch

```
/opt/ros/noetic/share/cob_twist_controller/launch/example.launch
/opt/ros/noetic/share/cob_voltage_control/ros/launch/csv_processing.launch
/opt/ros/noetic/share/cob_voltage_control/ros/launch/record_current_and_voltage.launch
```

```
labuser@lab-ubuntu-01:~$ roslaunch --files cob_gazebo robot.launch
while processing /opt/ros/noetic/share/cob_bringup/robots/!!NO_ROBOT_SET!!.xml:
Invalid roslaunch XML syntax: [Errno 2] No such file or directory:
'/opt/ros/noetic/share/cob_bringup/robots/!!NO_ROBOT_SET!!.xml'
labuser@lab-ubuntu-01:~$
```

```
labuser@lab-ubuntu-01:~$ head -100 /opt/ros/noetic/share/cob_gazebo/launch/robot.launch
<?xml version="1.0"?>
<launch>
```

```
  <!-- define arguments -->
  <arg name="robot" default="$(optenv ROBOT !!NO_ROBOT_SET!!)"/>
  <arg name="robot_env" default="$(optenv ROBOT_ENV !!NO_ROBOT_ENV_SET!!)"/>
  <arg name="pkg_robot_config" default="$(find cob_default_robot_config)"/>
  <arg name="paused" default="false"/>
  <arg name="initial_config" default=""/>

  <!-- upload default configuration parameters -->
  <include file="$(find cob_bringup)/robots/$(arg robot).xml" >
  <arg name="sim" value="true" />
  </include>

  <!-- push robot_description to factory and spawn robot in gazebo -->
  <node name="spawn_gazebo_model" pkg="gazebo_ros" type="spawn_model"
args="-urdf -param robot_description -model robot -z 0.1 $(arg initial_config) -wait $(arg
robot_env)" respawn="false" output="screen" />

</launch>
labuser@lab-ubuntu-01:~$
```

labuser@lab-ubuntu-01:~\$ head -50 /opt/ros/noetic/share/cob_bringup/robots/cob4-4.launch

```
<?xml version="1.0"?>
```

```
<launch>
```

```
  <arg name="env-script" default="$(find cob_bringup)/env.sh"/>
```

```
  <include file="$(find cob_bringup)/robots/cob4-4.xml">
```

```
    <arg name="cob4-4-b1" value="cob4-4-b1"/>
```

```
    <arg name="env-script" value="$(arg env-script)"/>
```

```
  </include>
```

```
</launch>
```

labuser@lab-ubuntu-01:~\$

labuser@lab-ubuntu-01:~\$ head -50 /opt/ros/noetic/share/cob_bringup/robots/cob4-4.launch

```
<?xml version="1.0"?>
```

```
<launch>
```

```
  <arg name="env-script" default="$(find cob_bringup)/env.sh"/>
```

```
  <include file="$(find cob_bringup)/robots/cob4-4.xml">
```

```
    <arg name="cob4-4-b1" value="cob4-4-b1"/>
```

```
    <arg name="env-script" value="$(arg env-script)"/>
```

```
  </include>
```

```
</launch>
```

labuser@lab-ubuntu-01:~\$

labuser@lab-ubuntu-01:~\$ ls /opt/ros/noetic/share/cob_bringup/robots/

base_solo.launch cob4-11.launch cob4-13.xml cob4-18.launch cob4-20.xml

cob4-24.launch cob4-25.xml cob4-30.launch cob4-3.xml cob4-5.launch cob4-6.xml

cob4-8.launch cob4-9.xml raw3-1.xml raw3-5.launch raw-mini.xml

cob4-10.launch cob4-11.xml cob4-16.launch cob4-18.xml cob4-23.launch

cob4-24.xml cob4-29.launch cob4-30.xml cob4-4.launch cob4-5.xml cob4-7.launch

cob4-8.xml component_solo_canopen.launch raw3-3.launch raw3-5.xml

```

cob4-10.xml  cob4-13.launch  cob4-16.xml          cob4-20.launch  cob4-23.xml
cob4-25.launch  cob4-29.xml          cob4-3.launch  cob4-4.xml  cob4-6.launch  cob4-7.xml
cob4-9.launch  raw3-1.launch          raw3-3.xml    raw-mini.launch
labuser@lab-ubuntu-01:~$ ls /opt/ros/noetic/share/cob_default_env_config
cmake  envs  package.xml  test_env_config.launch  upload_navigation_goals.launch
upload_object_locations.launch
labuser@lab-ubuntu-01:~$ grep -n "robot_env\|gazebo\|sim"
/opt/ros/noetic/share/cob_bringup/robots/cob4-4.xml
8:      <arg name="sim" default="false"/>
23:    <group unless="$(arg sim)">
33:        <arg name="sim" value="$(arg sim)"/>
37:        <arg name="sim" value="$(arg sim)"/>
40:    <!-- start common nodes used in hardware and simulation -->
53:    <!-- simulation only -->
54:    <include if="$(arg sim)" file="$(find cob_bringup)/tools/fake_diagnostics.launch">
labuser@lab-ubuntu-01:~$ head -100 /opt/ros/noetic/share/cob_bringup/robots/cob4-4.xml
<?xml version="1.0"?>
<launch>

```

```

    <!-- args -->
    <arg name="robot" value="cob4-4"/>
    <arg name="cob4-4-b1" default="localhost"/>
    <arg name="env-script" default="$(find cob_bringup)/env.sh"/>
    <arg name="sim" default="false"/>

    <!-- upload robot description -->
    <include file="$(find cob_hardware_config)/upload_robot.launch">
    <arg name="robot" value="$(arg robot)"/>
    </include>

    <!-- upload default configuration parameters -->
    <include file="$(find cob_default_robot_config)/upload_param.launch">
    <arg name="robot" value="$(arg robot)"/>
    </include>

    <group>
    <machine name="$(anon b1)" address="$(arg cob4-4-b1)" env-loader="$(arg
env-script)" default="true" timeout="30"/>

    <!-- start hardware -->
    <group unless="$(arg sim)">
    <include file="$(find cob_bringup)/tools/pc_monitor.launch">
        <arg name="robot" value="$(arg robot)"/>
        <arg name="pc" value="$(arg cob4-4-b1)"/>
        <arg name="enable_ntp_monitoring" value="false"/>
    </include>
    </group>

```



```

</include>
</group>

<include file="$(find cob_bringup)/drivers/bms.launch">
<arg name="robot" value="$(arg robot)"/>
<arg name="sim" value="$(arg sim)"/>
</include>
<include file="$(find cob_bringup)/components/cob4_base.launch">
<arg name="robot" value="$(arg robot)"/>
<arg name="sim" value="$(arg sim)"/>
</include>

<!-- start common nodes used in hardware and simulation -->
<include file="$(find cob_bringup)/tools/diagnostics_aggregator.launch"/>
<include file="$(find cob_bringup)/tools/teleop.launch">
<arg name="robot" value="$(arg robot)"/>
</include>
<include file="$(find cob_script_server)/launch/script_server.launch"/>
<!--include file="$(find cob_bringup)/tools/base_collision_observer.launch">
<arg name="robot" value="$(arg robot)"/>
</include-->
<include file="$(find cob_bringup)/tools/robot_state_publisher.launch">
<arg name="robot" value="$(arg robot)"/>
</include>

<!-- simulation only -->
<include if="$(arg sim)" file="$(find cob_bringup)/tools/fake_diagnostics.launch">
<arg name="fake_diagnostics" value="-b1, joy, safety"/>
</include>
</group>

<machine name="$(anon b1)" address="$(arg cob4-4-b1)" env-loader="$(arg
env-script)" default="true" timeout="30"/>

</launch>
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$

```

```
labuser@lab-ubuntu-01:~$ ls /opt/ros/noetic/share/cob_default_env_config/envs
```

automotive-assembly-line empty ipa-apartment ipa-kitchen ipa-office ipa-production-plant
izs-campus izs-carpark-top

labuser@lab-ubuntu-01:~\$

labuser@lab-ubuntu-01:~\$ grep -R "ROBOT_ENV" /opt/ros/noetic/share/cob_* 2>/dev/null

/opt/ros/noetic/share/cob_android_script_server/launch/script_server.launch: <arg
name="robot_env" default="\$(optenv ROBOT_ENV !!NO_ROBOT_ENV_SET!!)"/>

/opt/ros/noetic/share/cob_bringup/tools/dashboard.launch: <arg name="robot_env"
default="\$(optenv ROBOT_ENV !!NO_ROBOT_ENV_SET!!)"/>

/opt/ros/noetic/share/cob_default_env_config/test_env_config.launch: <arg
name="robot_env" default="\$(optenv ROBOT_ENV !!NO_ROBOT_ENV_SET!!)"/>

/opt/ros/noetic/share/cob_default_env_config/upload_object_locations.launch: <arg
name="robot_env" default="\$(optenv ROBOT_ENV !!NO_ROBOT_ENV_SET!!)"/>

/opt/ros/noetic/share/cob_default_env_config/upload_navigation_goals.launch: <arg
name="robot_env" default="\$(optenv ROBOT_ENV !!NO_ROBOT_ENV_SET!!)"/>

/opt/ros/noetic/share/cob_gazebo/launch/robot.launch: <arg name="robot_env"
default="\$(optenv ROBOT_ENV !!NO_ROBOT_ENV_SET!!)"/>

labuser@lab-ubuntu-01:~\$

labuser@lab-ubuntu-01:~\$ grep -R "robot_env" /opt/ros/noetic/share/cob_* 2>/dev/null

/opt/ros/noetic/share/cob_android_script_server/launch/script_server.launch: <arg
name="robot_env" default="\$(optenv ROBOT_ENV !!NO_ROBOT_ENV_SET!!)"/>

/opt/ros/noetic/share/cob_android_script_server/launch/script_server.launch: <rosparam
command="load" ns="nav_buttons" file="\$(arg pkg_env_config)/envs/\$(arg
robot_env)/command_gui_buttons.yaml"/>

/opt/ros/noetic/share/cob_bringup/tools/dashboard.launch: <arg name="robot_env"
default="\$(optenv ROBOT_ENV !!NO_ROBOT_ENV_SET!!)"/>

/opt/ros/noetic/share/cob_bringup/tools/dashboard.launch: <rosparam command="load"
ns="control_buttons" file="\$(arg pkg_env_config)/envs/\$(arg
robot_env)/command_gui_buttons.yaml"/>

/opt/ros/noetic/share/cob_bringup/tools/dashboard.launch: <rosparam command="load"
ns="nav_buttons" file="\$(arg pkg_env_config)/envs/\$(arg robot_env)/nav_buttons.yaml"/>

/opt/ros/noetic/share/cob_default_env_config/test_env_config.launch: <arg
name="robot_env" default="\$(optenv ROBOT_ENV !!NO_ROBOT_ENV_SET!!)"/>

/opt/ros/noetic/share/cob_default_env_config/test_env_config.launch: <rosparam
command="load" ns="nirvana" file="\$(find cob_default_env_config)/envs/\$(arg
robot_env)/map.yaml"/>

/opt/ros/noetic/share/cob_default_env_config/test_env_config.launch: <rosparam
command="load" ns="nirvana" file="\$(find cob_default_env_config)/envs/\$(arg
robot_env)/loc_map.yaml"/>

/opt/ros/noetic/share/cob_default_env_config/test_env_config.launch: <!--rosparam
command="load" ns="nirvana" file="\$(find cob_default_env_config)/envs/\$(arg
robot_env)/line_maps/laserline.yaml"/>

/opt/ros/noetic/share/cob_default_env_config/test_env_config.launch: <rosparam
command="load" ns="nirvana" file="\$(find cob_default_env_config)/envs/\$(arg
robot_env)/point_maps/lasermarker.yaml"/>

```

/opt/ros/noetic/share/cob_default_env_config/test_env_config.launch:    <rosparam
command="load" ns="nirvana" file="$(find cob_default_env_config)/envs/$(arg
robot_env)/polygon_maps/restriction_polygon.yaml"/>
/opt/ros/noetic/share/cob_default_env_config/test_env_config.launch:    <rosparam
command="load" ns="nirvana" file="$(find cob_default_env_config)/envs/$(arg
robot_env)/pose_maps/lasercorner.yaml"/>
/opt/ros/noetic/share/cob_default_env_config/test_env_config.launch:    <rosparam
command="load" ns="nirvana" file="$(find cob_default_env_config)/envs/$(arg
robot_env)/path_segments.yaml"/>
/opt/ros/noetic/share/cob_default_env_config/upload_object_locations.launch:    <arg
name="robot_env" default="$(optenv ROBOT_ENV !!NO_ROBOT_ENV_SET!!)"/>
/opt/ros/noetic/share/cob_default_env_config/upload_object_locations.launch:    <rosparam
command="load" ns="objects" file="$(find cob_default_env_config)/envs/$(arg
robot_env)/object_locations.yaml"/>
/opt/ros/noetic/share/cob_default_env_config/upload_object_locations.launch:    <rosparam
command="load" ns="object_groups" file="$(find cob_default_env_config)/envs/$(arg
robot_env)/object_groups.yaml"/>
/opt/ros/noetic/share/cob_default_env_config/upload_navigation_goals.launch:    <arg
name="robot_env" default="$(optenv ROBOT_ENV !!NO_ROBOT_ENV_SET!!)"/>
/opt/ros/noetic/share/cob_default_env_config/upload_navigation_goals.launch:    <rosparam
command="load" ns="/script_server/base" file="$(find cob_default_env_config)/envs/$(arg
robot_env)/navigation_goals.yaml"/>
/opt/ros/noetic/share/cob_gazebo/launch/robot.launch:    <arg name="robot_env"
default="$(optenv ROBOT_ENV !!NO_ROBOT_ENV_SET!!)"/>
/opt/ros/noetic/share/cob_gazebo/launch/robot.launch:    <node
name="spawn_gazebo_model" pkg="gazebo_ros" type="spawn_model" args="-urdf -param
robot_description -model robot -z 0.1 $(arg initial_config) -wait $(arg robot_env)"
respawn="false" output="screen" />
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$

```

```

labuser@lab-ubuntu-01:~$ export ROBOT=cob4-4
labuser@lab-ubuntu-01:~$ export ROBOT_ENV=ipa-office
labuser@lab-ubuntu-01:~$ roslaunch cob_gazebo robot.launch
... logging to
/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-4687.
log
Checking log directory for disk usage. This may take a while.
Press Ctrl-C to interrupt
Done checking log file disk usage. Usage is <1GB.

```

xacro: in-order processing became default in ROS Melodic. You can drop the option.
started roslaunch server <http://lab-ubuntu-01:38237/>

SUMMARY

=====

CLEAR PARAMETERS

* /bms/

PARAMETERS

- * /base/halt_detector/diagnostic_recovery_trigger: []
- * /base/halt_detector/recover_after_stuck_timeout: 3.0
- * /base/halt_detector/stop_threshold: 0.001
- * /base/halt_detector/stop_timeout: 10.0
- * /base/halt_detector/stuck_threshold: 0.174533
- * /base/halt_detector/stuck_timeout: 2.0
- * /base/joint_names: ['fl_caster_rotat...
- * /base/joint_state_controller/publish_rate: 50
- * /base/joint_state_controller/type: joint_state_contr...
- * /base/odometry_controller/child_frame_id: base_footprint
- * /base/odometry_controller/cov_pose: 0.1
- * /base/odometry_controller/cov_twist: 0.1
- * /base/odometry_controller/frame_id: odom_combined
- * /base/odometry_controller/publish_rate: 50
- * /base/odometry_controller/type: cob_omni_drive_co...
- * /base/odometry_controller/wheels: [{'steer': 'fl_ca...
- * /base/twist_controller/defaults/steer_ctrl/d_phi_max: 10.0
- * /base/twist_controller/defaults/steer_ctrl/damp: 2.75
- * /base/twist_controller/defaults/steer_ctrl/dd_phi_max: 40.0
- * /base/twist_controller/defaults/steer_ctrl/spring: 15.0
- * /base/twist_controller/defaults/steer_ctrl/virt_mass: 0.1
- * /base/twist_controller/pub_divider: 1
- * /base/twist_controller/required_drive_mode: 3
- * /base/twist_controller/timeout: 0.5
- * /base/twist_controller/type: cob_omni_drive_co...
- * /base/twist_controller/wheels: [{'steer': 'fl_ca...
- * /base/twist_mux/locks: [{'name': 'pause_...
- * /base/twist_mux/topics: [{'name': 'collis...
- * /base/velocity_smoother/accel_lim_vx: 0.5
- * /base/velocity_smoother/accel_lim_vy: 0.5
- * /base/velocity_smoother/accel_lim_w: 0.5
- * /base/velocity_smoother/decel_factor: 1.5
- * /base/velocity_smoother/decel_factor_safe: 2.5

- * /base/velocity_smoother/frequency: 50.0
- * /base/velocity_smoother/robot_feedback: 1
- * /base/velocity_smoother/speed_lim_vx: 1.2
- * /base/velocity_smoother/speed_lim_vy: 1.0
- * /base/velocity_smoother/speed_lim_w: 1.2
- * /bms/bms_id_to_poll: 512
- * /bms/can_device: can1
- * /bms/diagnostics: [{ 'id': 2, 'field...
- * /bms/poll_frequency_hz: 20
- * /bms/topics: ['current', 'volt...
- * /diagnostic_aggregator/analyzers/io/analyzers/joystick/contains: ['joy']
- * /diagnostic_aggregator/analyzers/io/analyzers/joystick/path: Joystick
- * /diagnostic_aggregator/analyzers/io/analyzers/joystick/type: diagnostic_aggreg...
- * /diagnostic_aggregator/analyzers/io/path: IO
- * /diagnostic_aggregator/analyzers/io/type: diagnostic_aggreg...
- * /diagnostic_aggregator/analyzers/motors/analyzers/base/contains: ['base/']
- * /diagnostic_aggregator/analyzers/motors/analyzers/base/path: base
- * /diagnostic_aggregator/analyzers/motors/analyzers/base/type: diagnostic_aggreg...
- * /diagnostic_aggregator/analyzers/motors/path: Actuators
- * /diagnostic_aggregator/analyzers/motors/type: diagnostic_aggreg...
- * /diagnostic_aggregator/analyzers/sensors/analyzers/bms/contains: ['bms']
- * /diagnostic_aggregator/analyzers/sensors/analyzers/bms/path: bms
- * /diagnostic_aggregator/analyzers/sensors/analyzers/bms/type: diagnostic_aggreg...
- * /diagnostic_aggregator/analyzers/sensors/path: Sensors
- * /diagnostic_aggregator/analyzers/sensors/type: diagnostic_aggreg...
- * /diagnostic_aggregator/other_as_errors: True
- * /joint_state_publisher/rate: 100
- * /joint_state_publisher/source_list: ['/base/joint_sta...
- * /joy_node_lab_ubuntu_01_4687_5655366149840568312/deadzone: 0.12
- * /joy_node_lab_ubuntu_01_4687_5655366149840568312/dev: /dev/joypad
- * /robot_description: <?xml version="1....
- * /roscdistro: noetic
- * /rosversion: 1.17.4
- * /script_server/base/service_ns: /base/driver
- * /script_server/base/topic_name: /base/twist_mux/c...
- * /teleop/teleop_node_lab_ubuntu_01_4687_7511819879343591480/apply_ramp: False
- * /teleop/teleop_node_lab_ubuntu_01_4687_7511819879343591480/axis_vx: 1
- * /teleop/teleop_node_lab_ubuntu_01_4687_7511819879343591480/axis_vy: 0
- * /teleop/teleop_node_lab_ubuntu_01_4687_7511819879343591480/axis_yaw: 3
- *

/teleop/teleop_node_lab_ubuntu_01_4687_7511819879343591480/components/base/twist_ma
x_acc: [0.5, 0.5, 0.7]

```

*
/teleop/teleop_node_lab_ubuntu_01_4687_7511819879343591480/components/base/twist_ma
x_velocity: [0.3, 0.2, 0.3]
*
/teleop/teleop_node_lab_ubuntu_01_4687_7511819879343591480/components/base/twist_saf
ety_topic_name: /base/collision_v...
*
/teleop/teleop_node_lab_ubuntu_01_4687_7511819879343591480/components/base/twist_topi
c_name: /base/twist_mux/c...
* /teleop/teleop_node_lab_ubuntu_01_4687_7511819879343591480/deadman_button: 5
* /teleop/teleop_node_lab_ubuntu_01_4687_7511819879343591480/init_button: 7
* /teleop/teleop_node_lab_ubuntu_01_4687_7511819879343591480/joy_num_axes: 6
* /teleop/teleop_node_lab_ubuntu_01_4687_7511819879343591480/joy_num_modes: 1
* /teleop/teleop_node_lab_ubuntu_01_4687_7511819879343591480/run_factor: 2.0
* /teleop/teleop_node_lab_ubuntu_01_4687_7511819879343591480/safety_button: 4

```

NODES

```

/
  bms (cob_bms_driver/fake_bms.py)
  dashboard_aggregator (cob_dashboard/cob_dashboard_aggregator.py)
  diagnostic_aggregator (diagnostic_aggregator/aggregator_node)
  fake_diagnostics_lab_ubuntu_01_4687_6653625250638953206
(cob_helper_tools/fake_diagnostics.py)
  joint_state_publisher (joint_state_publisher/joint_state_publisher)
  joy_node_lab_ubuntu_01_4687_5655366149840568312 (joy/joy_node)
  robot_state_publisher (robot_state_publisher/robot_state_publisher)
  script_server (cob_script_server/script_server.py)
  spawn_gazebo_model (gazebo_ros/spawn_model)
/base/
  base_controller_spawner (controller_manager/controller_manager)
  fake_driver (cob_helper_tools/fake_driver.py)
  halt_detector (cob_base_controller_utils/cob_halt_detector)
  twist_marker (twist_mux/twist_marker)
  twist_mux (twist_mux/twist_mux)
  velocity_smoother (cob_base_velocity_smoother/velocity_smoother)
/bms/
  power_state_aggregator (cob_bms_driver/power_state_aggregator.py)
/teleop/
  teleop_node_lab_ubuntu_01_4687_7511819879343591480 (cob_teleop/cob_teleop)

```

auto-starting new master

process[master]: started with pid [4699]

ROS_MASTER_URI=http://localhost:11311

setting /run_id to 9adb3ed0-5eab-11f1-b340-0390f703237c
process[rosout-1]: started with pid [4709]
started core service [/rosout]
process[bms-2]: started with pid [4717]
process[bms/power_state_aggregator-3]: started with pid [4718]
process[base/halt_detector-4]: started with pid [4719]
process[base/fake_driver-5]: started with pid [4720]
process[base/base_controller_spawner-6]: started with pid [4725]
[INFO] [1780422656.716094421]: waitForService: Service [/base/driver/halt] has not been advertised, waiting...
process[base/twist_mux-7]: started with pid [4727]
process[base/twist_marker-8]: started with pid [4728]
[INFO] [1780422656.727693178]: Topic handler 'collision_velocity_filter' subscribed to topic 'twist_mux/command_safe': timeout = 0.250000s, priority = 10
process[base/velocity_smoother-9]: started with pid [4733]
[INFO] [1780422656.729518515]: Topic handler 'navigation' subscribed to topic 'twist_mux/command_navigation': timeout = 0.250000s, priority = 20
[INFO] [1780422656.732455719]: Topic handler 'syncmm' subscribed to topic 'twist_mux/command_syncmm': timeout = 0.500000s, priority = 60
process[diagnostic_aggregator-10]: started with pid [4739]
[INFO] [1780422656.735158439]: Topic handler 'script_server' subscribed to topic 'twist_mux/command_script_server': timeout = 0.500000s, priority = 70
process[dashboard_aggregator-11]: started with pid [4742]
[INFO] [1780422656.738737721]: Topic handler 'teleop_keyboard' subscribed to topic 'twist_mux/command_teleop_keyboard': timeout = 0.500000s, priority = 80
[INFO] [1780422656.741152711]: Topic handler 'teleop_android' subscribed to topic 'twist_mux/command_teleop_android': timeout = 0.500000s, priority = 90
[INFO] [1780422656.743619758]: Topic handler 'teleop_joystick' subscribed to topic 'twist_mux/command_teleop_joy': timeout = 0.250000s, priority = 100
process[joy_node_lab_ubuntu_01_4687_5655366149840568312-12]: started with pid [4747]
[INFO] [1780422656.747357102]: Topic handler 'pause_navigation' subscribed to topic 'twist_mux/locks/pause_navigation': timeout = None, priority = 21
process[teleop/teleop_node_lab_ubuntu_01_4687_7511819879343591480-13]: started with pid [4748]
[INFO] [1780422656.751552123]: Topic handler 'pause_teleop' subscribed to topic 'twist_mux/locks/pause_teleop': timeout = None, priority = 101
[INFO] [1780422656.752285883]: Reconfigure request : 1.200000 1.000000 1.200000 0.500000 0.500000 0.500000 1.500000 2.500000
[INFO] [1780422656.754975612]: Topic handler 'pause_all' subscribed to topic 'twist_mux/locks/pause_all': timeout = None, priority = 255
process[script_server-14]: started with pid [4757]
process[robot_state_publisher-15]: started with pid [4759]
process[joint_state_publisher-16]: started with pid [4765]

```
process[fake_diagnostics_lab_ubuntu_01_4687_6653625250638953206-17]: started with pid [4768]
process[spawn_gazebo_model-18]: started with pid [4769]
[ERROR] [1780422656.785078515]: Couldn't open joystick /dev/joystick. Will retry every second.
[INFO] [1780422657.094913739]: waitForService: Service [/base/driver/halt] is now available.
[INFO] [1780422657.094940]: fake_driver running
[INFO] [1780422657.098153505]: waitForService: Service [/base/controller_manager/switch_controller] has not been advertised, waiting...
[INFO] [1780422657.107600]: power state aggregator running
[INFO] [1780422657.189079]: Waiting for model ipa-office before proceeding.
[INFO] [1780422658.293864]: script_server is running
```

```
labuser@lab-ubuntu-01:~$ find /opt/ros/noetic/share -name "*.world"
/opt/ros/noetic/share/stage/worlds/autolab.world
/opt/ros/noetic/share/stage/worlds/roomba.world
/opt/ros/noetic/share/stage/worlds/lsp_test.world
/opt/ros/noetic/share/stage/worlds/fasr.world
/opt/ros/noetic/share/stage/worlds/wifi.world
/opt/ros/noetic/share/stage/worlds/pioneer_walle.world
/opt/ros/noetic/share/stage/worlds/sensor_noise_module_demo.world
/opt/ros/noetic/share/stage/worlds/fasr_plan.world
/opt/ros/noetic/share/stage/worlds/camera.world
/opt/ros/noetic/share/stage/worlds/SFU.world
/opt/ros/noetic/share/stage/worlds/simple.world
/opt/ros/noetic/share/stage/worlds/large.world
/opt/ros/noetic/share/stage/worlds/uoa_robotics_lab.world
/opt/ros/noetic/share/stage/worlds/mbicp.world
/opt/ros/noetic/share/stage/worlds/pioneer_flocking.world
/opt/ros/noetic/share/stage/worlds/wifi/wifi_itu.world
/opt/ros/noetic/share/stage/worlds/wifi/wifi.world
/opt/ros/noetic/share/stage/worlds/wifi/wifi_simple.world
/opt/ros/noetic/share/stage/worlds/wifi/commando.world
/opt/ros/noetic/share/stage/worlds/wifi/hosp_wifi.world
/opt/ros/noetic/share/stage/worlds/wifi/hosp_wifi_5.world
/opt/ros/noetic/share/stage/worlds/wifi/wifi_ray.world
/opt/ros/noetic/share/stage/worlds/wifi/wifi_logdistance.world
/opt/ros/noetic/share/stage/worlds/fasr2.world
/opt/ros/noetic/share/stage/worlds/sensor_noise_demo.world
/opt/ros/noetic/share/stage/worlds/everything.world
/opt/ros/noetic/share/stage_ros/world/willow-four-erratics-multisensor.world
```



```

/opt/ros/noetic/share/stage_ros/world/intense.world
/opt/ros/noetic/share/stage_ros/world/willow-erratic.world
/opt/ros/noetic/share/stage_ros/world/willow-four-erratics.world
/opt/ros/noetic/share/stage_ros/world/erratic-inc.world
/opt/ros/noetic/share/gazebo_plugins/test/p3d_test/worlds/single_pendulum.world
/opt/ros/noetic/share/gazebo_plugins/test/p3d_test/worlds/3_double_pendulums.world
/opt/ros/noetic/share/gazebo_plugins/test/p3d_test/worlds/double_pendulum.world
/opt/ros/noetic/share/gazebo_plugins/test/p3d_test/worlds/3_single_pendulums.world
/opt/ros/noetic/share/gazebo_plugins/test/bumper_test/gazebo_ros_bumper.world
/opt/ros/noetic/share/gazebo_plugins/test/set_model_state_test/set_model_state_test_p2dx.world
/opt/ros/noetic/share/gazebo_plugins/test/test_worlds/test_lasers.world
/opt/ros/noetic/share/gazebo_plugins/test/test_worlds/bumper_test.world
/opt/ros/noetic/share/gazebo_plugins/test/test_worlds/gazebo_ros_block_laser_clipping.world
/opt/ros/noetic/share/gazebo_plugins/test/test_worlds/gazebo_ros_trimesh_collision.world
/opt/ros/noetic/share/gazebo_plugins/test/test_worlds/elevator.world
/opt/ros/noetic/share/gazebo_plugins/test/test_worlds/gazebo_ros_laser.world
/opt/ros/noetic/share/gazebo_plugins/test/test_worlds/gazebo_ros_depth_camera.world
/opt/ros/noetic/share/gazebo_plugins/test/test_worlds/gazebo_ros_range.world
/opt/ros/noetic/share/gazebo_plugins/test/test_worlds/gazebo_ros_camera.world
/opt/ros/noetic/share/gazebo_plugins/test/test_worlds/gazebo_ros_gpu_laser.world
/opt/ros/noetic/share/gazebo_plugins/test/test_worlds/gazebo_ros_block_laser.world
/opt/ros/noetic/share/gazebo_plugins/test/test_worlds/trisphere_cycle_wheel_slip.world
/opt/ros/noetic/share/gazebo_plugins/test/camera/triggered_camera.world
/opt/ros/noetic/share/gazebo_plugins/test/camera/depth_camera.world
/opt/ros/noetic/share/gazebo_plugins/test/camera/camera.world
/opt/ros/noetic/share/gazebo_plugins/test/camera/multicamera.world
/opt/ros/noetic/share/gazebo_plugins/test/camera/distortion_pincushion.world
/opt/ros/noetic/share/gazebo_plugins/test/camera/distortion_barrel.world
/opt/ros/noetic/share/gazebo_plugins/test/camera/camera16bit.world
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ grep -R "ipa-office" /opt/ros/noetic/share 2>/dev/null
/opt/ros/noetic/share/cob_default_env_config/cmake/cob_default_env_config-extras.cmake:set(
cob_default_env_config_ENVLIST
empty;ipa-apartment;ipa-kitchen;ipa-office;ipa-production-plant;izs-campus;izs-carpark-top)
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$

```

```

*****

```

```

labuser@lab-ubuntu-01:~$ find /opt/ros/noetic/share/cob_default_env_config/envs/ipa-office
-type f
/opt/ros/noetic/share/cob_default_env_config/envs/ipa-office/map.pgm

```

```

/opt/ros/noetic/share/cob_default_env_config/envs/ipa-office/line_maps/lineMap_IPA4thFloor.yaml
/opt/ros/noetic/share/cob_default_env_config/envs/ipa-office/nav_buttons.yaml
/opt/ros/noetic/share/cob_default_env_config/envs/ipa-office/navigation_goals.yaml
/opt/ros/noetic/share/cob_default_env_config/envs/ipa-office/loc_map.yaml
/opt/ros/noetic/share/cob_default_env_config/envs/ipa-office/object_groups.yaml
/opt/ros/noetic/share/cob_default_env_config/envs/ipa-office/polygon_maps/restriction_polygon.yaml
/opt/ros/noetic/share/cob_default_env_config/envs/ipa-office/pose_maps/corner_map.yaml
/opt/ros/noetic/share/cob_default_env_config/envs/ipa-office/map.yaml
/opt/ros/noetic/share/cob_default_env_config/envs/ipa-office/point_maps/reflMarkerMap_IPA4thFloor.yaml
/opt/ros/noetic/share/cob_default_env_config/envs/ipa-office/path_segments.yaml
/opt/ros/noetic/share/cob_default_env_config/envs/ipa-office/command_gui_buttons.yaml
/opt/ros/noetic/share/cob_default_env_config/envs/ipa-office/object_locations.yaml
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ find /opt/ros/noetic/share/cob_default_env_config/envs/empty -type f
/opt/ros/noetic/share/cob_default_env_config/envs/empty/map.pgm
/opt/ros/noetic/share/cob_default_env_config/envs/empty/nav_buttons.yaml
/opt/ros/noetic/share/cob_default_env_config/envs/empty/navigation_goals.yaml
/opt/ros/noetic/share/cob_default_env_config/envs/empty/loc_map.yaml
/opt/ros/noetic/share/cob_default_env_config/envs/empty/object_groups.yaml
/opt/ros/noetic/share/cob_default_env_config/envs/empty/map.yaml
/opt/ros/noetic/share/cob_default_env_config/envs/empty/path_segments.yaml
/opt/ros/noetic/share/cob_default_env_config/envs/empty/command_gui_buttons.yaml
/opt/ros/noetic/share/cob_default_env_config/envs/empty/object_locations.yaml
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ find /opt/ros/noetic/share -type d -name "models"
/opt/ros/noetic/share/gazebo_plugins/Media/models
/opt/ros/noetic/share/gazebo_plugins/Media/models/chair/models
/opt/ros/noetic/share/rviz/ogre_media/models
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ find ~/.gazebo -type d 2>/dev/null
/home/labuser/.gazebo
/home/labuser/.gazebo/server-11345
/home/labuser/.gazebo/client-11345
labuser@lab-ubuntu-01:~$

```

```

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```

```

labuser@lab-ubuntu-01:~$ find /home/labuser -type d -name "cob_simulation" 2>/dev/null
/home/labuser/catkin_ws/build/catkin_generated/metapackages/cob_simulation
/home/labuser/catkin_ws/build/cob_simulation

```

```

/home/labuser/catkin_ws/build/cob_simulation/cob_simulation
/home/labuser/catkin_ws/build/cob_simulation/cob_simulation/catkin_generated/stamps/cob_si
mulation
/home/labuser/catkin_ws/src/cob_simulation
/home/labuser/catkin_ws/src/cob_simulation/cob_simulation
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ find ~ -type f -name "setup.bash" 2>/dev/null
/home/labuser/catkin_ws/build/catkin_generated/installspace/setup.bash
/home/labuser/catkin_ws/devel/setup.bash
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$

```

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```

labuser@lab-ubuntu-01:~$ source /opt/ros/noetic/setup.bash
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ source ~/catkin_ws/devel/setup.bash
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ source ~/catkin_ws/devel/setup.bash
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ echo $ROS_PACKAGE_PATH
/home/labuser/catkin_ws/src:/opt/ros/noetic/share
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ rospack find cob_simulation
[rospack] Error: package 'cob_simulation' not found
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ source /opt/ros/noetic/setup.bash
labuser@lab-ubuntu-01:~$ source ~/catkin_ws/devel/setup.bash
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ echo $ROS_PACKAGE_PATH
/home/labuser/catkin_ws/src:/opt/ros/noetic/share
labuser@lab-ubuntu-01:~$ rospack find cob_simulation
[rospack] Error: package 'cob_simulation' not found
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ roscd cob_simulation
labuser@lab-ubuntu-01:~/catkin_ws/src/cob_simulation/cob_simulation$ ls
CHANGELOG.rst CMakeLists.txt package.xml
labuser@lab-ubuntu-01:~/catkin_ws/src/cob_simulation/cob_simulation$
labuser@lab-ubuntu-01:~/catkin_ws/src/cob_simulation/cob_simulation$ find $(rospack find
cob_simulation) -name "*.launch"

```

```
[rospack] Error: package 'cob_simulation' not found
labuser@lab-ubuntu-01:~/catkin_ws/src/cob_simulation/cob_simulation$
labuser@lab-ubuntu-01:~/catkin_ws/src/cob_simulation/cob_simulation$
```

```
labuser@lab-ubuntu-01:~/catkin_ws/src/cob_simulation/cob_simulation$ cd
~/catkin_ws/src/cob_simulation
labuser@lab-ubuntu-01:~/catkin_ws/src/cob_simulation$ ls
cob_bringup_sim  cob_gazebo  cob_gazebo_objects  cob_gazebo_tools  cob_gazebo_worlds
cob_simulation  LICENSE  README.md
labuser@lab-ubuntu-01:~/catkin_ws/src/cob_simulation$
labuser@lab-ubuntu-01:~/catkin_ws/src/cob_simulation$ find ~/catkin_ws/src/cob_simulation
-name "*.launch"
/home/labuser/catkin_ws/src/cob_simulation/cob_bringup_sim/launch/robot.launch
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/world.launch
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/world_joint_state_publisher.launch
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/display_world.launch
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_objects/launch/spawn_object_urdf.launch
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo/launch/robot.launch
labuser@lab-ubuntu-01:~/catkin_ws/src/cob_simulation$
labuser@lab-ubuntu-01:~/catkin_ws/src/cob_simulation$ find ~/catkin_ws/src/cob_simulation
-maxdepth 2 -name "package.xml"
/home/labuser/catkin_ws/src/cob_simulation/cob_bringup_sim/package.xml
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/package.xml
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_tools/package.xml
/home/labuser/catkin_ws/src/cob_simulation/cob_simulation/package.xml
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_objects/package.xml
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo/package.xml
labuser@lab-ubuntu-01:~/catkin_ws/src/cob_simulation$
labuser@lab-ubuntu-01:~/catkin_ws/src/cob_simulation$
```

```
labuser@lab-ubuntu-01:~/catkin_ws/src/cob_simulation$ head -100
~/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/world.launch
<?xml version="1.0"?>
<launch>

  <!-- define arguments -->
  <arg name="robot_env" default="$(optenv ROBOT_ENV !!NO_ROBOT_ENV_SET!!)" />
```

```

<arg name="pkg_env_config" default="$(find cob_default_env_config)"/>
<arg name="pkg_gazebo_worlds" default="$(find cob_gazebo_worlds)"/>
<arg name="world_name" default="worlds/empty.world"/>

<arg name="debug" default="false" />
<arg name="headless" default="false" />
<arg name="paused" default="false"/>
<arg name="use_sim_time" default="true"/>
<arg name="gui" default="true"/>

<env name="GAZEBO_MODEL_PATH" value="$(arg
pkg_gazebo_worlds)/Media/models:$(optenv GAZEBO_MODEL_PATH)"/>

<!-- upload default configuration parameters -->
<include file="$(arg pkg_env_config)/upload_navigation_goals.launch">
<arg name="robot_env" value="$(arg robot_env)" />
</include>
<include file="$(arg pkg_env_config)/upload_object_locations.launch">
<arg name="robot_env" value="$(arg robot_env)" />
</include>

<!-- start gazebo with empty world -->
<include file="$(find gazebo_ros)/launch/empty_world.launch">
<arg name="world_name" value="$(arg world_name)"/>
<arg name="debug" value="$(arg debug)" />
<arg name="headless" value="$(arg headless)"/>
<arg name="paused" value="$(arg paused)" />
<arg name="use_sim_time" value="$(arg use_sim_time)" />
<arg name="gui" value="$(arg gui)" />
</include>

<!-- spawn environment and start environment-specific nodes -->
<include file="$(arg pkg_gazebo_worlds)/launch/$(arg robot_env).xml">
<arg name="robot_env" value="$(arg robot_env)" />
</include>

</launch>
labuser@lab-ubuntu-01:~/catkin_ws/src/cob_simulation$
labuser@lab-ubuntu-01:~/catkin_ws/src/cob_simulation$
labuser@lab-ubuntu-01:~/catkin_ws/src/cob_simulation$ head -100
~/catkin_ws/src/cob_simulation/cob_bringup_sim/launch/robot.launch
<?xml version="1.0"?>
<launch>

```

```

<!-- define arguments -->
<arg name="robot" default="$(optenv ROBOT !!NO_ROBOT_SET!!)"/>
<arg name="robot_env" default="$(optenv ROBOT_ENV !!NO_ROBOT_ENV_SET!!)"/>
<arg name="pkg_robot_config" default="$(find cob_default_robot_config)"/>
<arg name="pkg_env_config" default="$(find cob_default_env_config)"/>
<arg name="pkg_gazebo_worlds" default="$(find cob_gazebo_worlds)"/>
<arg name="world_name" default="worlds/empty.world"/>
<arg name="initial_config" default=""/>

<arg name="paused" default="false"/>
<arg name="use_sim_time" default="true"/>
<arg name="gui" default="true"/>

<!-- startup simulated world -->
<include file="$(find cob_gazebo_worlds)/launch/world.launch">
<arg name="robot_env" value="$(arg robot_env)" />
<arg name="pkg_env_config" value="$(arg pkg_env_config)" />
<arg name="pkg_gazebo_worlds" value="$(arg pkg_gazebo_worlds)" />
<arg name="world_name" value="$(arg world_name)"/>
<arg name="paused" value="$(arg paused)" />
<arg name="use_sim_time" value="$(arg use_sim_time)" />
<arg name="gui" value="$(arg gui)" />
</include>

<!-- startup simulated robot -->
<include file="$(find cob_gazebo)/launch/robot.launch" >
<arg name="robot" value="$(arg robot)" />
<arg name="robot_env" value="$(arg robot_env)" />
<arg name="pkg_robot_config" value="$(arg pkg_robot_config)" />
<arg name="paused" value="$(arg paused)" />
<arg name="initial_config" value="$(arg initial_config)"/>
</include>

</launch>

```

```

labuser@lab-ubuntu-01:~/catkin_ws/src/cob_simulation$
labuser@lab-ubuntu-01:~/catkin_ws/src/cob_simulation$

```

```

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```

```

labuser@lab-ubuntu-01:~$ source /opt/ros/noetic/setup.bash
labuser@lab-ubuntu-01:~$ source ~/catkin_ws/devel/setup.bash
labuser@lab-ubuntu-01:~$ export ROBOT=cob4-4

```

```
labuser@lab-ubuntu-01:~$ export ROBOT_ENV=ipa-office
labuser@lab-ubuntu-01:~$ roslaunch cob_bringup_sim robot.launch
... logging to
/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-6295.
log
Checking log directory for disk usage. This may take a while.
Press Ctrl-C to interrupt
Done checking log file disk usage. Usage is <1GB.
```

xacro: in-order processing became default in ROS Melodic. You can drop the option.
xacro: in-order processing became default in ROS Melodic. You can drop the option.
started roslaunch server http://lab-ubuntu-01:35303/

SUMMARY

=====

CLEAR PARAMETERS

- * /bms/

PARAMETERS

- * /base/halt_detector/diagnostic_recovery_trigger: []
- * /base/halt_detector/recover_after_stuck_timeout: 3.0
- * /base/halt_detector/stop_threshold: 0.001
- * /base/halt_detector/stop_timeout: 10.0
- * /base/halt_detector/stuck_threshold: 0.174533
- * /base/halt_detector/stuck_timeout: 2.0
- * /base/joint_names: ['fl_caster_rotat...
- * /base/joint_state_controller/publish_rate: 50
- * /base/joint_state_controller/type: joint_state_contr...
- * /base/odometry_controller/child_frame_id: base_footprint
- * /base/odometry_controller/cov_pose: 0.1
- * /base/odometry_controller/cov_twist: 0.1
- * /base/odometry_controller/frame_id: odom_combined
- * /base/odometry_controller/publish_rate: 50
- * /base/odometry_controller/type: cob_omni_drive_co...
- * /base/odometry_controller/wheels: [{'steer': 'fl_ca...
- * /base/twist_controller/defaults/steer_ctrl/d_phi_max: 10.0
- * /base/twist_controller/defaults/steer_ctrl/damp: 2.75
- * /base/twist_controller/defaults/steer_ctrl/dd_phi_max: 40.0
- * /base/twist_controller/defaults/steer_ctrl/spring: 15.0
- * /base/twist_controller/defaults/steer_ctrl/virt_mass: 0.1
- * /base/twist_controller/pub_divider: 1
- * /base/twist_controller/required_drive_mode: 3
- * /base/twist_controller/timeout: 0.5

- * /base/twist_controller/type: cob_omni_drive_co...
- * /base/twist_controller/wheels: [{'steer': 'fl_ca...
- * /base/twist_mux/locks: [{'name': 'pause_...
- * /base/twist_mux/topics: [{'name': 'collis...
- * /base/velocity_smoother/accel_lim_vx: 0.5
- * /base/velocity_smoother/accel_lim_vy: 0.5
- * /base/velocity_smoother/accel_lim_w: 0.5
- * /base/velocity_smoother/decel_factor: 1.5
- * /base/velocity_smoother/decel_factor_safe: 2.5
- * /base/velocity_smoother/frequency: 50.0
- * /base/velocity_smoother/robot_feedback: 1
- * /base/velocity_smoother/speed_lim_vx: 1.2
- * /base/velocity_smoother/speed_lim_vy: 1.0
- * /base/velocity_smoother/speed_lim_w: 1.2
- * /bms/bms_id_to_poll: 512
- * /bms/can_device: can1
- * /bms/diagnostics: [{'id': 2, 'field...
- * /bms/poll_frequency_hz: 20
- * /bms/topics: ['current', 'volt...
- * /diagnostic_aggregator/analyzers/io/analyzers/joystick/contains: ['joy']
- * /diagnostic_aggregator/analyzers/io/analyzers/joystick/path: Joystick
- * /diagnostic_aggregator/analyzers/io/analyzers/joystick/type: diagnostic_aggreg...
- * /diagnostic_aggregator/analyzers/io/path: IO
- * /diagnostic_aggregator/analyzers/io/type: diagnostic_aggreg...
- * /diagnostic_aggregator/analyzers/motors/analyzers/base/contains: ['base']
- * /diagnostic_aggregator/analyzers/motors/analyzers/base/path: base
- * /diagnostic_aggregator/analyzers/motors/analyzers/base/type: diagnostic_aggreg...
- * /diagnostic_aggregator/analyzers/motors/path: Actuators
- * /diagnostic_aggregator/analyzers/motors/type: diagnostic_aggreg...
- * /diagnostic_aggregator/analyzers/sensors/analyzers/bms/contains: ['bms']
- * /diagnostic_aggregator/analyzers/sensors/analyzers/bms/path: bms
- * /diagnostic_aggregator/analyzers/sensors/analyzers/bms/type: diagnostic_aggreg...
- * /diagnostic_aggregator/analyzers/sensors/path: Sensors
- * /diagnostic_aggregator/analyzers/sensors/type: diagnostic_aggreg...
- * /diagnostic_aggregator/other_as_errors: True
- * /gazebo/enable_ros_network: True
- * /joint_state_publisher/rate: 100
- * /joint_state_publisher/source_list: ['/base/joint_sta...
- * /joy_node_lab_ubuntu_01_6295_2017253021453234181/deadzone: 0.12
- * /joy_node_lab_ubuntu_01_6295_2017253021453234181/dev: /dev/joypad
- * /robot_description: <?xml version="1....
- * /roscdistro: noetic
- * /rosversion: 1.17.4
- * /script_server/base/service_ns: /base/driver


```

* /script_server/base/topic_name: /base/twist_mux/c...
* /teleop/teleop_node_lab_ubuntu_01_6295_2758085363752191969/apply_ramp: False
* /teleop/teleop_node_lab_ubuntu_01_6295_2758085363752191969/axis_vx: 1
* /teleop/teleop_node_lab_ubuntu_01_6295_2758085363752191969/axis_vy: 0
* /teleop/teleop_node_lab_ubuntu_01_6295_2758085363752191969/axis_yaw: 3
*
/teleop/teleop_node_lab_ubuntu_01_6295_2758085363752191969/components/base/twist_max_acc: [0.5, 0.5, 0.7]
*
/teleop/teleop_node_lab_ubuntu_01_6295_2758085363752191969/components/base/twist_max_velocity: [0.3, 0.2, 0.3]
*
/teleop/teleop_node_lab_ubuntu_01_6295_2758085363752191969/components/base/twist_safety_topic_name: /base/collision_v...
*
/teleop/teleop_node_lab_ubuntu_01_6295_2758085363752191969/components/base/twist_topic_name: /base/twist_mux/c...
* /teleop/teleop_node_lab_ubuntu_01_6295_2758085363752191969/deadman_button: 5
* /teleop/teleop_node_lab_ubuntu_01_6295_2758085363752191969/init_button: 7
* /teleop/teleop_node_lab_ubuntu_01_6295_2758085363752191969/joy_num_axes: 6
* /teleop/teleop_node_lab_ubuntu_01_6295_2758085363752191969/joy_num_modes: 1
* /teleop/teleop_node_lab_ubuntu_01_6295_2758085363752191969/run_factor: 2.0
* /teleop/teleop_node_lab_ubuntu_01_6295_2758085363752191969/safety_button: 4
* /use_sim_time: True
* /world_description: <?xml version="1....

```

NODES

```

/
  bms (cob_bms_driver/fake_bms.py)
  dashboard_aggregator (cob_dashboard/cob_dashboard_aggregator.py)
  diagnostic_aggregator (diagnostic_aggregator/aggregator_node)
  fake_diagnostics_lab_ubuntu_01_6295_3108807645079097714
(cob_helper_tools/fake_diagnostics.py)
  gazebo (gazebo_ros/gzserver)
  gazebo_gui (gazebo_ros/gzclient)
  joint_state_publisher (joint_state_publisher/joint_state_publisher)
  joy_node_lab_ubuntu_01_6295_2017253021453234181 (joy/joy_node)
  robot_state_publisher (robot_state_publisher/robot_state_publisher)
  script_server (cob_script_server/script_server.py)
  spawn_gazebo_model (gazebo_ros/spawn_model)
  spawn_urdf_world (gazebo_ros/spawn_model)
/base/
  base_controller_spawner (controller_manager/controller_manager)
  fake_driver (cob_helper_tools/fake_driver.py)

```

```
    halt_detector (cob_base_controller_utils/cob_halt_detector)
    twist_mux (twist_mux/twist_mux)
    twist_mux (twist_mux/twist_mux)
    velocity_smoother (cob_base_velocity_smoother/velocity_smoother)
/bms/
    power_state_aggregator (cob_bms_driver/power_state_aggregator.py)
/teleop/
    teleop_node_lab_ubuntu_01_6295_2758085363752191969 (cob_teleop/cob_teleop)
```

ROS_MASTER_URI=http://localhost:11311

running rosparam delete /objects

ERROR: parameter [/objects] is not set

process[gazebo-1]: started with pid [6319]

process[gazebo_gui-2]: started with pid [6323]

process[spawn_urdf_world-3]: started with pid [6328]

process[bms-4]: started with pid [6330]

process[bms/power_state_aggregator-5]: started with pid [6331]

process[base/halt_detector-6]: started with pid [6332]

process[base/fake_driver-7]: started with pid [6333]

process[base/base_controller_spawner-8]: started with pid [6338]

process[base/twist_mux-9]: started with pid [6340]

[INFO] [1780423529.860192450]: waitForService: Service

/base/controller_manager/switch_controller has not been advertised, waiting...

process[base/twist_marker-10]: started with pid [6343]

process[base/velocity_smoother-11]: started with pid [6348]

[INFO] [1780423529.873583126]: Topic handler 'collision_velocity_filter' subscribed to topic

'twist_mux/command_safe': timeout = 0.250000s, priority = 10

process[diagnostic_aggregator-12]: started with pid [6354]

[INFO] [1780423529.879270180]: Topic handler 'navigation' subscribed to topic

'twist_mux/command_navigation': timeout = 0.250000s, priority = 20

process[dashboard_aggregator-13]: started with pid [6355]

process[joy_node_lab_ubuntu_01_6295_2017253021453234181-14]: started with pid [6361]

[INFO] [1780423529.890698864]: Topic handler 'syncmm' subscribed to topic

'twist_mux/command_syncmm': timeout = 0.500000s, priority = 60

process[teleop/teleop_node_lab_ubuntu_01_6295_2758085363752191969-15]: started with pid [6366]

process[script_server-16]: started with pid [6369]

process[robot_state_publisher-17]: started with pid [6373]

[INFO] [1780423529.908444095]: Topic handler 'script_server' subscribed to topic

'twist_mux/command_script_server': timeout = 0.500000s, priority = 70

process[joint_state_publisher-18]: started with pid [6378]

[INFO] [1780423529.923416378]: Topic handler 'teleop_keyboard' subscribed to topic

'twist_mux/command_teleop_keyboard': timeout = 0.500000s, priority = 80

process[fake_diagnostics_lab_ubuntu_01_6295_3108807645079097714-19]: started with pid [6388]
[WARN] [1780423529.940528552]: Shutdown request received.
process[spawn_gazebo_model-20]: started with pid [6389]
[INFO] [1780423529.948386991]: Reconfigure request : 1.200000 1.000000 1.200000 0.500000 0.500000 0.500000 1.500000 2.500000
[WARN] [1780423529.959496570]: Reason given for shutdown: [[/base/twist_marker] Reason: new node registered with same name]
[INFO] [1780423529.970828477]: Topic handler 'teleop_android' subscribed to topic 'twist_mux/command_teleop_android': timeout = 0.500000s, priority = 90
[WARN] [1780423529.986130970]: Shutdown request received.
[WARN] [1780423530.019895181]: Shutdown request received.
[WARN] [1780423530.020490229]: Reason given for shutdown: [[/diagnostic_aggregator] Reason: new node registered with same name]
[INFO] [1780423530.026572851]: Topic handler 'teleop_joystick' subscribed to topic 'twist_mux/command_teleop_joy': timeout = 0.250000s, priority = 100
[ERROR] [1780423530.057781345]: Couldn't open joystick /dev/joypad. Will retry every second.
[WARN] [1780423530.067444969]: Reason given for shutdown: [[/robot_state_publisher] Reason: new node registered with same name]
[INFO] [1780423530.098957513]: Topic handler 'pause_navigation' subscribed to topic 'twist_mux/locks/pause_navigation': timeout = None, priority = 21
[INFO] [1780423530.135183922]: Topic handler 'pause_teleop' subscribed to topic 'twist_mux/locks/pause_teleop': timeout = None, priority = 101
[INFO] [1780423530.152211587]: Topic handler 'pause_all' subscribed to topic 'twist_mux/locks/pause_all': timeout = None, priority = 255
[INFO] [1780423530.199822, 0.000000]: fake_driver running
[INFO] [1780423530.222605, 0.000000]: power state aggregator running
[base/twist_marker-10] process has finished cleanly
log file:
/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/base-twist_marker-10*.log
[robot_state_publisher-17] process has finished cleanly
log file:
/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/robot_state_publisher-17*.log
[INFO] [1780423530.311017, 0.000000]: Waiting for model ipa-office before proceeding.
[INFO] [1780423530.321314, 0.000000]: Loading model XML from ros parameter world_description
[INFO] [1780423530.328120, 0.000000]: Waiting for service /gazebo/spawn_urdf_model
[diagnostic_aggregator-12] process has finished cleanly
log file:
/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/diagnostic_aggregator-12*.log
[gazebo-1] process has died [pid 6319, exit code 255, cmd /opt/ros/noetic/lib/gazebo_ros/gzserver -e ode worlds/empty.world __name:=gazebo __log:=/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/gazebo-1.log].
log file: /home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/gazebo-1*.log

```
[INFO] [1780423530.566938637]: Finished loading Gazebo ROS API Plugin.
[INFO] [1780423530.567818843]: waitForService: Service [/gazebo_gui/set_physics_properties]
has not been advertised, waiting...
[INFO] [1780423531.408501, 0.000000]: script_server is running
```

*****+

```
labuser@lab-ubuntu-01:~$ tail -100
/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/gazebo-1.log
tail: cannot open
'/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/gazebo-1.log' for reading: No
such file or directory
labuser@lab-ubuntu-01:~$ grep -i "error|fatal|exception"
/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/gazebo-1.log
grep: /home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/gazebo-1.log: No such
file or directory
labuser@lab-ubuntu-01:~$ find ~/catkin_ws/src/cob_simulation/cob_gazebo_worlds -name
"*.world"
labuser@lab-ubuntu-01:~$ ls ~/catkin_ws/src/cob_simulation/cob_gazebo_worlds/worlds
ls: cannot access '/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/worlds': No
such file or directory
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ find ~/catkin_ws/src/cob_simulation/cob_gazebo_worlds -maxdepth
3 -type f
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/CMakeLists.txt
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/urdf/ipa-production-plant/ipa-p
roduction-plant.property.xacro
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/urdf/ipa-production-plant/walls
.urdf.xacro
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/urdf/ipa-production-plant/sens
ors.urdf.xacro
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/urdf/ipa-production-plant/ipa-p
roduction-plant.urdf.xacro
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/urdf/common.xacro
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/urdf/materials.urdf.xacro
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/urdf/izs-carpark-top/izs-carpar
k-top.urdf.xacro
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/urdf/izs-carpark-top/world_con
troller.yaml
```

/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/urdf/ipa-office/ipa-office.urdf.xacro
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/urdf/ipa-apartment/landing.urdf.xacro
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/urdf/ipa-apartment/wall_ipa.urdf.xacro
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/urdf/ipa-apartment/apartment.urdf.xacro
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/urdf/ipa-apartment/elevator.urdf.xacro
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/urdf/ipa-apartment/ipa-apartment.urdf.xacro
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/urdf/ipa-apartment/office.urdf.xacro
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/urdf/ipa-apartment/kitchen.urdf.xacro
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/urdf/ipa-apartment/cabinets.urdf.xacro
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/urdf/ipa-apartment/world_controller.yaml
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/urdf/empty/empty.urdf.xacro
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/urdf/ipa-kitchen/ipa-kitchen.urdf.xacro
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/urdf/ipa-kitchen/world_controller.yaml
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/urdf/automotive-assembly-line/automotive-assembly-line.urdf.xacro
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/urdf/automotive-assembly-line/world_controller.yaml
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/ipa-office.xml
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/ipa-production-plant.xml
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/izs-campus.xml
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/world.launch
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/world_joint_state_publisher.launch
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/ipa-kitchen.xml
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/ipa-apartment.xml
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/empty.xml
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/display_world.launch
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/world_urdf.xml
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/izs-carpark-top.xml
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/automotive-assembly-line.xml

```

/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/CHANGELOG.rst
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/Media/models/ipa-office_walls
.dae
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/Media/models/kitchen_plank.d
ae
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/Media/models/boards_before_
bedroom.dae
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/scripts/elevator.py
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/package.xml
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/test/urdf_worlds.test
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ ls -l ~/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch
total 48
-rw-rw-r-- 1 labuser labuser 791 Nov 14 2025 automotive-assembly-line.xml
-rw-rw-r-- 1 labuser labuser 756 Nov 14 2025 display_world.launch
-rw-rw-r-- 1 labuser labuser 362 Nov 14 2025 empty.xml
-rw-rw-r-- 1 labuser labuser 999 Nov 14 2025 ipa-apartment.xml
-rw-rw-r-- 1 labuser labuser 668 Nov 14 2025 ipa-kitchen.xml
-rw-rw-r-- 1 labuser labuser 367 Nov 14 2025 ipa-office.xml
-rw-rw-r-- 1 labuser labuser 932 Nov 14 2025 ipa-production-plant.xml
-rw-rw-r-- 1 labuser labuser 372 Nov 14 2025 izs-campus.xml
-rw-rw-r-- 1 labuser labuser 372 Nov 14 2025 izs-carpark-top.xml
-rw-rw-r-- 1 labuser labuser 428 Nov 14 2025 world_joint_state_publisher.launch
-rw-rw-r-- 1 labuser labuser 1572 Nov 14 2025 world.launch
-rw-rw-r-- 1 labuser labuser 1738 Nov 14 2025 world_urdf.xml
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ ls -l ~/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c
total 576
-rw-rw-r-- 1 labuser labuser 1604 Jun 2 20:05 base-base_controller_spawner-6.log
-rw-rw-r-- 1 labuser labuser 1118 Jun 2 20:05 base-base_controller_spawner-8.log
-rw-rw-r-- 1 labuser labuser 1865 Jun 2 20:05 base-fake_driver-5.log
-rw-rw-r-- 1 labuser labuser 1166 Jun 2 20:05 base-fake_driver-7.log
-rw-rw-r-- 1 labuser labuser 0 Jun 2 20:05 base-twist_marker-10-stdout.log
-rw-rw-r-- 1 labuser labuser 0 Jun 2 19:50 base-twist_marker-8-stdout.log
-rw-rw-r-- 1 labuser labuser 3057 Jun 2 20:05 bms-2.log
-rw-rw-r-- 1 labuser labuser 1761 Jun 2 20:05 bms-4.log
-rw-rw-r-- 1 labuser labuser 3195 Jun 2 20:05 bms-power_state_aggregator-3.log
-rw-rw-r-- 1 labuser labuser 2090 Jun 2 20:05 bms-power_state_aggregator-5.log
-rw-rw-r-- 1 labuser labuser 2351 Jun 2 20:05 dashboard_aggregator-11.log
-rw-rw-r-- 1 labuser labuser 85 Jun 2 20:05 dashboard_aggregator-11-stdout.log
-rw-rw-r-- 1 labuser labuser 1110 Jun 2 20:05 dashboard_aggregator-13.log
-rw-rw-r-- 1 labuser labuser 0 Jun 2 20:05 dashboard_aggregator-13-stdout.log

```

```

-rw-rw-r-- 1 labuser labuser  0 Jun  2 19:50 diagnostic_aggregator-10-stdout.log
-rw-rw-r-- 1 labuser labuser  0 Jun  2 20:05 diagnostic_aggregator-12-stdout.log
-rw-rw-r-- 1 labuser labuser 1351 Jun  2 20:05
fake_diagnostics_lab_ubuntu_01_4687_6653625250638953206-17.log
-rw-rw-r-- 1 labuser labuser  0 Jun  2 19:50
fake_diagnostics_lab_ubuntu_01_4687_6653625250638953206-17-stdout.log
-rw-rw-r-- 1 labuser labuser 1145 Jun  2 20:05
fake_diagnostics_lab_ubuntu_01_6295_3108807645079097714-19.log
-rw-rw-r-- 1 labuser labuser  0 Jun  2 20:05
fake_diagnostics_lab_ubuntu_01_6295_3108807645079097714-19-stdout.log
-rw-rw-r-- 1 labuser labuser 1822 Jun  2 20:05 joint_state_publisher-16.log
-rw-rw-r-- 1 labuser labuser 1111 Jun  2 20:05 joint_state_publisher-18.log
-rw-rw-r-- 1 labuser labuser  0 Jun  2 19:50
joy_node_lab_ubuntu_01_4687_5655366149840568312-12-stdout.log
-rw-rw-r-- 1 labuser labuser  0 Jun  2 20:05
joy_node_lab_ubuntu_01_6295_2017253021453234181-14-stdout.log
-rw-rw-r-- 1 labuser labuser 214232 Jun  2 20:05 master.log
-rw-rw-r-- 1 labuser labuser 130483 Jun  2 20:05 roslaunch-lab-ubuntu-01-4687.log
-rw-rw-r-- 1 labuser labuser 142021 Jun  2 20:05 roslaunch-lab-ubuntu-01-6295.log
-rw-rw-r-- 1 labuser labuser  153 Jun  2 19:50 rosout-1-stdout.log
-rw-rw-r-- 1 labuser labuser  5044 Jun  2 20:05 rosout.log
-rw-rw-r-- 1 labuser labuser  3135 Jun  2 20:05 script_server-14.log
-rw-rw-r-- 1 labuser labuser  1946 Jun  2 20:05 script_server-16.log
-rw-rw-r-- 1 labuser labuser  1662 Jun  2 20:05 spawn_gazebo_model-18.log
-rw-rw-r-- 1 labuser labuser  1196 Jun  2 20:05 spawn_gazebo_model-20.log
-rw-rw-r-- 1 labuser labuser  1286 Jun  2 20:05 spawn_urdf_world-3.log
labuser@lab-ubuntu-01:~$

```

```

labuser@lab-ubuntu-01:~$ ls ~/catkin_ws/devel
env.sh lib local_setup.bash local_setup.fish local_setup.sh local_setup.zsh setup.bash
setup.fish setup.sh _setup_util.py setup.zsh share
labuser@lab-ubuntu-01:~$ ls ~/catkin_ws/build
atomic_configure bin catkin catkin_generated CATKIN_IGNORE catkin_make.cache
CMakeCache.txt CMakeFiles cmake_install.cmake cob_demo cob_simulation
cob_supported_robots CTestConfiguration.ini CTestCustom.cmake CTestTestfile.cmake gtest
Makefile test_results
labuser@lab-ubuntu-01:~$ cd ~/catkin_ws
labuser@lab-ubuntu-01:~/catkin_ws$ catkin_make
Base path: /home/labuser/catkin_ws
Source space: /home/labuser/catkin_ws/src
Build space: /home/labuser/catkin_ws/build

```

Devel space: /home/labuser/catkin_ws/devel

Install space: /home/labuser/catkin_ws/install

####

Running command: "make cmake_check_build_system" in

"/home/labuser/catkin_ws/build"

####

####

Running command: "make -j16 -l16" in "/home/labuser/catkin_ws/build"

####

labuser@lab-ubuntu-01:~/catkin_ws\$ source /opt/ros/noetic/setup.bash

labuser@lab-ubuntu-01:~/catkin_ws\$ source ~/catkin_ws/devel/setup.bash

labuser@lab-ubuntu-01:~/catkin_ws\$ echo \$CMAKE_PREFIX_PATH

/home/labuser/catkin_ws/devel:/opt/ros/noetic

labuser@lab-ubuntu-01:~/catkin_ws\$ rospack profile

Full tree crawl took 0.029829 seconds.

Directories marked with (*) contain no manifest. You may

want to delete these directories.

To get just of list of directories without manifests,

re-run the profile with --zombie-only

0.027005 /opt/ros/noetic/share

0.001302 /home/labuser/catkin_ws/src

0.000826 /home/labuser/catkin_ws/src/cob_simulation

0.000152 * /opt/ros/noetic/share/ament_index

0.000091 * /opt/ros/noetic/share/ament_index/resource_index

0.000017 * /opt/ros/noetic/share/ament_index/resource_index/packages

labuser@lab-ubuntu-01:~/catkin_ws\$ rospack find cob_gazebo_worlds

/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds

labuser@lab-ubuntu-01:~/catkin_ws\$ find ~/catkin_ws/src/cob_simulation -name

CATKIN_IGNORE

labuser@lab-ubuntu-01:~/catkin_ws\$ head -30

~/catkin_ws/src/cob_simulation/cob_gazebo_worlds/package.xml

<package format="2">

<name>cob_gazebo_worlds</name>

<version>0.7.8</version>

<description>This package provides some worlds for gazebo simulation.</description>

<license>Apache 2.0</license>

<url type="website">http://ros.org/wiki/cob_gazebo_worlds</url>

<!-- <url type="bugtracker"></url> -->

<maintainer email="jba@ipa.fhg.de">Jannik Abbenseth</maintainer>

<author email="fxm@ipa.fhg.de">Felix Messmer</author>


```
<author email="nhg@ipa.fhg.de">Nadia Hammoudeh Garcia</author>
<author email="fmw@ipa.fhg.de">Florian Weisshardt</author>
```

```
<buildtool_depend>catkin</buildtool_depend>
```

```
<exec_depend>cob_default_env_config</exec_depend>
<exec_depend>controller_manager</exec_depend>
<exec_depend>gazebo_msgs</exec_depend>
<exec_depend>gazebo_ros_control</exec_depend>
<exec_depend>gazebo_ros</exec_depend>
<exec_depend>joint_state_controller</exec_depend>
<exec_depend>joint_state_publisher</exec_depend>
<exec_depend>position_controllers</exec_depend>
<exec_depend>robot_state_publisher</exec_depend>
<exec_depend>rospy</exec_depend>
<exec_depend>std_msgs</exec_depend>
<exec_depend>tf</exec_depend>
<exec_depend>tf2_ros</exec_depend>
```

```
labuser@lab-ubuntu-01:~/catkin_ws$ head -30
```

```
~/catkin_ws/src/cob_simulation/cob_simulation/package.xml
```

```
<package format="2">
  <name>cob_simulation</name>
  <version>0.7.8</version>
  <description>The cob_simulation stack includes packages to work with Care-O-bot within
simulation environments, e.g. gazebo.</description>
```

```
<maintainer email="richard.bormann@ipa.fraunhofer.de">Richard Bormann</maintainer>
```

```
<license>Apache 2.0</license>
```

```
<url type="website">http://ros.org/wiki/cob_simulation</url>
```

```
<!-- <url type="bugtracker"></url> -->
```

```
<buildtool_depend>catkin</buildtool_depend>
<exec_depend>cob_bringup_sim</exec_depend>
<exec_depend>cob_gazebo</exec_depend>
<exec_depend>cob_gazebo_objects</exec_depend>
<exec_depend>cob_gazebo_tools</exec_depend>
<exec_depend>cob_gazebo_worlds</exec_depend>
```

```
<export>
  <metapackage/>
```

```
</export>
```

```
</package>
```

```
labuser@lab-ubuntu-01:~/catkin_ws$
```

```
*****+
```

```
labuser@lab-ubuntu-01:~/catkin_ws$ ls
```

```
~/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch  
automotive-assembly-line.xml display_world.launch empty.xml ipa-apartment.xml  
ipa-kitchen.xml ipa-office.xml ipa-production-plant.xml iza-campus.xml iza-carpark-top.xml  
world_joint_state_publisher.launch world.launch world_urdf.xml
```

```
labuser@lab-ubuntu-01:~/catkin_ws$ head -100
```

```
~/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/empty.xml
```

```
<?xml version="1.0"?>
```

```
<launch>
```

```
  <arg name="robot_env" default="empty"/>
```

```
  <arg name="publish_world_state" value="false"/>
```

```
  <!-- spawn environment -->
```

```
  <include file="$(find cob_gazebo_worlds)/launch/world_urdf.xml">
```

```
    <arg name="robot_env" value="$(arg robot_env)"/>
```

```
    <arg name="publish_world_state" value="$(arg publish_world_state)"/>
```

```
  </include>
```

```
</launch>
```

```
labuser@lab-ubuntu-01:~/catkin_ws$ head -100
```

```
~/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/ipa-office.xml
```

```
<?xml version="1.0"?>
```

```
<launch>
```

```
  <arg name="robot_env" default="ipa-office"/>
```

```
  <arg name="publish_world_state" value="false"/>
```

```
  <!-- spawn environment -->
```

```
  <include file="$(find cob_gazebo_worlds)/launch/world_urdf.xml">
```

```
    <arg name="robot_env" value="$(arg robot_env)"/>
```

```
    <arg name="publish_world_state" value="$(arg publish_world_state)"/>
```

```
  </include>
```

```
</launch>
```

```
labuser@lab-ubuntu-01:~/catkin_ws$ ls /opt/ros/noetic/share/cob_bringup/robots/*.xml
```

```
/opt/ros/noetic/share/cob_bringup/robots/cob4-10.xml
```

```
/opt/ros/noetic/share/cob_bringup/robots/cob4-20.xml
```

```
/opt/ros/noetic/share/cob_bringup/robots/cob4-30.xml
```

```

/opt/ros/noetic/share/cob_bringup/robots/cob4-7.xml
/opt/ros/noetic/share/cob_bringup/robots/raw3-5.xml
/opt/ros/noetic/share/cob_bringup/robots/cob4-11.xml
/opt/ros/noetic/share/cob_bringup/robots/cob4-23.xml
/opt/ros/noetic/share/cob_bringup/robots/cob4-3.xml
/opt/ros/noetic/share/cob_bringup/robots/cob4-8.xml
/opt/ros/noetic/share/cob_bringup/robots/raw-mini.xml
/opt/ros/noetic/share/cob_bringup/robots/cob4-13.xml
/opt/ros/noetic/share/cob_bringup/robots/cob4-24.xml
/opt/ros/noetic/share/cob_bringup/robots/cob4-4.xml
/opt/ros/noetic/share/cob_bringup/robots/cob4-9.xml
/opt/ros/noetic/share/cob_bringup/robots/cob4-16.xml
/opt/ros/noetic/share/cob_bringup/robots/cob4-25.xml
/opt/ros/noetic/share/cob_bringup/robots/cob4-5.xml
/opt/ros/noetic/share/cob_bringup/robots/raw3-1.xml
/opt/ros/noetic/share/cob_bringup/robots/cob4-18.xml
/opt/ros/noetic/share/cob_bringup/robots/cob4-29.xml
/opt/ros/noetic/share/cob_bringup/robots/cob4-6.xml
/opt/ros/noetic/share/cob_bringup/robots/raw3-3.xml
labuser@lab-ubuntu-01:~/catkin_ws$
labuser@lab-ubuntu-01:~/catkin_ws$

```

*****+

```

labuser@lab-ubuntu-01:~$ gazebo --verbose
Gazebo multi-robot simulator, version 11.15.1
Copyright (C) 2012 Open Source Robotics Foundation.
Released under the Apache 2 License.
http://gazebosim.org

```

```

#      # ##### ##### ### ##### #####
##     # #   #   #   # #   # #
# #   # #   #   #   # #   #
# #   #   #   #   # #   #####
# #   #   #   #   # #   #
#     ## #   #   #   # #   #
#     # #####   #   ### #####

```

This version of Gazebo, now called Gazebo classic, reaches end-of-life in January 2025. Users are highly encouraged to migrate to the new Gazebo using our migration guides (https://gazebosim.org/docs/latest/gazebo_classic_migration/)

[Msg] Waiting for master.
Gazebo multi-robot simulator, version 11.15.1
Copyright (C) 2012 Open Source Robotics Foundation.
Released under the Apache 2 License.
<http://gazebo-sim.org>

```
#      # ##### ##### ### ##### #####
##    # #   #   #   # #   # #
# # # #   #   #   # #   #
# # # #   #   #   # #   #####
# # # #   #   #   # #   #
#      ## #   #   #   # #   #
#      # #####   #   ### ##### #####
```

This version of Gazebo, now called Gazebo classic, reaches end-of-life in January 2025. Users are highly encouraged to migrate to the new Gazebo using our migration guides (https://gazebo-sim.org/docs/latest/gazebo_classic_migration/)

[Msg] Waiting for master.
[Msg] Connected to gazebo master @ <http://127.0.0.1:11345>
[Msg] Publicized address: 10.131.33.33
[Msg] Loading world file [/usr/share/gazebo-11/worlds/empty.world]
[Msg] Connected to gazebo master @ <http://127.0.0.1:11345>
[Msg] Publicized address: 10.131.33.33

```
labuser@lab-ubuntu-01:~$ tail -100
~/ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/spawn_urdf_world-3.log
[rospy.client][INFO] 2026-06-02 20:05:30,213: init_node, name[/spawn_urdf_world], pid[6328]
[xmllrpc][INFO] 2026-06-02 20:05:30,214: XML-RPC server binding to 0.0.0.0:0
[xmllrpc][INFO] 2026-06-02 20:05:30,214: Started XML-RPC server [http://lab-ubuntu-01:45909/]
[rospy.init][INFO] 2026-06-02 20:05:30,214: ROS Slave URI: [http://lab-ubuntu-01:45909/]
[rospy.impl.masterslave][INFO] 2026-06-02 20:05:30,214: _ready: http://lab-ubuntu-01:45909/
[rospy.registration][INFO] 2026-06-02 20:05:30,215: Registering with master node
http://localhost:11311
[xmllrpc][INFO] 2026-06-02 20:05:30,215: xml rpc node: starting XML-RPC server
[rospy.init][INFO] 2026-06-02 20:05:30,314: registered with master
```

```
[rospy.rosout][INFO] 2026-06-02 20:05:30,315: initializing /rosout core topic
[rospy.rosout][INFO] 2026-06-02 20:05:30,317: connected to core topic /rosout
[rospy.simtime][INFO] 2026-06-02 20:05:30,318: initializing /clock core topic
[rospy.simtime][INFO] 2026-06-02 20:05:30,319: connected to core topic /clock
[rosout][INFO] 2026-06-02 20:05:30,321: Loading model XML from ros parameter
world_description
[rosout][INFO] 2026-06-02 20:05:30,328: Waiting for service /gazebo/spawn_urdf_model
[rospy.internal][INFO] 2026-06-02 20:05:30,677: topic[/rosout] adding connection to [/rosout],
count 0
[rospy.core][INFO] 2026-06-02 20:15:13,715: signal_shutdown [signal-2]
[rospy.internal][INFO] 2026-06-02 20:15:13,716: topic[/rosout] removing connection to /rosout
[rospy.impl.masterslave][INFO] 2026-06-02 20:15:13,716: signal-2
[rospy.core][INFO] 2026-06-02 20:15:14,429: signal_shutdown [atexit]
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ tail -100
~/ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/spawn_gazebo_model-20.log
[rospy.client][INFO] 2026-06-02 20:05:30,301: init_node, name[/spawn_gazebo_model],
pid[6389]
[xmlrpc][INFO] 2026-06-02 20:05:30,301: XML-RPC server binding to 0.0.0.0:0
[xmlrpc][INFO] 2026-06-02 20:05:30,301: Started XML-RPC server [http://lab-ubuntu-01:38857/]
[rospy.impl.masterslave][INFO] 2026-06-02 20:05:30,301: _ready: http://lab-ubuntu-01:38857/
[rospy.registration][INFO] 2026-06-02 20:05:30,302: Registering with master node
http://localhost:11311
[rospy.init][INFO] 2026-06-02 20:05:30,302: ROS Slave URI: [http://lab-ubuntu-01:38857/]
[xmlrpc][INFO] 2026-06-02 20:05:30,302: xml rpc node: starting XML-RPC server
[rospy.init][INFO] 2026-06-02 20:05:30,302: registered with master
[rospy.rosout][INFO] 2026-06-02 20:05:30,303: initializing /rosout core topic
[rospy.rosout][INFO] 2026-06-02 20:05:30,305: connected to core topic /rosout
[rospy.simtime][INFO] 2026-06-02 20:05:30,306: initializing /clock core topic
[rospy.simtime][INFO] 2026-06-02 20:05:30,307: connected to core topic /clock
[rosout][INFO] 2026-06-02 20:05:30,310: Waiting for model ipa-office before proceeding.
[rospy.internal][INFO] 2026-06-02 20:05:30,677: topic[/rosout] adding connection to [/rosout],
count 0
[rospy.core][INFO] 2026-06-02 20:15:13,010: signal_shutdown [signal-2]
[rospy.internal][INFO] 2026-06-02 20:15:13,011: topic[/rosout] removing connection to /rosout
[rospy.impl.masterslave][INFO] 2026-06-02 20:15:13,011: signal-2
[rospy.core][INFO] 2026-06-02 20:15:13,812: signal_shutdown [atexit]
labuser@lab-ubuntu-01:~$ grep -i "error|fatal|exception|failed"
~/ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/spawn_urdf_world-3.log
labuser@lab-ubuntu-01:~$ grep -i "error|fatal|exception|failed"
~/ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/spawn_gazebo_model-20.log
labuser@lab-ubuntu-01:~$ grep -i "error|fatal|exception|failed"
~/ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-6295.log | tail -50
```

```
[roslaunch][INFO] 2026-06-02 20:05:28,827: roslaunch env is environ({'SHELL': '/bin/bash',
'ROS_VERSION': '1', 'SESSION_MANAGER':
'local/lab-ubuntu-01:@/tmp/.ICE-unix/2147,unix/lab-ubuntu-01:/tmp/.ICE-unix/2147',
'QT_ACCESSIBILITY': '1', 'COLORTERM': 'truecolor', 'XDG_CONFIG_DIRS':
'/etc/xdg/xdg-ubuntu:/etc/xdg', 'XDG_MENU_PREFIX': 'gnome-',
'GNOME_DESKTOP_SESSION_ID': 'this-is-deprecated', 'PKG_CONFIG_PATH':
'/home/labuser/catkin_ws/devel/lib/pkgconfig:/opt/ros/noetic/lib/pkgconfig:/opt/ros/noetic/lib/x86_
64-linux-gnu/pkgconfig', 'ROS_PYTHON_VERSION': '3', 'LC_ADDRESS': 'de_DE.UTF-8',
'GNOME_SHELL_SESSION_MODE': 'ubuntu', 'LC_NAME': 'de_DE.UTF-8',
'SSH_AUTH_SOCK': '/run/user/1000/keyring/ssh', 'XMODIFIERS': '@im=ibus',
'DESKTOP_SESSION': 'ubuntu', 'LC_MONETARY': 'de_DE.UTF-8', 'SSH_AGENT_PID': '2099',
'ROS_PACKAGE_PATH': '/home/labuser/catkin_ws/src:/opt/ros/noetic/share',
'GTK_MODULES': 'gail:atk-bridge', 'ROSLISP_PACKAGE_DIRECTORIES':
'/home/labuser/catkin_ws/devel/share/common-lisp', 'PWD': '/home/labuser', 'LOGNAME':
'labuser', 'XDG_SESSION_DESKTOP': 'ubuntu', 'XDG_SESSION_TYPE': 'x11',
'GPG_AGENT_INFO': '/run/user/1000/gnupg/S.gpg-agent:0:1', 'XAUTHORITY':
'/run/user/1000/gdm/Xauthority', 'GJS_DEBUG_TOPICS': 'JS ERROR;JS LOG',
'WINDOWPATH': '2', 'HOME': '/home/labuser', 'USERNAME': 'labuser', 'IM_CONFIG_PHASE':
'1', 'LC_PAPER': 'de_DE.UTF-8', 'LANG': 'en_US.UTF-8', 'ROS_ETC_DIR':
'/opt/ros/noetic/etc/ros', 'LS_COLORS':
'rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;3
1;01:mi=00:su=37;41:sg=30;43:ca=30;41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.t
gz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:
*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lr
z=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tb
z=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:
*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;3
1:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.jpg=01;35:*.jpeg=01;35:*.
mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.t
ga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;
35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;3
5:*.webm=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:
*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;
35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=0
1;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.
midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=
00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:', 'XDG_CURRENT_DESKTOP':
'ubuntu:GNOME', 'VTE_VERSION': '6003', 'CMAKE_PREFIX_PATH':
'/home/labuser/catkin_ws/devel:/opt/ros/noetic', 'GNOME_TERMINAL_SCREEN':
'/org/gnome/Terminal/screen/1849a7e0_8ec5_432a_b712_e503c1ee208a', 'INVOCATION_ID':
'45652184429345298a08c97e5db93533', 'MANAGERPID': '1997', 'GJS_DEBUG_OUTPUT':
'stderr', 'ROBOT_ENV': 'ipa-office', 'ROBOT': 'cob4-4', 'LESSCLOSE': '/usr/bin/lesspipe %s %s',
'XDG_SESSION_CLASS': 'user', 'PYTHONPATH': '/opt/ros/noetic/lib/python3/dist-packages',
'TERM': 'xterm-256color', 'LC_IDENTIFICATION': 'de_DE.UTF-8', 'LESSOPEN': '|
```

```
/usr/bin/lesspipe %s', 'USER': 'labuser', 'GNOME_TERMINAL_SERVICE': ':1.105', 'DISPLAY':  
'1', 'SHLVL': '1', 'LC_TELEPHONE': 'de_DE.UTF-8', 'ROS_MASTER_URI':  
'http://localhost:11311', 'QT_IM_MODULE': 'ibus', 'LC_MEASUREMENT': 'de_DE.UTF-8',  
'LD_LIBRARY_PATH':  
'/home/labuser/catkin_ws/devel/lib:/opt/ros/noetic/lib:/opt/ros/noetic/lib/x86_64-linux-gnu',  
'XDG_RUNTIME_DIR': '/run/user/1000', 'LC_TIME': 'de_DE.UTF-8', 'JOURNAL_STREAM':  
'8:54340', 'XDG_DATA_DIRS':  
'/usr/share/ubuntu:/usr/local/share:/usr/share:/var/lib/snapd/desktop', 'PATH':  
'/opt/ros/noetic/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/  
games:/snap/bin', 'GDMSESSION': 'ubuntu', 'DBUS_SESSION_BUS_ADDRESS':  
'unix:path=/run/user/1000/bus', 'ROS_ROOT': '/opt/ros/noetic/share/ros', 'ROS_DISTRO':  
'noetic', 'LC_NUMERIC': 'de_DE.UTF-8', '_': '/opt/ros/noetic/bin/roslaunch',  
'ROS_LOG_FILENAME':  
'/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-6295  
.log'})  
[roslaunch][INFO] 2026-06-02 20:05:29,793: process[gazebo-1]: env[{'SHELL': '/bin/bash',  
'ROS_VERSION': '1', 'SESSION_MANAGER':  
'local/lab-ubuntu-01:@/tmp/.ICE-unix/2147,unix/lab-ubuntu-01:/tmp/.ICE-unix/2147',  
'QT_ACCESSIBILITY': '1', 'COLORTERM': 'truecolor', 'XDG_CONFIG_DIRS':  
'/etc/xdg/xdg-ubuntu:/etc/xdg', 'XDG_MENU_PREFIX': 'gnome-',  
'GNOME_DESKTOP_SESSION_ID': 'this-is-deprecated', 'PKG_CONFIG_PATH':  
'/home/labuser/catkin_ws/devel/lib/pkgconfig:/opt/ros/noetic/lib/pkgconfig:/opt/ros/noetic/lib/x86_  
64-linux-gnu/pkgconfig', 'ROS_PYTHON_VERSION': '3', 'LC_ADDRESS': 'de_DE.UTF-8',  
'GNOME_SHELL_SESSION_MODE': 'ubuntu', 'LC_NAME': 'de_DE.UTF-8',  
'SSH_AUTH_SOCK': '/run/user/1000/keyring/ssh', 'XMODIFIERS': '@im=ibus',  
'DESKTOP_SESSION': 'ubuntu', 'LC_MONETARY': 'de_DE.UTF-8', 'SSH_AGENT_PID': '2099',  
'ROS_PACKAGE_PATH': '/home/labuser/catkin_ws/src:/opt/ros/noetic/share',  
'GTK_MODULES': 'gail:atk-bridge', 'ROSLISP_PACKAGE_DIRECTORIES':  
'/home/labuser/catkin_ws/devel/share/common-lisp', 'PWD': '/home/labuser', 'LOGNAME':  
'labuser', 'XDG_SESSION_DESKTOP': 'ubuntu', 'XDG_SESSION_TYPE': 'x11',  
'GPG_AGENT_INFO': '/run/user/1000/gnupg/S.gpg-agent:0:1', 'XAUTHORITY':  
'/run/user/1000/gdm/Xauthority', 'GJS_DEBUG_TOPICS': 'JS ERROR;JS LOG',  
'WINDOWPATH': '2', 'HOME': '/home/labuser', 'USERNAME': 'labuser', 'IM_CONFIG_PHASE':  
'1', 'LC_PAPER': 'de_DE.UTF-8', 'LANG': 'en_US.UTF-8', 'ROS_ETC_DIR':  
'/opt/ros/noetic/etc/ros', 'LS_COLORS':  
'rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;3  
1;01:mi=00:su=37;41:sg=30;43:ca=30;41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.t  
gz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:  
*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lr  
z=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tb  
z=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:  
*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;3  
1:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.jpg=01;35:*.jpeg=01;35:*.  
mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.t
```

ga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:.', 'XDG_CURRENT_DESKTOP': 'ubuntu:GNOME', 'VTE_VERSION': '6003', 'CMAKE_PREFIX_PATH': '/home/labuser/catkin_ws/devel:/opt/ros/noetic', 'GNOME_TERMINAL_SCREEN': '/org/gnome/Terminal/screen/1849a7e0_8ec5_432a_b712_e503c1ee208a', 'INVOCATION_ID': '45652184429345298a08c97e5db93533', 'MANAGERPID': '1997', 'GJS_DEBUG_OUTPUT': 'stderr', 'ROBOT_ENV': 'ipa-office', 'ROBOT': 'cob4-4', 'LESSCLOSE': '/usr/bin/lesspipe %s %s', 'XDG_SESSION_CLASS': 'user', 'PYTHONPATH': '/opt/ros/noetic/lib/python3/dist-packages', 'TERM': 'xterm-256color', 'LC_IDENTIFICATION': 'de_DE.UTF-8', 'LESSOPEN': '| /usr/bin/lesspipe %s', 'USER': 'labuser', 'GNOME_TERMINAL_SERVICE': ':1.105', 'DISPLAY': ':1', 'SHLVL': '1', 'LC_TELEPHONE': 'de_DE.UTF-8', 'ROS_MASTER_URI': 'http://localhost:11311', 'QT_IM_MODULE': 'ibus', 'LC_MEASUREMENT': 'de_DE.UTF-8', 'LD_LIBRARY_PATH': '/home/labuser/catkin_ws/devel/lib:/opt/ros/noetic/lib:/opt/ros/noetic/lib/x86_64-linux-gnu', 'XDG_RUNTIME_DIR': '/run/user/1000', 'LC_TIME': 'de_DE.UTF-8', 'JOURNAL_STREAM': '8:54340', 'XDG_DATA_DIRS': '/usr/share/ubuntu:/usr/local/share:/usr/share:/var/lib/snapd/desktop', 'PATH': '/opt/ros/noetic/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin', 'GDMSESSION': 'ubuntu', 'DBUS_SESSION_BUS_ADDRESS': 'unix:path=/run/user/1000/bus', 'ROS_ROOT': '/opt/ros/noetic/share/ros', 'ROS_DISTRO': 'noetic', 'LC_NUMERIC': 'de_DE.UTF-8', '_': '/opt/ros/noetic/bin/roslaunch', 'ROS_LOG_FILENAME': '/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-6295.log', 'GAZEBO_MODEL_PATH': '/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/Media/models:')[roslaunch][INFO] 2026-06-02 20:05:29,827: process[gazebo_gui-2]: env[{'SHELL': '/bin/bash', 'ROS_VERSION': '1', 'SESSION_MANAGER': 'local/lab-ubuntu-01:@/tmp/.ICE-unix/2147,unix/lab-ubuntu-01:/tmp/.ICE-unix/2147', 'QT_ACCESSIBILITY': '1', 'COLORTERM': 'truecolor', 'XDG_CONFIG_DIRS': '/etc/xdg/xdg-ubuntu:/etc/xdg', 'XDG_MENU_PREFIX': 'gnome-', 'GNOME_DESKTOP_SESSION_ID': 'this-is-deprecated', 'PKG_CONFIG_PATH': '/home/labuser/catkin_ws/devel/lib/pkgconfig:/opt/ros/noetic/lib/pkgconfig:/opt/ros/noetic/lib/x86_64-linux-gnu/pkgconfig', 'ROS_PYTHON_VERSION': '3', 'LC_ADDRESS': 'de_DE.UTF-8', 'GNOME_SHELL_SESSION_MODE': 'ubuntu', 'LC_NAME': 'de_DE.UTF-8', 'SSH_AUTH_SOCK': '/run/user/1000/keyring/ssh', 'XMODIFIERS': '@im=ibus', 'DESKTOP_SESSION': 'ubuntu', 'LC_MONETARY': 'de_DE.UTF-8', 'SSH_AGENT_PID': '2099', 'ROS_PACKAGE_PATH': '/home/labuser/catkin_ws/src:/opt/ros/noetic/share', 'GTK_MODULES': 'gail:atk-bridge', 'ROSLISP_PACKAGE_DIRECTORIES':

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'WINDOWPATH': '2', 'HOME': '/home/labuser', 'USERNAME': 'labuser', 'IM_CONFIG_PHASE':
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1;01:mi=00:su=37;41:sg=30;43:ca=30;41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.t
gz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:
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z=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tztst=01;31:*.bz2=01;31:*.bz=01;31:*.tb
z=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:
.sar=01;31:.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;3
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mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.t
ga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;
35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;3
5:*.webm=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:
.nuv=01;35:.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;
35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=0
1;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.
midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=
00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:.', 'XDG_CURRENT_DESKTOP':
'ubuntu:GNOME', 'VTE_VERSION': '6003', 'CMAKE_PREFIX_PATH':
'/home/labuser/catkin_ws/devel:/opt/ros/noetic', 'GNOME_TERMINAL_SCREEN':
'/org/gnome/Terminal/screen/1849a7e0_8ec5_432a_b712_e503c1ee208a', 'INVOCATION_ID':
'45652184429345298a08c97e5db93533', 'MANAGERPID': '1997', 'GJS_DEBUG_OUTPUT':
'stderr', 'ROBOT_ENV': 'ipa-office', 'ROBOT': 'cob4-4', 'LESSCLOSE': '/usr/bin/lesspipe %s %s',
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'TERM': 'xterm-256color', 'LC_IDENTIFICATION': 'de_DE.UTF-8', 'LESSOPEN': '|
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'http://localhost:11311', 'QT_IM_MODULE': 'ibus', 'LC_MEASUREMENT': 'de_DE.UTF-8',
'LD_LIBRARY_PATH':
'/home/labuser/catkin_ws/devel/lib:/opt/ros/noetic/lib:/opt/ros/noetic/lib/x86_64-linux-gnu',
'XDG_RUNTIME_DIR': '/run/user/1000', 'LC_TIME': 'de_DE.UTF-8', 'JOURNAL_STREAM':
'8:54340', 'XDG_DATA_DIRS':
'/usr/share/ubuntu:/usr/local/share:/usr/share:/var/lib/snapd/desktop', 'PATH':
'/opt/ros/noetic/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/
games:/snap/bin', 'GDMSESSION': 'ubuntu', 'DBUS_SESSION_BUS_ADDRESS':
'unix:path=/run/user/1000/bus', 'ROS_ROOT': '/opt/ros/noetic/share/ros', 'ROS_DISTRO':
'noetic', 'LC_NUMERIC': 'de_DE.UTF-8', ' _': '/opt/ros/noetic/bin/roslaunch',
'ROS_LOG_FILENAME':

/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-6295.log', 'GAZEBO_MODEL_PATH':
'/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/Media/models:']
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'/bin/bash', 'ROS_VERSION': '1', 'SESSION_MANAGER':
'local/lab-ubuntu-01:@/tmp/.ICE-unix/2147,unix/lab-ubuntu-01:/tmp/.ICE-unix/2147',
'QT_ACCESSIBILITY': '1', 'COLORTERM': 'truecolor', 'XDG_CONFIG_DIRS':
'/etc/xdg/xdg-ubuntu:/etc/xdg', 'XDG_MENU_PREFIX': 'gnome-',
'GNOME_DESKTOP_SESSION_ID': 'this-is-deprecated', 'PKG_CONFIG_PATH':
'/home/labuser/catkin_ws/devel/lib/pkgconfig:/opt/ros/noetic/lib/pkgconfig:/opt/ros/noetic/lib/x86_64-linux-gnu/pkgconfig', 'ROS_PYTHON_VERSION': '3', 'LC_ADDRESS': 'de_DE.UTF-8',
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'DESKTOP_SESSION': 'ubuntu', 'LC_MONETARY': 'de_DE.UTF-8', 'SSH_AGENT_PID': '2099',
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'GTK_MODULES': 'gail:atk-bridge', 'ROSLISP_PACKAGE_DIRECTORIES':
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'labuser', 'XDG_SESSION_DESKTOP': 'ubuntu', 'XDG_SESSION_TYPE': 'x11',
'GPG_AGENT_INFO': '/run/user/1000/gnupg/S.gpg-agent:0:1', 'XAUTHORITY':
'/run/user/1000/gdm/Xauthority', 'GJS_DEBUG_TOPICS': 'JS ERROR;JS LOG',
'WINDOWPATH': '2', 'HOME': '/home/labuser', 'USERNAME': 'labuser', 'IM_CONFIG_PHASE':
'1', 'LC_PAPER': 'de_DE.UTF-8', 'LANG': 'en_US.UTF-8', 'ROS_ETC_DIR':
'/opt/ros/noetic/etc/ros', 'LS_COLORS':
'rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;3
1;01:mi=00:su=37;41:sg=30;43:ca=30;41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.t
gz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:
.tlz=01;31:.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lr
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z=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:
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1:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.jpg=01;35:*.jpeg=01;35:*.
mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.t
ga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;
35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;3
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'ubuntu:GNOME', 'VTE_VERSION': '6003', 'CMAKE_PREFIX_PATH':
'/home/labuser/catkin_ws/devel:/opt/ros/noetic', 'GNOME_TERMINAL_SCREEN':
'/org/gnome/Terminal/screen/1849a7e0_8ec5_432a_b712_e503c1ee208a', 'INVOCATION_ID':
'45652184429345298a08c97e5db93533', 'MANAGERPID': '1997', 'GJS_DEBUG_OUTPUT':

'stderr', 'ROBOT_ENV': 'ipa-office', 'ROBOT': 'cob4-4', 'LESSCLOSE': '/usr/bin/lesspipe %s %s',
'XDG_SESSION_CLASS': 'user', 'PYTHONPATH': '/opt/ros/noetic/lib/python3/dist-packages',
'TERM': 'xterm-256color', 'LC_IDENTIFICATION': 'de_DE.UTF-8', 'LESSOPEN': '|
/usr/bin/lesspipe %s', 'USER': 'labuser', 'GNOME_TERMINAL_SERVICE': ':1.105', 'DISPLAY':
':1', 'SHLVL': '1', 'LC_TELEPHONE': 'de_DE.UTF-8', 'ROS_MASTER_URI':
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'LD_LIBRARY_PATH':
'/home/labuser/catkin_ws/devel/lib:/opt/ros/noetic/lib:/opt/ros/noetic/lib/x86_64-linux-gnu',
'XDG_RUNTIME_DIR': '/run/user/1000', 'LC_TIME': 'de_DE.UTF-8', 'JOURNAL_STREAM':
'8:54340', 'XDG_DATA_DIRS':
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'/opt/ros/noetic/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/
games:/snap/bin', 'GDMSESSION': 'ubuntu', 'DBUS_SESSION_BUS_ADDRESS':
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'noetic', 'LC_NUMERIC': 'de_DE.UTF-8', '_': '/opt/ros/noetic/bin/roslaunch',
'ROS_LOG_FILENAME':
'/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-6295
.log', 'GAZEBO_MODEL_PATH':
'/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/Media/models:}}
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'QT_ACCESSIBILITY': '1', 'COLORTERM': 'truecolor', 'XDG_CONFIG_DIRS':
'/etc/xdg/xdg-ubuntu:/etc/xdg', 'XDG_MENU_PREFIX': 'gnome-',
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'/home/labuser/catkin_ws/devel/lib/pkgconfig:/opt/ros/noetic/lib/pkgconfig:/opt/ros/noetic/lib/x86_
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'DESKTOP_SESSION': 'ubuntu', 'LC_MONETARY': 'de_DE.UTF-8', 'SSH_AGENT_PID': '2099',
'ROS_PACKAGE_PATH': '/home/labuser/catkin_ws/src:/opt/ros/noetic/share',
'GTK_MODULES': 'gail:atk-bridge', 'ROSLISP_PACKAGE_DIRECTORIES':
'/home/labuser/catkin_ws/devel/share/common-lisp', 'PWD': '/home/labuser', 'LOGNAME':
'labuser', 'XDG_SESSION_DESKTOP': 'ubuntu', 'XDG_SESSION_TYPE': 'x11',
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'/run/user/1000/gdm/Xauthority', 'GJS_DEBUG_TOPICS': 'JS ERROR;JS LOG',
'WINDOWPATH': '2', 'HOME': '/home/labuser', 'USERNAME': 'labuser', 'IM_CONFIG_PHASE':
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'/opt/ros/noetic/etc/ros', 'LS_COLORS':
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1;01:mi=00:su=37;41:sg=30;43:ca=30;41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.t
gz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:
.tlz=01;31:.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lr
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z=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:', 'XDG_CURRENT_DESKTOP': 'ubuntu:GNOME', 'VTE_VERSION': '6003', 'CMAKE_PREFIX_PATH': '/home/labuser/catkin_ws/devel:/opt/ros/noetic', 'GNOME_TERMINAL_SCREEN': '/org/gnome/Terminal/screen/1849a7e0_8ec5_432a_b712_e503c1ee208a', 'INVOCATION_ID': '45652184429345298a08c97e5db93533', 'MANAGERPID': '1997', 'GJS_DEBUG_OUTPUT': 'stderr', 'ROBOT_ENV': 'ipa-office', 'ROBOT': 'cob4-4', 'LESSCLOSE': '/usr/bin/lesspipe %s %s', 'XDG_SESSION_CLASS': 'user', 'PYTHONPATH': '/opt/ros/noetic/lib/python3/dist-packages', 'TERM': 'xterm-256color', 'LC_IDENTIFICATION': 'de_DE.UTF-8', 'LESSOPEN': '| /usr/bin/lesspipe %s', 'USER': 'labuser', 'GNOME_TERMINAL_SERVICE': ':1.105', 'DISPLAY': ':1', 'SHLVL': '1', 'LC_TELEPHONE': 'de_DE.UTF-8', 'ROS_MASTER_URI': 'http://localhost:11311', 'QT_IM_MODULE': 'ibus', 'LC_MEASUREMENT': 'de_DE.UTF-8', 'LD_LIBRARY_PATH': '/home/labuser/catkin_ws/devel/lib:/opt/ros/noetic/lib:/opt/ros/noetic/lib/x86_64-linux-gnu', 'XDG_RUNTIME_DIR': '/run/user/1000', 'LC_TIME': 'de_DE.UTF-8', 'JOURNAL_STREAM': '8:54340', 'XDG_DATA_DIRS': '/usr/share/ubuntu:/usr/local/share:/usr/share:/var/lib/snapd/desktop', 'PATH': '/opt/ros/noetic/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin', 'GDMSESSION': 'ubuntu', 'DBUS_SESSION_BUS_ADDRESS': 'unix:path=/run/user/1000/bus', 'ROS_ROOT': '/opt/ros/noetic/share/ros', 'ROS_DISTRO': 'noetic', 'LC_NUMERIC': 'de_DE.UTF-8', '_': '/opt/ros/noetic/bin/roslaunch', 'ROS_LOG_FILENAME': '/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-6295.log'}} [roslaunch][INFO] 2026-06-02 20:05:29,838: process[bms/power_state_aggregator-5]: env[{'SHELL': '/bin/bash', 'ROS_VERSION': '1', 'SESSION_MANAGER': 'local/lab-ubuntu-01:@/tmp/.ICE-unix/2147,unix/lab-ubuntu-01:/tmp/.ICE-unix/2147', 'QT_ACCESSIBILITY': '1', 'COLORTERM': 'truecolor', 'XDG_CONFIG_DIRS': '/etc/xdg/xdg-ubuntu:/etc/xdg', 'XDG_MENU_PREFIX': 'gnome-', 'GNOME_DESKTOP_SESSION_ID': 'this-is-deprecated', 'PKG_CONFIG_PATH': '/home/labuser/catkin_ws/devel/lib/pkgconfig:/opt/ros/noetic/lib/pkgconfig:/opt/ros/noetic/lib/x86_64-linux-gnu/pkgconfig', 'ROS_PYTHON_VERSION': '3', 'LC_ADDRESS': 'de_DE.UTF-8', 'GNOME_SHELL_SESSION_MODE': 'ubuntu', 'LC_NAME': 'de_DE.UTF-8', 'SSH_AUTH_SOCK': '/run/user/1000/keyring/ssh', 'XMODIFIERS': '@im=ibus',

'DESKTOP_SESSION': 'ubuntu', 'LC_MONETARY': 'de_DE.UTF-8', 'SSH_AGENT_PID': '2099',
'ROS_PACKAGE_PATH': '/home/labuser/catkin_ws/src:/opt/ros/noetic/share',
'GTK_MODULES': 'gail:atk-bridge', 'ROSLISP_PACKAGE_DIRECTORIES':
'/home/labuser/catkin_ws/devel/share/common-lisp', 'PWD': '/home/labuser', 'LOGNAME':
'labuser', 'XDG_SESSION_DESKTOP': 'ubuntu', 'XDG_SESSION_TYPE': 'x11',
'GPG_AGENT_INFO': '/run/user/1000/gnupg/S.gpg-agent:0:1', 'XAUTHORITY':
'/run/user/1000/gdm/Xauthority', 'GJS_DEBUG_TOPICS': 'JS ERROR;JS LOG',
'WINDOWPATH': '2', 'HOME': '/home/labuser', 'USERNAME': 'labuser', 'IM_CONFIG_PHASE':
'1', 'LC_PAPER': 'de_DE.UTF-8', 'LANG': 'en_US.UTF-8', 'ROS_ETC_DIR':
'/opt/ros/noetic/etc/ros', 'LS_COLORS':
'rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;3
1;01:mi=00:su=37;41:sg=30;43:ca=30;41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.t
gz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:
.tlz=01;31:.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lr
z=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tb
z=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:
.sar=01;31:.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;3
1:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.jpg=01;35:*.jpeg=01;35:*.
mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.t
ga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;
35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;3
5:*.webm=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:
.nuv=01;35:.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;
35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=0
1;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.
midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=
00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:.', 'XDG_CURRENT_DESKTOP':
'ubuntu:GNOME', 'VTE_VERSION': '6003', 'CMAKE_PREFIX_PATH':
'/home/labuser/catkin_ws/devel:/opt/ros/noetic', 'GNOME_TERMINAL_SCREEN':
'/org/gnome/Terminal/screen/1849a7e0_8ec5_432a_b712_e503c1ee208a', 'INVOCATION_ID':
'45652184429345298a08c97e5db93533', 'MANAGERPID': '1997', 'GJS_DEBUG_OUTPUT':
'stderr', 'ROBOT_ENV': 'ipa-office', 'ROBOT': 'cob4-4', 'LESSCLOSE': '/usr/bin/lesspipe %s %s',
'XDG_SESSION_CLASS': 'user', 'PYTHONPATH': '/opt/ros/noetic/lib/python3/dist-packages',
'TERM': 'xterm-256color', 'LC_IDENTIFICATION': 'de_DE.UTF-8', 'LESSOPEN': '|
'/usr/bin/lesspipe %s', 'USER': 'labuser', 'GNOME_TERMINAL_SERVICE': ':1.105', 'DISPLAY':
'.:1', 'SHLVL': '1', 'LC_TELEPHONE': 'de_DE.UTF-8', 'ROS_MASTER_URI':
'http://localhost:11311', 'QT_IM_MODULE': 'ibus', 'LC_MEASUREMENT': 'de_DE.UTF-8',
'LD_LIBRARY_PATH':
'/home/labuser/catkin_ws/devel/lib:/opt/ros/noetic/lib:/opt/ros/noetic/lib/x86_64-linux-gnu',
'XDG_RUNTIME_DIR': '/run/user/1000', 'LC_TIME': 'de_DE.UTF-8', 'JOURNAL_STREAM':
'8:54340', 'XDG_DATA_DIRS':
'/usr/share/ubuntu:/usr/local/share:/usr/share:/var/lib/snapd/desktop', 'PATH':
'/opt/ros/noetic/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/
games:/snap/bin', 'GDMSESSION': 'ubuntu', 'DBUS_SESSION_BUS_ADDRESS':

'unix:path=/run/user/1000/bus', 'ROS_ROOT': '/opt/ros/noetic/share/ros', 'ROS_DISTRO':
'noetic', 'LC_NUMERIC': 'de_DE.UTF-8', '_': '/opt/ros/noetic/bin/roslaunch',
'ROS_LOG_FILENAME':
'/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-6295
.log', 'ROS_NAMESPACE': '/bms']]
[roslaunch][INFO] 2026-06-02 20:05:29,842: process[base/halt_detector-6]: env[{'SHELL':
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'local/lab-ubuntu-01:@/tmp/.ICE-unix/2147,unix/lab-ubuntu-01:/tmp/.ICE-unix/2147',
'QT_ACCESSIBILITY': '1', 'COLORTERM': 'truecolor', 'XDG_CONFIG_DIRS':
'/etc/xdg/xdg-ubuntu:/etc/xdg', 'XDG_MENU_PREFIX': 'gnome-',
'GNOME_DESKTOP_SESSION_ID': 'this-is-deprecated', 'PKG_CONFIG_PATH':
'/home/labuser/catkin_ws/devel/lib/pkgconfig:/opt/ros/noetic/lib/pkgconfig:/opt/ros/noetic/lib/x86_
64-linux-gnu/pkgconfig', 'ROS_PYTHON_VERSION': '3', 'LC_ADDRESS': 'de_DE.UTF-8',
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'SSH_AUTH_SOCK': '/run/user/1000/keyring/ssh', 'XMODIFIERS': '@im=ibus',
'DESKTOP_SESSION': 'ubuntu', 'LC_MONETARY': 'de_DE.UTF-8', 'SSH_AGENT_PID': '2099',
'ROS_PACKAGE_PATH': '/home/labuser/catkin_ws/src:/opt/ros/noetic/share',
'GTK_MODULES': 'gail:atk-bridge', 'ROSLISP_PACKAGE_DIRECTORIES':
'/home/labuser/catkin_ws/devel/share/common-lisp', 'PWD': '/home/labuser', 'LOGNAME':
'labuser', 'XDG_SESSION_DESKTOP': 'ubuntu', 'XDG_SESSION_TYPE': 'x11',
'GPG_AGENT_INFO': '/run/user/1000/gnupg/S.gpg-agent:0:1', 'XAUTHORITY':
'/run/user/1000/gdm/Xauthority', 'GJS_DEBUG_TOPICS': 'JS ERROR;JS LOG',
'WINDOWPATH': '2', 'HOME': '/home/labuser', 'USERNAME': 'labuser', 'IM_CONFIG_PHASE':
'1', 'LC_PAPER': 'de_DE.UTF-8', 'LANG': 'en_US.UTF-8', 'ROS_ETC_DIR':
'/opt/ros/noetic/etc/ros', 'LS_COLORS':
'rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;3
1;01:mi=00:su=37;41:sg=30;43:ca=30;41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.t
gz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lzh=01;31:*.lzma=01;31:
.tlz=01;31:.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lr
z=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tb
z=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:
.sar=01;31:.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;3
1:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.jpg=01;35:*.jpeg=01;35:*.
mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.t
ga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;
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5:*.webm=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:
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35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=0
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midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=
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'/home/labuser/catkin_ws/devel:/opt/ros/noetic', 'GNOME_TERMINAL_SCREEN':

'/org/gnome/Terminal/screen/1849a7e0_8ec5_432a_b712_e503c1ee208a', 'INVOCATION_ID':
'45652184429345298a08c97e5db93533', 'MANAGERPID': '1997', 'GJS_DEBUG_OUTPUT':
'stderr', 'ROBOT_ENV': 'ipa-office', 'ROBOT': 'cob4-4', 'LESSCLOSE': '/usr/bin/lesspipe %s %s',
'XDG_SESSION_CLASS': 'user', 'PYTHONPATH': '/opt/ros/noetic/lib/python3/dist-packages',
'TERM': 'xterm-256color', 'LC_IDENTIFICATION': 'de_DE.UTF-8', 'LESSOPEN': '|
'/usr/bin/lesspipe %s', 'USER': 'labuser', 'GNOME_TERMINAL_SERVICE': ':1.105', 'DISPLAY':
'1', 'SHLVL': '1', 'LC_TELEPHONE': 'de_DE.UTF-8', 'ROS_MASTER_URI':
'http://localhost:11311', 'QT_IM_MODULE': 'ibus', 'LC_MEASUREMENT': 'de_DE.UTF-8',
'LD_LIBRARY_PATH':
'/home/labuser/catkin_ws/devel/lib:/opt/ros/noetic/lib:/opt/ros/noetic/lib/x86_64-linux-gnu',
'XDG_RUNTIME_DIR': '/run/user/1000', 'LC_TIME': 'de_DE.UTF-8', 'JOURNAL_STREAM':
'8:54340', 'XDG_DATA_DIRS':
'/usr/share/ubuntu:/usr/local/share:/usr/share:/var/lib/snapd/desktop', 'PATH':
'/opt/ros/noetic/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/
games:/snap/bin', 'GDMSESSION': 'ubuntu', 'DBUS_SESSION_BUS_ADDRESS':
'unix:path=/run/user/1000/bus', 'ROS_ROOT': '/opt/ros/noetic/share/ros', 'ROS_DISTRO':
'noetic', 'LC_NUMERIC': 'de_DE.UTF-8', '_': '/opt/ros/noetic/bin/roslaunch',
'ROS_LOG_FILENAME':
'/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-6295
.log', 'ROS_NAMESPACE': '/base'}}
[roslaunch][INFO] 2026-06-02 20:05:29,846: process[base/fake_driver-7]: env[{'SHELL':
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'QT_ACCESSIBILITY': '1', 'COLORTERM': 'truecolor', 'XDG_CONFIG_DIRS':
'/etc/xdg/xdg-ubuntu:/etc/xdg', 'XDG_MENU_PREFIX': 'gnome-',
'GNOME_DESKTOP_SESSION_ID': 'this-is-deprecated', 'PKG_CONFIG_PATH':
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64-linux-gnu/pkgconfig', 'ROS_PYTHON_VERSION': '3', 'LC_ADDRESS': 'de_DE.UTF-8',
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'SSH_AUTH_SOCK': '/run/user/1000/keyring/ssh', 'XMODIFIERS': '@im=ibus',
'DESKTOP_SESSION': 'ubuntu', 'LC_MONETARY': 'de_DE.UTF-8', 'SSH_AGENT_PID': '2099',
'ROS_PACKAGE_PATH': '/home/labuser/catkin_ws/src:/opt/ros/noetic/share',
'GTK_MODULES': 'gail:atk-bridge', 'ROSLISP_PACKAGE_DIRECTORIES':
'/home/labuser/catkin_ws/devel/share/common-lisp', 'PWD': '/home/labuser', 'LOGNAME':
'labuser', 'XDG_SESSION_DESKTOP': 'ubuntu', 'XDG_SESSION_TYPE': 'x11',
'GPG_AGENT_INFO': '/run/user/1000/gnupg/S.gpg-agent:0:1', 'XAUTHORITY':
'/run/user/1000/gdm/Xauthority', 'GJS_DEBUG_TOPICS': 'JS ERROR;JS LOG',
'WINDOWPATH': '2', 'HOME': '/home/labuser', 'USERNAME': 'labuser', 'IM_CONFIG_PHASE':
'1', 'LC_PAPER': 'de_DE.UTF-8', 'LANG': 'en_US.UTF-8', 'ROS_ETC_DIR':
'/opt/ros/noetic/etc/ros', 'LS_COLORS':
'rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;3
1;01:mi=00:su=37;41:sg=30;43:ca=30;41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.t
gz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:
.tlz=01;31:.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lr

z=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:'. 'XDG_CURRENT_DESKTOP':
'ubuntu:GNOME', 'VTE_VERSION': '6003', 'CMAKE_PREFIX_PATH':
'/home/labuser/catkin_ws/devel:/opt/ros/noetic', 'GNOME_TERMINAL_SCREEN':
'/org/gnome/Terminal/screen/1849a7e0_8ec5_432a_b712_e503c1ee208a', 'INVOCATION_ID':
'45652184429345298a08c97e5db93533', 'MANAGERPID': '1997', 'GJS_DEBUG_OUTPUT':
'stderr', 'ROBOT_ENV': 'ipa-office', 'ROBOT': 'cob4-4', 'LESSCLOSE': '/usr/bin/lesspipe %s %s',
'XDG_SESSION_CLASS': 'user', 'PYTHONPATH': '/opt/ros/noetic/lib/python3/dist-packages',
'TERM': 'xterm-256color', 'LC_IDENTIFICATION': 'de_DE.UTF-8', 'LESSOPEN': '|
'/usr/bin/lesspipe %s', 'USER': 'labuser', 'GNOME_TERMINAL_SERVICE': ':1.105', 'DISPLAY':
'1', 'SHLVL': '1', 'LC_TELEPHONE': 'de_DE.UTF-8', 'ROS_MASTER_URI':
'http://localhost:11311', 'QT_IM_MODULE': 'ibus', 'LC_MEASUREMENT': 'de_DE.UTF-8',
'LD_LIBRARY_PATH':
'/home/labuser/catkin_ws/devel/lib:/opt/ros/noetic/lib:/opt/ros/noetic/lib/x86_64-linux-gnu',
'XDG_RUNTIME_DIR': '/run/user/1000', 'LC_TIME': 'de_DE.UTF-8', 'JOURNAL_STREAM':
'8:54340', 'XDG_DATA_DIRS':
'/usr/share/ubuntu:/usr/local/share:/usr/share:/var/lib/snapd/desktop', 'PATH':
'/opt/ros/noetic/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/
games:/snap/bin', 'GDMSESSION': 'ubuntu', 'DBUS_SESSION_BUS_ADDRESS':
'unix:path=/run/user/1000/bus', 'ROS_ROOT': '/opt/ros/noetic/share/ros', 'ROS_DISTRO':
'noetic', 'LC_NUMERIC': 'de_DE.UTF-8', '_': '/opt/ros/noetic/bin/roslaunch',
'ROS_LOG_FILENAME':
'/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-6295
.log', 'ROS_NAMESPACE': '/base'}]
[roslaunch][INFO] 2026-06-02 20:05:29,851: process[base/base_controller_spawner-8]:
env[{'SHELL': '/bin/bash', 'ROS_VERSION': '1', 'SESSION_MANAGER':
'local/lab-ubuntu-01:@/tmp/.ICE-unix/2147,unix/lab-ubuntu-01:/tmp/.ICE-unix/2147',
'QT_ACCESSIBILITY': '1', 'COLORTERM': 'truecolor', 'XDG_CONFIG_DIRS':
'/etc/xdg/xdg-ubuntu:/etc/xdg', 'XDG_MENU_PREFIX': 'gnome-'
'GNOME_DESKTOP_SESSION_ID': 'this-is-deprecated', 'PKG_CONFIG_PATH':
'/home/labuser/catkin_ws/devel/lib/pkgconfig:/opt/ros/noetic/lib/pkgconfig:/opt/ros/noetic/lib/x86_64-linux-gnu/pkgconfig', 'ROS_PYTHON_VERSION': '3', 'LC_ADDRESS': 'de_DE.UTF-8',
'GNOME_SHELL_SESSION_MODE': 'ubuntu', 'LC_NAME': 'de_DE.UTF-8',

'SSH_AUTH_SOCK': '/run/user/1000/keyring/ssh', 'XMODIFIERS': '@im=ibus',
'DESKTOP_SESSION': 'ubuntu', 'LC_MONETARY': 'de_DE.UTF-8', 'SSH_AGENT_PID': '2099',
'ROS_PACKAGE_PATH': '/home/labuser/catkin_ws/src:/opt/ros/noetic/share',
'GTK_MODULES': 'gail:atk-bridge', 'ROSLISP_PACKAGE_DIRECTORIES':
'/home/labuser/catkin_ws/devel/share/common-lisp', 'PWD': '/home/labuser', 'LOGNAME':
'labuser', 'XDG_SESSION_DESKTOP': 'ubuntu', 'XDG_SESSION_TYPE': 'x11',
'GPG_AGENT_INFO': '/run/user/1000/gnupg/S.gpg-agent:0:1', 'XAUTHORITY':
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'WINDOWPATH': '2', 'HOME': '/home/labuser', 'USERNAME': 'labuser', 'IM_CONFIG_PHASE':
'1', 'LC_PAPER': 'de_DE.UTF-8', 'LANG': 'en_US.UTF-8', 'ROS_ETC_DIR':
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.tlz=01;31:.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lr
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z=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:
.sar=01;31:.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;3
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35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;3
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35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=0
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midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=
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'ubuntu:GNOME', 'VTE_VERSION': '6003', 'CMAKE_PREFIX_PATH':
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'/org/gnome/Terminal/screen/1849a7e0_8ec5_432a_b712_e503c1ee208a', 'INVOCATION_ID':
'45652184429345298a08c97e5db93533', 'MANAGERPID': '1997', 'GJS_DEBUG_OUTPUT':
'stderr', 'ROBOT_ENV': 'ipa-office', 'ROBOT': 'cob4-4', 'LESSCLOSE': '/usr/bin/lesspipe %s %s',
'XDG_SESSION_CLASS': 'user', 'PYTHONPATH': '/opt/ros/noetic/lib/python3/dist-packages',
'TERM': 'xterm-256color', 'LC_IDENTIFICATION': 'de_DE.UTF-8', 'LESSOPEN': '|
'/usr/bin/lesspipe %s', 'USER': 'labuser', 'GNOME_TERMINAL_SERVICE': ':1.105', 'DISPLAY':
' :1', 'SHLVL': '1', 'LC_TELEPHONE': 'de_DE.UTF-8', 'ROS_MASTER_URI':
'http://localhost:11311', 'QT_IM_MODULE': 'ibus', 'LC_MEASUREMENT': 'de_DE.UTF-8',
'LD_LIBRARY_PATH':
'/home/labuser/catkin_ws/devel/lib:/opt/ros/noetic/lib:/opt/ros/noetic/lib/x86_64-linux-gnu',
'XDG_RUNTIME_DIR': '/run/user/1000', 'LC_TIME': 'de_DE.UTF-8', 'JOURNAL_STREAM':
'8:54340', 'XDG_DATA_DIRS':
'/usr/share/ubuntu:/usr/local/share:/usr/share:/var/lib/snapd/desktop', 'PATH':
'/opt/ros/noetic/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/

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games:/snap/bin', 'GDMSESSION': 'ubuntu', 'DBUS_SESSION_BUS_ADDRESS':  
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'noetic', 'LC_NUMERIC': 'de_DE.UTF-8', '_': '/opt/ros/noetic/bin/roslaunch',  
'ROS_LOG_FILENAME':  
'/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-6295  
.log', 'ROS_NAMESPACE': '/base'}}]  
[roslaunch][INFO] 2026-06-02 20:05:29,855: process[base/twist_mux-9]: env[{'SHELL':  
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'QT_ACCESSIBILITY': '1', 'COLORTERM': 'truecolor', 'XDG_CONFIG_DIRS':  
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'GNOME_DESKTOP_SESSION_ID': 'this-is-deprecated', 'PKG_CONFIG_PATH':  
'/home/labuser/catkin_ws/devel/lib/pkgconfig:/opt/ros/noetic/lib/pkgconfig:/opt/ros/noetic/lib/x86_  
64-linux-gnu/pkgconfig', 'ROS_PYTHON_VERSION': '3', 'LC_ADDRESS': 'de_DE.UTF-8',  
'GNOME_SHELL_SESSION_MODE': 'ubuntu', 'LC_NAME': 'de_DE.UTF-8',  
'SSH_AUTH_SOCK': '/run/user/1000/keyring/ssh', 'XMODIFIERS': '@im=ibus',  
'DESKTOP_SESSION': 'ubuntu', 'LC_MONETARY': 'de_DE.UTF-8', 'SSH_AGENT_PID': '2099',  
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'GTK_MODULES': 'gail:atk-bridge', 'ROSLISP_PACKAGE_DIRECTORIES':  
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'labuser', 'XDG_SESSION_DESKTOP': 'ubuntu', 'XDG_SESSION_TYPE': 'x11',  
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'/run/user/1000/gdm/Xauthority', 'GJS_DEBUG_TOPICS': 'JS ERROR;JS LOG',  
'WINDOWPATH': '2', 'HOME': '/home/labuser', 'USERNAME': 'labuser', 'IM_CONFIG_PHASE':  
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35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;3  
5:*.webm=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:  
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35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=0  
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midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=  
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'ubuntu:GNOME', 'VTE_VERSION': '6003', 'CMAKE_PREFIX_PATH':
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'45652184429345298a08c97e5db93533', 'MANAGERPID': '1997', 'GJS_DEBUG_OUTPUT':
'stderr', 'ROBOT_ENV': 'ipa-office', 'ROBOT': 'cob4-4', 'LESSCLOSE': '/usr/bin/lesspipe %s %s',
'XDG_SESSION_CLASS': 'user', 'PYTHONPATH': '/opt/ros/noetic/lib/python3/dist-packages',
'TERM': 'xterm-256color', 'LC_IDENTIFICATION': 'de_DE.UTF-8', 'LESSOPEN': '|
'/usr/bin/lesspipe %s', 'USER': 'labuser', 'GNOME_TERMINAL_SERVICE': ':1.105', 'DISPLAY':
'1', 'SHLVL': '1', 'LC_TELEPHONE': 'de_DE.UTF-8', 'ROS_MASTER_URI':
'http://localhost:11311', 'QT_IM_MODULE': 'ibus', 'LC_MEASUREMENT': 'de_DE.UTF-8',
'LD_LIBRARY_PATH':
'/home/labuser/catkin_ws/devel/lib:/opt/ros/noetic/lib:/opt/ros/noetic/lib/x86_64-linux-gnu',
'XDG_RUNTIME_DIR': '/run/user/1000', 'LC_TIME': 'de_DE.UTF-8', 'JOURNAL_STREAM':
'8:54340', 'XDG_DATA_DIRS':
'/usr/share/ubuntu:/usr/local/share:/usr/share:/var/lib/snapd/desktop', 'PATH':
'/opt/ros/noetic/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/
games:/snap/bin', 'GDMSESSION': 'ubuntu', 'DBUS_SESSION_BUS_ADDRESS':
'unix:path=/run/user/1000/bus', 'ROS_ROOT': '/opt/ros/noetic/share/ros', 'ROS_DISTRO':
'noetic', 'LC_NUMERIC': 'de_DE.UTF-8', '_': '/opt/ros/noetic/bin/roslaunch',
'ROS_LOG_FILENAME':
'/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-6295
.log', 'ROS_NAMESPACE': '/base'}]
[roslaunch][INFO] 2026-06-02 20:05:29,860: process[base/twist_marker-10]: env[{'SHELL':
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'QT_ACCESSIBILITY': '1', 'COLORTERM': 'truecolor', 'XDG_CONFIG_DIRS':
'/etc/xdg/xdg-ubuntu:/etc/xdg', 'XDG_MENU_PREFIX': 'gnome-'
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64-linux-gnu/pkgconfig', 'ROS_PYTHON_VERSION': '3', 'LC_ADDRESS': 'de_DE.UTF-8',
'GNOME_SHELL_SESSION_MODE': 'ubuntu', 'LC_NAME': 'de_DE.UTF-8',
'SSH_AUTH_SOCK': '/run/user/1000/keyring/ssh', 'XMODIFIERS': '@im=ibus',
'DESKTOP_SESSION': 'ubuntu', 'LC_MONETARY': 'de_DE.UTF-8', 'SSH_AGENT_PID': '2099',
'ROS_PACKAGE_PATH': '/home/labuser/catkin_ws/src:/opt/ros/noetic/share',
'GTK_MODULES': 'gail:atk-bridge', 'ROSLISP_PACKAGE_DIRECTORIES':
'/home/labuser/catkin_ws/devel/share/common-lisp', 'PWD': '/home/labuser', 'LOGNAME':
'labuser', 'XDG_SESSION_DESKTOP': 'ubuntu', 'XDG_SESSION_TYPE': 'x11',
'GPG_AGENT_INFO': '/run/user/1000/gnupg/S.gpg-agent:0:1', 'XAUTHORITY':
'/run/user/1000/gdm/Xauthority', 'GJS_DEBUG_TOPICS': 'JS ERROR;JS LOG',
'WINDOWPATH': '2', 'HOME': '/home/labuser', 'USERNAME': 'labuser', 'IM_CONFIG_PHASE':
'1', 'LC_PAPER': 'de_DE.UTF-8', 'LANG': 'en_US.UTF-8', 'ROS_ETC_DIR':
'/opt/ros/noetic/etc/ros', 'LS_COLORS':
'rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;3
1;01:mi=00:su=37;41:sg=30;43:ca=30;41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.t
gz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:

.tlz=01;31:.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lzo=01;31:*.lxz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:'. 'XDG_CURRENT_DESKTOP':
'ubuntu:GNOME', 'VTE_VERSION': '6003', 'CMAKE_PREFIX_PATH':
'/home/labuser/catkin_ws/devel:/opt/ros/noetic', 'GNOME_TERMINAL_SCREEN':
'/org/gnome/Terminal/screen/1849a7e0_8ec5_432a_b712_e503c1ee208a', 'INVOCATION_ID':
'45652184429345298a08c97e5db93533', 'MANAGERPID': '1997', 'GJS_DEBUG_OUTPUT':
'stderr', 'ROBOT_ENV': 'ipa-office', 'ROBOT': 'cob4-4', 'LESSCLOSE': '/usr/bin/lesspipe %s %s',
'XDG_SESSION_CLASS': 'user', 'PYTHONPATH': '/opt/ros/noetic/lib/python3/dist-packages',
'TERM': 'xterm-256color', 'LC_IDENTIFICATION': 'de_DE.UTF-8', 'LESSOPEN': '|
'/usr/bin/lesspipe %s', 'USER': 'labuser', 'GNOME_TERMINAL_SERVICE': ':1.105', 'DISPLAY':
'1', 'SHLVL': '1', 'LC_TELEPHONE': 'de_DE.UTF-8', 'ROS_MASTER_URI':
'http://localhost:11311', 'QT_IM_MODULE': 'ibus', 'LC_MEASUREMENT': 'de_DE.UTF-8',
'LD_LIBRARY_PATH':
'/home/labuser/catkin_ws/devel/lib:/opt/ros/noetic/lib:/opt/ros/noetic/lib/x86_64-linux-gnu',
'XDG_RUNTIME_DIR': '/run/user/1000', 'LC_TIME': 'de_DE.UTF-8', 'JOURNAL_STREAM':
'8:54340', 'XDG_DATA_DIRS':
'/usr/share/ubuntu:/usr/local/share:/usr/share:/var/lib/snapd/desktop', 'PATH':
'/opt/ros/noetic/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/
games:/snap/bin', 'GDMSESSION': 'ubuntu', 'DBUS_SESSION_BUS_ADDRESS':
'unix:path=/run/user/1000/bus', 'ROS_ROOT': '/opt/ros/noetic/share/ros', 'ROS_DISTRO':
'noetic', 'LC_NUMERIC': 'de_DE.UTF-8', '_': '/opt/ros/noetic/bin/roslaunch',
'ROS_LOG_FILENAME':
'/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-6295
.log', 'ROS_NAMESPACE': '/base']]
[roslaunch][INFO] 2026-06-02 20:05:29,865: process[base/velocity_smoother-11]: env[{'SHELL':
'/bin/bash', 'ROS_VERSION': '1', 'SESSION_MANAGER':
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'QT_ACCESSIBILITY': '1', 'COLORTERM': 'truecolor', 'XDG_CONFIG_DIRS':
'/etc/xdg/xdg-ubuntu:/etc/xdg', 'XDG_MENU_PREFIX': 'gnome-',
'GNOME_DESKTOP_SESSION_ID': 'this-is-deprecated', 'PKG_CONFIG_PATH':
'/home/labuser/catkin_ws/devel/lib/pkgconfig:/opt/ros/noetic/lib/pkgconfig:/opt/ros/noetic/lib/x86_64-linux-gnu/pkgconfig', 'ROS_PYTHON_VERSION': '3', 'LC_ADDRESS': 'de_DE.UTF-8',

'GNOME_SHELL_SESSION_MODE': 'ubuntu', 'LC_NAME': 'de_DE.UTF-8',
'SSH_AUTH_SOCK': '/run/user/1000/keyring/ssh', 'XMODIFIERS': '@im=ibus',
'DESKTOP_SESSION': 'ubuntu', 'LC_MONETARY': 'de_DE.UTF-8', 'SSH_AGENT_PID': '2099',
'ROS_PACKAGE_PATH': '/home/labuser/catkin_ws/src:/opt/ros/noetic/share',
'GTK_MODULES': 'gail:atk-bridge', 'ROSLISP_PACKAGE_DIRECTORIES':
'/home/labuser/catkin_ws/devel/share/common-lisp', 'PWD': '/home/labuser', 'LOGNAME':
'labuser', 'XDG_SESSION_DESKTOP': 'ubuntu', 'XDG_SESSION_TYPE': 'x11',
'GPG_AGENT_INFO': '/run/user/1000/gnupg/S.gpg-agent:0:1', 'XAUTHORITY':
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'WINDOWPATH': '2', 'HOME': '/home/labuser', 'USERNAME': 'labuser', 'IM_CONFIG_PHASE':
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gz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:
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z=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:
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ga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;
35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;3
5:*.webm=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:
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35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=0
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midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=
00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:.', 'XDG_CURRENT_DESKTOP':
'ubuntu:GNOME', 'VTE_VERSION': '6003', 'CMAKE_PREFIX_PATH':
'/home/labuser/catkin_ws/devel:/opt/ros/noetic', 'GNOME_TERMINAL_SCREEN':
'/org/gnome/Terminal/screen/1849a7e0_8ec5_432a_b712_e503c1ee208a', 'INVOCATION_ID':
'45652184429345298a08c97e5db93533', 'MANAGERPID': '1997', 'GJS_DEBUG_OUTPUT':
'stderr', 'ROBOT_ENV': 'ipa-office', 'ROBOT': 'cob4-4', 'LESSCLOSE': '/usr/bin/lesspipe %s %s',
'XDG_SESSION_CLASS': 'user', 'PYTHONPATH': '/opt/ros/noetic/lib/python3/dist-packages',
'TERM': 'xterm-256color', 'LC_IDENTIFICATION': 'de_DE.UTF-8', 'LESSOPEN': '|
'/usr/bin/lesspipe %s', 'USER': 'labuser', 'GNOME_TERMINAL_SERVICE': ':1.105', 'DISPLAY':
'1', 'SHLVL': '1', 'LC_TELEPHONE': 'de_DE.UTF-8', 'ROS_MASTER_URI':
'http://localhost:11311', 'QT_IM_MODULE': 'ibus', 'LC_MEASUREMENT': 'de_DE.UTF-8',
'LD_LIBRARY_PATH':
'/home/labuser/catkin_ws/devel/lib:/opt/ros/noetic/lib:/opt/ros/noetic/lib/x86_64-linux-gnu',
'XDG_RUNTIME_DIR': '/run/user/1000', 'LC_TIME': 'de_DE.UTF-8', 'JOURNAL_STREAM':
'8:54340', 'XDG_DATA_DIRS':
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games:/snap/bin', 'GDMSESSION': 'ubuntu', 'DBUS_SESSION_BUS_ADDRESS':
'unix:path=/run/user/1000/bus', 'ROS_ROOT': '/opt/ros/noetic/share/ros', 'ROS_DISTRO':
'noetic', 'LC_NUMERIC': 'de_DE.UTF-8', '_': '/opt/ros/noetic/bin/roslaunch',
'ROS_LOG_FILENAME':
'/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-6295
.log', 'ROS_NAMESPACE': '/base'}}
[roslaunch][INFO] 2026-06-02 20:05:29,871: process[diagnostic_aggregator-12]: env[{'SHELL':
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'local/lab-ubuntu-01:@/tmp/.ICE-unix/2147,unix/lab-ubuntu-01:/tmp/.ICE-unix/2147',
'QT_ACCESSIBILITY': '1', 'COLORTERM': 'truecolor', 'XDG_CONFIG_DIRS':
'/etc/xdg/xdg-ubuntu:/etc/xdg', 'XDG_MENU_PREFIX': 'gnome-',
'GNOME_DESKTOP_SESSION_ID': 'this-is-deprecated', 'PKG_CONFIG_PATH':
'/home/labuser/catkin_ws/devel/lib/pkgconfig:/opt/ros/noetic/lib/pkgconfig:/opt/ros/noetic/lib/x86_
64-linux-gnu/pkgconfig', 'ROS_PYTHON_VERSION': '3', 'LC_ADDRESS': 'de_DE.UTF-8',
'GNOME_SHELL_SESSION_MODE': 'ubuntu', 'LC_NAME': 'de_DE.UTF-8',
'SSH_AUTH_SOCK': '/run/user/1000/keyring/ssh', 'XMODIFIERS': '@im=ibus',
'DESKTOP_SESSION': 'ubuntu', 'LC_MONETARY': 'de_DE.UTF-8', 'SSH_AGENT_PID': '2099',
'ROS_PACKAGE_PATH': '/home/labuser/catkin_ws/src:/opt/ros/noetic/share',
'GTK_MODULES': 'gail:atk-bridge', 'ROSLISP_PACKAGE_DIRECTORIES':
'/home/labuser/catkin_ws/devel/share/common-lisp', 'PWD': '/home/labuser', 'LOGNAME':
'labuser', 'XDG_SESSION_DESKTOP': 'ubuntu', 'XDG_SESSION_TYPE': 'x11',
'GPG_AGENT_INFO': '/run/user/1000/gnupg/S.gpg-agent:0:1', 'XAUTHORITY':
'/run/user/1000/gdm/Xauthority', 'GJS_DEBUG_TOPICS': 'JS ERROR;JS LOG',
'WINDOWPATH': '2', 'HOME': '/home/labuser', 'USERNAME': 'labuser', 'IM_CONFIG_PHASE':
'1', 'LC_PAPER': 'de_DE.UTF-8', 'LANG': 'en_US.UTF-8', 'ROS_ETC_DIR':
'/opt/ros/noetic/etc/ros', 'LS_COLORS':
'rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;3
1;01:mi=00:su=37;41:sg=30;43:ca=30;41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.t
gz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:
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z=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:
*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;3
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ga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;
35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;3
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35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=0
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midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=
00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:.', 'XDG_CURRENT_DESKTOP':
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'ubuntu:GNOME', 'VTE_VERSION': '6003', 'CMAKE_PREFIX_PATH':
'/home/labuser/catkin_ws/devel:/opt/ros/noetic', 'GNOME_TERMINAL_SCREEN':
'/org/gnome/Terminal/screen/1849a7e0_8ec5_432a_b712_e503c1ee208a', 'INVOCATION_ID':
'45652184429345298a08c97e5db93533', 'MANAGERPID': '1997', 'GJS_DEBUG_OUTPUT':
'stderr', 'ROBOT_ENV': 'ipa-office', 'ROBOT': 'cob4-4', 'LESSCLOSE': '/usr/bin/lesspipe %s %s',
'XDG_SESSION_CLASS': 'user', 'PYTHONPATH': '/opt/ros/noetic/lib/python3/dist-packages',
'TERM': 'xterm-256color', 'LC_IDENTIFICATION': 'de_DE.UTF-8', 'LESSOPEN': '|
'/usr/bin/lesspipe %s', 'USER': 'labuser', 'GNOME_TERMINAL_SERVICE': ':1.105', 'DISPLAY':
'1', 'SHLVL': '1', 'LC_TELEPHONE': 'de_DE.UTF-8', 'ROS_MASTER_URI':
'http://localhost:11311', 'QT_IM_MODULE': 'ibus', 'LC_MEASUREMENT': 'de_DE.UTF-8',
'LD_LIBRARY_PATH':
'/home/labuser/catkin_ws/devel/lib:/opt/ros/noetic/lib:/opt/ros/noetic/lib/x86_64-linux-gnu',
'XDG_RUNTIME_DIR': '/run/user/1000', 'LC_TIME': 'de_DE.UTF-8', 'JOURNAL_STREAM':
'8:54340', 'XDG_DATA_DIRS':
'/usr/share/ubuntu:/usr/local/share:/usr/share:/var/lib/snapd/desktop', 'PATH':
'/opt/ros/noetic/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/
games:/snap/bin', 'GDMSESSION': 'ubuntu', 'DBUS_SESSION_BUS_ADDRESS':
'unix:path=/run/user/1000/bus', 'ROS_ROOT': '/opt/ros/noetic/share/ros', 'ROS_DISTRO':
'noetic', 'LC_NUMERIC': 'de_DE.UTF-8', '_': '/opt/ros/noetic/bin/roslaunch',
'ROS_LOG_FILENAME':
'/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-6295
.log'}}]
[roslaunch][INFO] 2026-06-02 20:05:29,876: process[dashboard_aggregator-13]: env[{'SHELL':
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'QT_ACCESSIBILITY': '1', 'COLORTERM': 'truecolor', 'XDG_CONFIG_DIRS':
'/etc/xdg/xdg-ubuntu:/etc/xdg', 'XDG_MENU_PREFIX': 'gnome-',
'GNOME_DESKTOP_SESSION_ID': 'this-is-deprecated', 'PKG_CONFIG_PATH':
'/home/labuser/catkin_ws/devel/lib/pkgconfig:/opt/ros/noetic/lib/pkgconfig:/opt/ros/noetic/lib/x86_
64-linux-gnu/pkgconfig', 'ROS_PYTHON_VERSION': '3', 'LC_ADDRESS': 'de_DE.UTF-8',
'GNOME_SHELL_SESSION_MODE': 'ubuntu', 'LC_NAME': 'de_DE.UTF-8',
'SSH_AUTH_SOCK': '/run/user/1000/keyring/ssh', 'XMODIFIERS': '@im=ibus',
'DESKTOP_SESSION': 'ubuntu', 'LC_MONETARY': 'de_DE.UTF-8', 'SSH_AGENT_PID': '2099',
'ROS_PACKAGE_PATH': '/home/labuser/catkin_ws/src:/opt/ros/noetic/share',
'GTK_MODULES': 'gail:atk-bridge', 'ROSLISP_PACKAGE_DIRECTORIES':
'/home/labuser/catkin_ws/devel/share/common-lisp', 'PWD': '/home/labuser', 'LOGNAME':
'labuser', 'XDG_SESSION_DESKTOP': 'ubuntu', 'XDG_SESSION_TYPE': 'x11',
'GPG_AGENT_INFO': '/run/user/1000/gnupg/S.gpg-agent:0:1', 'XAUTHORITY':
'/run/user/1000/gdm/Xauthority', 'GJS_DEBUG_TOPICS': 'JS ERROR;JS LOG',
'WINDOWPATH': '2', 'HOME': '/home/labuser', 'USERNAME': 'labuser', 'IM_CONFIG_PHASE':
'1', 'LC_PAPER': 'de_DE.UTF-8', 'LANG': 'en_US.UTF-8', 'ROS_ETC_DIR':
'/opt/ros/noetic/etc/ros', 'LS_COLORS':
'rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;3
1;01:mi=00:su=37;41:sg=30;43:ca=30;41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.t

gz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:'. 'XDG_CURRENT_DESKTOP': 'ubuntu:GNOME', 'VTE_VERSION': '6003', 'CMAKE_PREFIX_PATH': '/home/labuser/catkin_ws/devel:/opt/ros/noetic', 'GNOME_TERMINAL_SCREEN': '/org/gnome/Terminal/screen/1849a7e0_8ec5_432a_b712_e503c1ee208a', 'INVOCATION_ID': '45652184429345298a08c97e5db93533', 'MANAGERPID': '1997', 'GJS_DEBUG_OUTPUT': 'stderr', 'ROBOT_ENV': 'ipa-office', 'ROBOT': 'cob4-4', 'LESSCLOSE': '/usr/bin/lesspipe %s %s', 'XDG_SESSION_CLASS': 'user', 'PYTHONPATH': '/opt/ros/noetic/lib/python3/dist-packages', 'TERM': 'xterm-256color', 'LC_IDENTIFICATION': 'de_DE.UTF-8', 'LESSOPEN': '| /usr/bin/lesspipe %s', 'USER': 'labuser', 'GNOME_TERMINAL_SERVICE': ':1.105', 'DISPLAY': ':1', 'SHLVL': '1', 'LC_TELEPHONE': 'de_DE.UTF-8', 'ROS_MASTER_URI': 'http://localhost:11311', 'QT_IM_MODULE': 'ibus', 'LC_MEASUREMENT': 'de_DE.UTF-8', 'LD_LIBRARY_PATH': '/home/labuser/catkin_ws/devel/lib:/opt/ros/noetic/lib:/opt/ros/noetic/lib/x86_64-linux-gnu', 'XDG_RUNTIME_DIR': '/run/user/1000', 'LC_TIME': 'de_DE.UTF-8', 'JOURNAL_STREAM': '8:54340', 'XDG_DATA_DIRS': '/usr/share/ubuntu:/usr/local/share:/usr/share:/var/lib/snapd/desktop', 'PATH': '/opt/ros/noetic/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin', 'GDMSESSION': 'ubuntu', 'DBUS_SESSION_BUS_ADDRESS': 'unix:path=/run/user/1000/bus', 'ROS_ROOT': '/opt/ros/noetic/share/ros', 'ROS_DISTRO': 'noetic', 'LC_NUMERIC': 'de_DE.UTF-8', '_': '/opt/ros/noetic/bin/roslaunch', 'ROS_LOG_FILENAME': '/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-6295.log']}]

[roslaunch][INFO] 2026-06-02 20:05:29,881:

process[joy_node_lab_ubuntu_01_6295_2017253021453234181-14]: env[{'SHELL': '/bin/bash', 'ROS_VERSION': '1', 'SESSION_MANAGER': 'local/lab-ubuntu-01:@/tmp/.ICE-unix/2147,unix/lab-ubuntu-01:/tmp/.ICE-unix/2147', 'QT_ACCESSIBILITY': '1', 'COLORTERM': 'truecolor', 'XDG_CONFIG_DIRS': '/etc/xdg/xdg-ubuntu:/etc/xdg', 'XDG_MENU_PREFIX': 'gnome-', 'GNOME_DESKTOP_SESSION_ID': 'this-is-deprecated', 'PKG_CONFIG_PATH':

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games:/snap/bin', 'GDMSESSION': 'ubuntu', 'DBUS_SESSION_BUS_ADDRESS':  
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'ROS_LOG_FILENAME':  
'/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-6295  
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'QT_ACCESSIBILITY': '1', 'COLORTERM': 'truecolor', 'XDG_CONFIG_DIRS':  
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'GNOME_DESKTOP_SESSION_ID': 'this-is-deprecated', 'PKG_CONFIG_PATH':  
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64-linux-gnu/pkgconfig', 'ROS_PYTHON_VERSION': '3', 'LC_ADDRESS': 'de_DE.UTF-8',  
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'GPG_AGENT_INFO': '/run/user/1000/gnupg/S.gpg-agent:0:1', 'XAUTHORITY':  
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'WINDOWPATH': '2', 'HOME': '/home/labuser', 'USERNAME': 'labuser', 'IM_CONFIG_PHASE':  
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1;01:mi=00:su=37;41:sg=30;43:ca=30;41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.t  
gz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:  
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5:*.webm=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:  
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35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=0
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'ubuntu:GNOME', 'VTE_VERSION': '6003', 'CMAKE_PREFIX_PATH':
'/home/labuser/catkin_ws/devel:/opt/ros/noetic', 'GNOME_TERMINAL_SCREEN':
'/org/gnome/Terminal/screen/1849a7e0_8ec5_432a_b712_e503c1ee208a', 'INVOCATION_ID':
'45652184429345298a08c97e5db93533', 'MANAGERPID': '1997', 'GJS_DEBUG_OUTPUT':
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'XDG_SESSION_CLASS': 'user', 'PYTHONPATH': '/opt/ros/noetic/lib/python3/dist-packages',
'TERM': 'xterm-256color', 'LC_IDENTIFICATION': 'de_DE.UTF-8', 'LESSOPEN': '|
'/usr/bin/lesspipe %s', 'USER': 'labuser', 'GNOME_TERMINAL_SERVICE': ':1.105', 'DISPLAY':
'1', 'SHLVL': '1', 'LC_TELEPHONE': 'de_DE.UTF-8', 'ROS_MASTER_URI':
'http://localhost:11311', 'QT_IM_MODULE': 'ibus', 'LC_MEASUREMENT': 'de_DE.UTF-8',
'LD_LIBRARY_PATH':
'/home/labuser/catkin_ws/devel/lib:/opt/ros/noetic/lib:/opt/ros/noetic/lib/x86_64-linux-gnu',
'XDG_RUNTIME_DIR': '/run/user/1000', 'LC_TIME': 'de_DE.UTF-8', 'JOURNAL_STREAM':
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games:/snap/bin', 'GDMSESSION': 'ubuntu', 'DBUS_SESSION_BUS_ADDRESS':
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'ROS_LOG_FILENAME':
'/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-6295
.log', 'ROS_NAMESPACE': '/teleop']]
[roslaunch][INFO] 2026-06-02 20:05:29,895: process[script_server-16]: env[{'SHELL':
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'QT_ACCESSIBILITY': '1', 'COLORTERM': 'truecolor', 'XDG_CONFIG_DIRS':
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64-linux-gnu/pkgconfig', 'ROS_PYTHON_VERSION': '3', 'LC_ADDRESS': 'de_DE.UTF-8',
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'DESKTOP_SESSION': 'ubuntu', 'LC_MONETARY': 'de_DE.UTF-8', 'SSH_AGENT_PID': '2099',
'ROS_PACKAGE_PATH': '/home/labuser/catkin_ws/src:/opt/ros/noetic/share',
'GTK_MODULES': 'gail:atk-bridge', 'ROSLISP_PACKAGE_DIRECTORIES':
'/home/labuser/catkin_ws/devel/share/common-lisp', 'PWD': '/home/labuser', 'LOGNAME':
'labuser', 'XDG_SESSION_DESKTOP': 'ubuntu', 'XDG_SESSION_TYPE': 'x11',
'GPG_AGENT_INFO': '/run/user/1000/gnupg/S.gpg-agent:0:1', 'XAUTHORITY':
'/run/user/1000/gdm/Xauthority', 'GJS_DEBUG_TOPICS': 'JS ERROR;JS LOG',
'WINDOWPATH': '2', 'HOME': '/home/labuser', 'USERNAME': 'labuser', 'IM_CONFIG_PHASE':
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/opt/ros/noetic/etc/ros', 'LS_COLORS':
'rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;3
1;01:mi=00:su=37;41:sg=30;43:ca=30;41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.t
gz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:
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midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=
00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:'. 'XDG_CURRENT_DESKTOP':
'ubuntu:GNOME', 'VTE_VERSION': '6003', 'CMAKE_PREFIX_PATH':
'/home/labuser/catkin_ws/devel:/opt/ros/noetic', 'GNOME_TERMINAL_SCREEN':
'/org/gnome/Terminal/screen/1849a7e0_8ec5_432a_b712_e503c1ee208a', 'INVOCATION_ID':
'45652184429345298a08c97e5db93533', 'MANAGERPID': '1997', 'GJS_DEBUG_OUTPUT':
'stderr', 'ROBOT_ENV': 'ipa-office', 'ROBOT': 'cob4-4', 'LESSCLOSE': '/usr/bin/lesspipe %s %s',
'XDG_SESSION_CLASS': 'user', 'PYTHONPATH': '/opt/ros/noetic/lib/python3/dist-packages',
'TERM': 'xterm-256color', 'LC_IDENTIFICATION': 'de_DE.UTF-8', 'LESSOPEN': '|
'/usr/bin/lesspipe %s', 'USER': 'labuser', 'GNOME_TERMINAL_SERVICE': ':1.105', 'DISPLAY':
' :1', 'SHLVL': '1', 'LC_TELEPHONE': 'de_DE.UTF-8', 'ROS_MASTER_URI':
'http://localhost:11311', 'QT_IM_MODULE': 'ibus', 'LC_MEASUREMENT': 'de_DE.UTF-8',
'LD_LIBRARY_PATH':
'/home/labuser/catkin_ws/devel/lib:/opt/ros/noetic/lib:/opt/ros/noetic/lib/x86_64-linux-gnu',
'XDG_RUNTIME_DIR': '/run/user/1000', 'LC_TIME': 'de_DE.UTF-8', 'JOURNAL_STREAM':
'8:54340', 'XDG_DATA_DIRS':
'/usr/share/ubuntu:/usr/local/share:/usr/share:/var/lib/snapd/desktop', 'PATH':
'/opt/ros/noetic/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/
games:/snap/bin', 'GDMSESSION': 'ubuntu', 'DBUS_SESSION_BUS_ADDRESS':
'unix:path=/run/user/1000/bus', 'ROS_ROOT': '/opt/ros/noetic/share/ros', 'ROS_DISTRO':
'noetic', 'LC_NUMERIC': 'de_DE.UTF-8', '_': '/opt/ros/noetic/bin/roslaunch',
'ROS_LOG_FILENAME':
'/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-6295
.log"]]
[roslaunch][INFO] 2026-06-02 20:05:29,901: process[robot_state_publisher-17]: env[{'SHELL':
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'GNOME_SHELL_SESSION_MODE': 'ubuntu', 'LC_NAME': 'de_DE.UTF-8',
'SSH_AUTH_SOCK': '/run/user/1000/keyring/ssh', 'XMODIFIERS': '@im=ibus',
'DESKTOP_SESSION': 'ubuntu', 'LC_MONETARY': 'de_DE.UTF-8', 'SSH_AGENT_PID': '2099',
'ROS_PACKAGE_PATH': '/home/labuser/catkin_ws/src:/opt/ros/noetic/share',
'GTK_MODULES': 'gail:atk-bridge', 'ROSLISP_PACKAGE_DIRECTORIES':
'/home/labuser/catkin_ws/devel/share/common-lisp', 'PWD': '/home/labuser', 'LOGNAME':
'labuser', 'XDG_SESSION_DESKTOP': 'ubuntu', 'XDG_SESSION_TYPE': 'x11',
'GPG_AGENT_INFO': '/run/user/1000/gnupg/S.gpg-agent:0:1', 'XAUTHORITY':
'/run/user/1000/gdm/Xauthority', 'GJS_DEBUG_TOPICS': 'JS ERROR;JS LOG',
'WINDOWPATH': '2', 'HOME': '/home/labuser', 'USERNAME': 'labuser', 'IM_CONFIG_PHASE':
'1', 'LC_PAPER': 'de_DE.UTF-8', 'LANG': 'en_US.UTF-8', 'ROS_ETC_DIR':
'/opt/ros/noetic/etc/ros', 'LS_COLORS':
'rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;3
1;01:mi=00:su=37;41:sg=30;43:ca=30;41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.t
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z=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:
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ga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;
35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;3
5:*.webm=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:
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35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=0
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midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=
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'ubuntu:GNOME', 'VTE_VERSION': '6003', 'CMAKE_PREFIX_PATH':
'/home/labuser/catkin_ws/devel:/opt/ros/noetic', 'GNOME_TERMINAL_SCREEN':
'/org/gnome/Terminal/screen/1849a7e0_8ec5_432a_b712_e503c1ee208a', 'INVOCATION_ID':
'45652184429345298a08c97e5db93533', 'MANAGERPID': '1997', 'GJS_DEBUG_OUTPUT':
'stderr', 'ROBOT_ENV': 'ipa-office', 'ROBOT': 'cob4-4', 'LESSCLOSE': '/usr/bin/lesspipe %s %s',
'XDG_SESSION_CLASS': 'user', 'PYTHONPATH': '/opt/ros/noetic/lib/python3/dist-packages',
'TERM': 'xterm-256color', 'LC_IDENTIFICATION': 'de_DE.UTF-8', 'LESSOPEN': '|
/usr/bin/lesspipe %s', 'USER': 'labuser', 'GNOME_TERMINAL_SERVICE': ':1.105', 'DISPLAY':
'1', 'SHLVL': '1', 'LC_TELEPHONE': 'de_DE.UTF-8', 'ROS_MASTER_URI':
'http://localhost:11311', 'QT_IM_MODULE': 'ibus', 'LC_MEASUREMENT': 'de_DE.UTF-8',
'LD_LIBRARY_PATH':

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'XDG_RUNTIME_DIR': '/run/user/1000', 'LC_TIME': 'de_DE.UTF-8', 'JOURNAL_STREAM':
'8:54340', 'XDG_DATA_DIRS':
'/usr/share/ubuntu:/usr/local/share:/usr/share:/var/lib/snapd/desktop', 'PATH':
'/opt/ros/noetic/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/
games:/snap/bin', 'GDMSESSION': 'ubuntu', 'DBUS_SESSION_BUS_ADDRESS':
'unix:path=/run/user/1000/bus', 'ROS_ROOT': '/opt/ros/noetic/share/ros', 'ROS_DISTRO':
'noetic', 'LC_NUMERIC': 'de_DE.UTF-8', '_': '/opt/ros/noetic/bin/roslaunch',
'ROS_LOG_FILENAME':
'/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-6295
.log'}}
[roslaunch][INFO] 2026-06-02 20:05:29,908: process[joint_state_publisher-18]: env[{'SHELL':
'/bin/bash', 'ROS_VERSION': '1', 'SESSION_MANAGER':
'local/lab-ubuntu-01:@/tmp/.ICE-unix/2147,unix/lab-ubuntu-01:/tmp/.ICE-unix/2147',
'QT_ACCESSIBILITY': '1', 'COLORTERM': 'truecolor', 'XDG_CONFIG_DIRS':
'/etc/xdg/xdg-ubuntu:/etc/xdg', 'XDG_MENU_PREFIX': 'gnome-',
'GNOME_DESKTOP_SESSION_ID': 'this-is-deprecated', 'PKG_CONFIG_PATH':
'/home/labuser/catkin_ws/devel/lib/pkgconfig:/opt/ros/noetic/lib/pkgconfig:/opt/ros/noetic/lib/x86_
64-linux-gnu/pkgconfig', 'ROS_PYTHON_VERSION': '3', 'LC_ADDRESS': 'de_DE.UTF-8',
'GNOME_SHELL_SESSION_MODE': 'ubuntu', 'LC_NAME': 'de_DE.UTF-8',
'SSH_AUTH_SOCK': '/run/user/1000/keyring/ssh', 'XMODIFIERS': '@im=ibus',
'DESKTOP_SESSION': 'ubuntu', 'LC_MONETARY': 'de_DE.UTF-8', 'SSH_AGENT_PID': '2099',
'ROS_PACKAGE_PATH': '/home/labuser/catkin_ws/src:/opt/ros/noetic/share',
'GTK_MODULES': 'gail:atk-bridge', 'ROSLISP_PACKAGE_DIRECTORIES':
'/home/labuser/catkin_ws/devel/share/common-lisp', 'PWD': '/home/labuser', 'LOGNAME':
'labuser', 'XDG_SESSION_DESKTOP': 'ubuntu', 'XDG_SESSION_TYPE': 'x11',
'GPG_AGENT_INFO': '/run/user/1000/gnupg/S.gpg-agent:0:1', 'XAUTHORITY':
'/run/user/1000/gdm/Xauthority', 'GJS_DEBUG_TOPICS': 'JS ERROR;JS LOG',
'WINDOWPATH': '2', 'HOME': '/home/labuser', 'USERNAME': 'labuser', 'IM_CONFIG_PHASE':
'1', 'LC_PAPER': 'de_DE.UTF-8', 'LANG': 'en_US.UTF-8', 'ROS_ETC_DIR':
'/opt/ros/noetic/etc/ros', 'LS_COLORS':
'rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;3
1;01:mi=00:su=37;41:sg=30;43:ca=30;41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.t
gz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:
*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lr
z=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tb
z=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:
*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;3
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mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.t
ga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;
35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;3
5:*.webm=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:
*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;
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35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:', 'XDG_CURRENT_DESKTOP':
'ubuntu:GNOME', 'VTE_VERSION': '6003', 'CMAKE_PREFIX_PATH':
'/home/labuser/catkin_ws/devel:/opt/ros/noetic', 'GNOME_TERMINAL_SCREEN':
'/org/gnome/Terminal/screen/1849a7e0_8ec5_432a_b712_e503c1ee208a', 'INVOCATION_ID':
'45652184429345298a08c97e5db93533', 'MANAGERPID': '1997', 'GJS_DEBUG_OUTPUT':
'stderr', 'ROBOT_ENV': 'ipa-office', 'ROBOT': 'cob4-4', 'LESSCLOSE': '/usr/bin/lesspipe %s %s',
'XDG_SESSION_CLASS': 'user', 'PYTHONPATH': '/opt/ros/noetic/lib/python3/dist-packages',
'TERM': 'xterm-256color', 'LC_IDENTIFICATION': 'de_DE.UTF-8', 'LESSOPEN': '|
'/usr/bin/lesspipe %s', 'USER': 'labuser', 'GNOME_TERMINAL_SERVICE': ':1.105', 'DISPLAY':
' :1', 'SHLVL': '1', 'LC_TELEPHONE': 'de_DE.UTF-8', 'ROS_MASTER_URI':
'http://localhost:11311', 'QT_IM_MODULE': 'ibus', 'LC_MEASUREMENT': 'de_DE.UTF-8',
'LD_LIBRARY_PATH':
'/home/labuser/catkin_ws/devel/lib:/opt/ros/noetic/lib:/opt/ros/noetic/lib/x86_64-linux-gnu',
'XDG_RUNTIME_DIR': '/run/user/1000', 'LC_TIME': 'de_DE.UTF-8', 'JOURNAL_STREAM':
'8:54340', 'XDG_DATA_DIRS':
'/usr/share/ubuntu:/usr/local/share:/usr/share:/var/lib/snapd/desktop', 'PATH':
'/opt/ros/noetic/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/
games:/snap/bin', 'GDMSESSION': 'ubuntu', 'DBUS_SESSION_BUS_ADDRESS':
'unix:path=/run/user/1000/bus', 'ROS_ROOT': '/opt/ros/noetic/share/ros', 'ROS_DISTRO':
'noetic', 'LC_NUMERIC': 'de_DE.UTF-8', '_': '/opt/ros/noetic/bin/roslaunch',
'ROS_LOG_FILENAME':
'/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-6295
.log"]]
[roslaunch][INFO] 2026-06-02 20:05:29,922:
process[fake_diagnostics_lab_ubuntu_01_6295_3108807645079097714-19]: env[{'SHELL':
'/bin/bash', 'ROS_VERSION': '1', 'SESSION_MANAGER':
'local/lab-ubuntu-01:@/tmp/.ICE-unix/2147,unix/lab-ubuntu-01:/tmp/.ICE-unix/2147',
'QT_ACCESSIBILITY': '1', 'COLORTERM': 'truecolor', 'XDG_CONFIG_DIRS':
'/etc/xdg/xdg-ubuntu:/etc/xdg', 'XDG_MENU_PREFIX': 'gnome-',
'GNOME_DESKTOP_SESSION_ID': 'this-is-deprecated', 'PKG_CONFIG_PATH':
'/home/labuser/catkin_ws/devel/lib/pkgconfig:/opt/ros/noetic/lib/pkgconfig:/opt/ros/noetic/lib/x86_
64-linux-gnu/pkgconfig', 'ROS_PYTHON_VERSION': '3', 'LC_ADDRESS': 'de_DE.UTF-8',
'GNOME_SHELL_SESSION_MODE': 'ubuntu', 'LC_NAME': 'de_DE.UTF-8',
'SSH_AUTH_SOCK': '/run/user/1000/keyring/ssh', 'XMODIFIERS': '@im=ibus',
'DESKTOP_SESSION': 'ubuntu', 'LC_MONETARY': 'de_DE.UTF-8', 'SSH_AGENT_PID': '2099',
'ROS_PACKAGE_PATH': '/home/labuser/catkin_ws/src:/opt/ros/noetic/share',
'GTK_MODULES': 'gail:atk-bridge', 'ROSLISP_PACKAGE_DIRECTORIES':
'/home/labuser/catkin_ws/devel/share/common-lisp', 'PWD': '/home/labuser', 'LOGNAME':
'labuser', 'XDG_SESSION_DESKTOP': 'ubuntu', 'XDG_SESSION_TYPE': 'x11',
'GPG_AGENT_INFO': '/run/user/1000/gnupg/S.gpg-agent:0:1', 'XAUTHORITY':
'/run/user/1000/gdm/Xauthority', 'GJS_DEBUG_TOPICS': 'JS ERROR;JS LOG',

'WINDOWPATH': '2', 'HOME': '/home/labuser', 'USERNAME': 'labuser', 'IM_CONFIG_PHASE': '1', 'LC_PAPER': 'de_DE.UTF-8', 'LANG': 'en_US.UTF-8', 'ROS_ETC_DIR': '/opt/ros/noetic/etc/ros', 'LS_COLORS':
'rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=30;41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tztst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:.', 'XDG_CURRENT_DESKTOP': 'ubuntu:GNOME', 'VTE_VERSION': '6003', 'CMAKE_PREFIX_PATH': '/home/labuser/catkin_ws/devel:/opt/ros/noetic', 'GNOME_TERMINAL_SCREEN': '/org/gnome/Terminal/screen/1849a7e0_8ec5_432a_b712_e503c1ee208a', 'INVOCATION_ID': '45652184429345298a08c97e5db93533', 'MANAGERPID': '1997', 'GJS_DEBUG_OUTPUT': 'stderr', 'ROBOT_ENV': 'ipa-office', 'ROBOT': 'cob4-4', 'LESSCLOSE': '/usr/bin/lesspipe %s %s', 'XDG_SESSION_CLASS': 'user', 'PYTHONPATH': '/opt/ros/noetic/lib/python3/dist-packages', 'TERM': 'xterm-256color', 'LC_IDENTIFICATION': 'de_DE.UTF-8', 'LESSOPEN': '| /usr/bin/lesspipe %s', 'USER': 'labuser', 'GNOME_TERMINAL_SERVICE': ':1.105', 'DISPLAY': ':1', 'SHLVL': '1', 'LC_TELEPHONE': 'de_DE.UTF-8', 'ROS_MASTER_URI': 'http://localhost:11311', 'QT_IM_MODULE': 'ibus', 'LC_MEASUREMENT': 'de_DE.UTF-8', 'LD_LIBRARY_PATH': '/home/labuser/catkin_ws/devel/lib:/opt/ros/noetic/lib:/opt/ros/noetic/lib/x86_64-linux-gnu', 'XDG_RUNTIME_DIR': '/run/user/1000', 'LC_TIME': 'de_DE.UTF-8', 'JOURNAL_STREAM': '8:54340', 'XDG_DATA_DIRS': '/usr/share/ubuntu:/usr/local/share:/usr/share:/var/lib/snapd/desktop', 'PATH': '/opt/ros/noetic/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin', 'GDMSESSION': 'ubuntu', 'DBUS_SESSION_BUS_ADDRESS': 'unix:path=/run/user/1000/bus', 'ROS_ROOT': '/opt/ros/noetic/share/ros', 'ROS_DISTRO': 'noetic', 'LC_NUMERIC': 'de_DE.UTF-8', '_': '/opt/ros/noetic/bin/roslaunch', 'ROS_LOG_FILENAME': '/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-6295.log']
[roslaunch][INFO] 2026-06-02 20:05:29,934: process[spawn_gazebo_model-20]: env[{'SHELL': '/bin/bash', 'ROS_VERSION': '1', 'SESSION_MANAGER':

'local/lab-ubuntu-01:@/tmp/.ICE-unix/2147,unix/lab-ubuntu-01:/tmp/.ICE-unix/2147',
'QT_ACCESSIBILITY': '1', 'COLORTERM': 'truecolor', 'XDG_CONFIG_DIRS':
'/etc/xdg/xdg-ubuntu:/etc/xdg', 'XDG_MENU_PREFIX': 'gnome-',
'GNOME_DESKTOP_SESSION_ID': 'this-is-deprecated', 'PKG_CONFIG_PATH':
'/home/labuser/catkin_ws/devel/lib/pkgconfig:/opt/ros/noetic/lib/pkgconfig:/opt/ros/noetic/lib/x86_64-linux-gnu/pkgconfig', 'ROS_PYTHON_VERSION': '3', 'LC_ADDRESS': 'de_DE.UTF-8',
'GNOME_SHELL_SESSION_MODE': 'ubuntu', 'LC_NAME': 'de_DE.UTF-8',
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'DESKTOP_SESSION': 'ubuntu', 'LC_MONETARY': 'de_DE.UTF-8', 'SSH_AGENT_PID': '2099',
'ROS_PACKAGE_PATH': '/home/labuser/catkin_ws/src:/opt/ros/noetic/share',
'GTK_MODULES': 'gail:atk-bridge', 'ROSLISP_PACKAGE_DIRECTORIES':
'/home/labuser/catkin_ws/devel/share/common-lisp', 'PWD': '/home/labuser', 'LOGNAME':
'labuser', 'XDG_SESSION_DESKTOP': 'ubuntu', 'XDG_SESSION_TYPE': 'x11',
'GPG_AGENT_INFO': '/run/user/1000/gnupg/S.gpg-agent:0:1', 'XAUTHORITY':
'/run/user/1000/gdm/Xauthority', 'GJS_DEBUG_TOPICS': 'JS ERROR;JS LOG',
'WINDOWPATH': '2', 'HOME': '/home/labuser', 'USERNAME': 'labuser', 'IM_CONFIG_PHASE':
'1', 'LC_PAPER': 'de_DE.UTF-8', 'LANG': 'en_US.UTF-8', 'ROS_ETC_DIR':
'/opt/ros/noetic/etc/ros', 'LS_COLORS':
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gz=01;31;*.arc=01;31;*.arj=01;31;*.taz=01;31;*.lha=01;31;*.lzh=01;31;*.lzma=01;31;
.tlz=01;31;.txz=01;31;*.tzo=01;31;*.t7z=01;31;*.zip=01;31;*.z=01;31;*.dz=01;31;*.gz=01;31;*.lr
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z=01;31;*.tbz2=01;31;*.tz=01;31;*.deb=01;31;*.rpm=01;31;*.jar=01;31;*.war=01;31;*.ear=01;31;
.sar=01;31;.rar=01;31;*.alz=01;31;*.ace=01;31;*.zoo=01;31;*.cpio=01;31;*.7z=01;31;*.rz=01;3
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ga=01;35;*.xbm=01;35;*.xpm=01;35;*.tif=01;35;*.tiff=01;35;*.png=01;35;*.svg=01;35;*.svgz=01;
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.nuv=01;35;.wmv=01;35;*.asf=01;35;*.rm=01;35;*.rmvb=01;35;*.flc=01;35;*.avi=01;35;*.fli=01;
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00;36;*.opus=00;36;*.spx=00;36;*.xspf=00;36;', 'XDG_CURRENT_DESKTOP':
'ubuntu:GNOME', 'VTE_VERSION': '6003', 'CMAKE_PREFIX_PATH':
'/home/labuser/catkin_ws/devel:/opt/ros/noetic', 'GNOME_TERMINAL_SCREEN':
'/org/gnome/Terminal/screen/1849a7e0_8ec5_432a_b712_e503c1ee208a', 'INVOCATION_ID':
'45652184429345298a08c97e5db93533', 'MANAGERPID': '1997', 'GJS_DEBUG_OUTPUT':
'stderr', 'ROBOT_ENV': 'ipa-office', 'ROBOT': 'cob4-4', 'LESSCLOSE': '/usr/bin/lesspipe %s %s',
'XDG_SESSION_CLASS': 'user', 'PYTHONPATH': '/opt/ros/noetic/lib/python3/dist-packages',
'TERM': 'xterm-256color', 'LC_IDENTIFICATION': 'de_DE.UTF-8', 'LESSOPEN': '|
'/usr/bin/lesspipe %s', 'USER': 'labuser', 'GNOME_TERMINAL_SERVICE': ':1.105', 'DISPLAY':
'1', 'SHLVL': '1', 'LC_TELEPHONE': 'de_DE.UTF-8', 'ROS_MASTER_URI':

```
'http://localhost:11311', 'QT_IM_MODULE': 'ibus', 'LC_MEASUREMENT': 'de_DE.UTF-8',
'LD_LIBRARY_PATH':
'/home/labuser/catkin_ws/devel/lib:/opt/ros/noetic/lib:/opt/ros/noetic/lib/x86_64-linux-gnu',
'XDG_RUNTIME_DIR': '/run/user/1000', 'LC_TIME': 'de_DE.UTF-8', 'JOURNAL_STREAM':
'8:54340', 'XDG_DATA_DIRS':
'/usr/share/ubuntu:/usr/local/share:/usr/share:/var/lib/snapd/desktop', 'PATH':
'/opt/ros/noetic/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/
games:/snap/bin', 'GDMSESSION': 'ubuntu', 'DBUS_SESSION_BUS_ADDRESS':
'unix:path=/run/user/1000/bus', 'ROS_ROOT': '/opt/ros/noetic/share/ros', 'ROS_DISTRO':
'noetic', 'LC_NUMERIC': 'de_DE.UTF-8', '_': '/opt/ros/noetic/bin/roslaunch',
'ROS_LOG_FILENAME':
'/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-6295
.log'}}
[roslaunch][ERROR] 2026-06-02 20:05:30,559: [gazebo-1] process has died [pid 6319, exit
code 255, cmd /opt/ros/noetic/lib/gazebo_ros/gzserver -e ode worlds/empty.world
__name:=gazebo
__log:=/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/gazebo-1.log].
[roslaunch][ERROR] 2026-06-02 20:15:28,744: [gazebo_gui-2] escalating to SIGTERM
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$
```

```
labuser@lab-ubuntu-01:~$ source /opt/ros/noetic/setup.bash
labuser@lab-ubuntu-01:~$ source ~/catkin_ws/devel/setup.bash
labuser@lab-ubuntu-01:~$ rospack find cob_gazebo
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo
labuser@lab-ubuntu-01:~$ rospack find cob_bringup_sim
/home/labuser/catkin_ws/src/cob_simulation/cob_bringup_sim
labuser@lab-ubuntu-01:~$ roslaunch cob_bringup_sim robot.launch
... logging to
/home/labuser/.ros/log/9adb3ed0-5eab-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-1458
0.log
Checking log directory for disk usage. This may take a while.
Press Ctrl-C to interrupt
Done checking log file disk usage. Usage is <1GB.
```

```
RLEException: error loading <rosparam> tag:
file does not exist
```

```
[/opt/ros/noetic/share/cob_default_env_config/envs/!!NO_ROBOT_ENV_SET!!/navigation_goals
.yaml]
XML is <rosparam command="load" ns="/script_server/base" file="$(find
cob_default_env_config)/envs/$(arg robot_env)/navigation_goals.yaml"/>
```

The traceback for the exception was written to the log file

labuser@lab-ubuntu-01:~\$

labuser@lab-ubuntu-01:~\$ source /opt/ros/noetic/setup.bash

labuser@lab-ubuntu-01:~\$ source ~/catkin_ws/devel/setup.bash

labuser@lab-ubuntu-01:~\$ export ROBOT=cob4-4

labuser@lab-ubuntu-01:~\$ export ROBOT_ENV=ipa-office

labuser@lab-ubuntu-01:~\$ echo \$ROBOT

cob4-4

labuser@lab-ubuntu-01:~\$ echo \$ROBOT_ENV

ipa-office

labuser@lab-ubuntu-01:~\$ roslaunch cob_bringup_sim robot.launch

... logging to

/home/labuser/.ros/log/26344c56-5eb1-11f1-b340-0390f703237c/roslaunch-lab-ubuntu-01-1481

3.log

Checking log directory for disk usage. This may take a while.

Press Ctrl-C to interrupt

Done checking log file disk usage. Usage is <1GB.

xacro: in-order processing became default in ROS Melodic. You can drop the option.

xacro: in-order processing became default in ROS Melodic. You can drop the option.

started roslaunch server http://lab-ubuntu-01:46439/

SUMMARY

=====

CLEAR PARAMETERS

* /bms/

PARAMETERS

* /base/halt_detector/diagnostic_recovery_trigger: []

* /base/halt_detector/recover_after_stuck_timeout: 3.0

* /base/halt_detector/stop_threshold: 0.001

* /base/halt_detector/stop_timeout: 10.0

* /base/halt_detector/stuck_threshold: 0.174533

* /base/halt_detector/stuck_timeout: 2.0

* /base/joint_names: ['fl_caster_rotat...

* /base/joint_state_controller/publish_rate: 50

* /base/joint_state_controller/type: joint_state_contr...

* /base/odometry_controller/child_frame_id: base_footprint

* /base/odometry_controller/cov_pose: 0.1

* /base/odometry_controller/cov_twist: 0.1

- * /base/odometry_controller/frame_id: odom_combined
- * /base/odometry_controller/publish_rate: 50
- * /base/odometry_controller/type: cob_omni_drive_co...
- * /base/odometry_controller/wheels: [{'steer': 'fl_ca...
- * /base/twist_controller/defaults/steer_ctrl/d_phi_max: 10.0
- * /base/twist_controller/defaults/steer_ctrl/damp: 2.75
- * /base/twist_controller/defaults/steer_ctrl/dd_phi_max: 40.0
- * /base/twist_controller/defaults/steer_ctrl/spring: 15.0
- * /base/twist_controller/defaults/steer_ctrl/virt_mass: 0.1
- * /base/twist_controller/pub_divider: 1
- * /base/twist_controller/required_drive_mode: 3
- * /base/twist_controller/timeout: 0.5
- * /base/twist_controller/type: cob_omni_drive_co...
- * /base/twist_controller/wheels: [{'steer': 'fl_ca...
- * /base/twist_mux/locks: [{'name': 'pause_...
- * /base/twist_mux/topics: [{'name': 'collis...
- * /base/velocity_smoother/accel_lim_vx: 0.5
- * /base/velocity_smoother/accel_lim_vy: 0.5
- * /base/velocity_smoother/accel_lim_w: 0.5
- * /base/velocity_smoother/decel_factor: 1.5
- * /base/velocity_smoother/decel_factor_safe: 2.5
- * /base/velocity_smoother/frequency: 50.0
- * /base/velocity_smoother/robot_feedback: 1
- * /base/velocity_smoother/speed_lim_vx: 1.2
- * /base/velocity_smoother/speed_lim_vy: 1.0
- * /base/velocity_smoother/speed_lim_w: 1.2
- * /bms/bms_id_to_poll: 512
- * /bms/can_device: can1
- * /bms/diagnostics: [{'id': 2, 'field...
- * /bms/poll_frequency_hz: 20
- * /bms/topics: ['current', 'volt...
- * /diagnostic_aggregator/analyzers/io/analyzers/joystick/contains: ['joy']
- * /diagnostic_aggregator/analyzers/io/analyzers/joystick/path: Joystick
- * /diagnostic_aggregator/analyzers/io/analyzers/joystick/type: diagnostic_aggreg...
- * /diagnostic_aggregator/analyzers/io/path: IO
- * /diagnostic_aggregator/analyzers/io/type: diagnostic_aggreg...
- * /diagnostic_aggregator/analyzers/motors/analyzers/base/contains: ['base/']
- * /diagnostic_aggregator/analyzers/motors/analyzers/base/path: base
- * /diagnostic_aggregator/analyzers/motors/analyzers/base/type: diagnostic_aggreg...
- * /diagnostic_aggregator/analyzers/motors/path: Actuators
- * /diagnostic_aggregator/analyzers/motors/type: diagnostic_aggreg...
- * /diagnostic_aggregator/analyzers/sensors/analyzers/bms/contains: ['bms']
- * /diagnostic_aggregator/analyzers/sensors/analyzers/bms/path: bms
- * /diagnostic_aggregator/analyzers/sensors/analyzers/bms/type: diagnostic_aggreg...

```

* /diagnostic_aggregator/analyzers/sensors/path: Sensors
* /diagnostic_aggregator/analyzers/sensors/type: diagnostic_aggreg...
* /diagnostic_aggregator/other_as_errors: True
* /gazebo/enable_ros_network: True
* /joint_state_publisher/rate: 100
* /joint_state_publisher/source_list: ['/base/joint_sta...
* /joy_node_lab_ubuntu_01_14813_2785014673887014938/deadzone: 0.12
* /joy_node_lab_ubuntu_01_14813_2785014673887014938/dev: /dev/joypad
* /robot_description: <?xml version="1....
* /roscdistro: noetic
* /rosversion: 1.17.4
* /script_server/base/service_ns: /base/driver
* /script_server/base/topic_name: /base/twist_mux/c...
* /teleop/teleop_node_lab_ubuntu_01_14813_5783477407708573307/apply_ramp: False
* /teleop/teleop_node_lab_ubuntu_01_14813_5783477407708573307/axis_vx: 1
* /teleop/teleop_node_lab_ubuntu_01_14813_5783477407708573307/axis_vy: 0
* /teleop/teleop_node_lab_ubuntu_01_14813_5783477407708573307/axis_yaw: 3
*
/teleop/teleop_node_lab_ubuntu_01_14813_5783477407708573307/components/base/twist_m
ax_acc: [0.5, 0.5, 0.7]
*
/teleop/teleop_node_lab_ubuntu_01_14813_5783477407708573307/components/base/twist_m
ax_velocity: [0.3, 0.2, 0.3]
*
/teleop/teleop_node_lab_ubuntu_01_14813_5783477407708573307/components/base/twist_sa
fety_topic_name: /base/collision_v...
*
/teleop/teleop_node_lab_ubuntu_01_14813_5783477407708573307/components/base/twist_to
pic_name: /base/twist_mux/c...
* /teleop/teleop_node_lab_ubuntu_01_14813_5783477407708573307/deadman_button: 5
* /teleop/teleop_node_lab_ubuntu_01_14813_5783477407708573307/init_button: 7
* /teleop/teleop_node_lab_ubuntu_01_14813_5783477407708573307/joy_num_axes: 6
* /teleop/teleop_node_lab_ubuntu_01_14813_5783477407708573307/joy_num_modes: 1
* /teleop/teleop_node_lab_ubuntu_01_14813_5783477407708573307/run_factor: 2.0
* /teleop/teleop_node_lab_ubuntu_01_14813_5783477407708573307/safety_button: 4
* /use_sim_time: True
* /world_description: <?xml version="1....

```

NODES

/

```

bms (cob_bms_driver/fake_bms.py)
dashboard_aggregator (cob_dashboard/cob_dashboard_aggregator.py)
diagnostic_aggregator (diagnostic_aggregator/aggregator_node)

```

```
fake_diagnostics_lab_ubuntu_01_14813_5532009427712672313
(cob_helper_tools/fake_diagnostics.py)
gazebo (gazebo_ros/gzserver)
gazebo_gui (gazebo_ros/gzclient)
joint_state_publisher (joint_state_publisher/joint_state_publisher)
joy_node_lab_ubuntu_01_14813_2785014673887014938 (joy/joy_node)
robot_state_publisher (robot_state_publisher/robot_state_publisher)
script_server (cob_script_server/script_server.py)
spawn_gazebo_model (gazebo_ros/spawn_model)
spawn_urdf_world (gazebo_ros/spawn_model)
/base/
base_controller_spawner (controller_manager/controller_manager)
fake_driver (cob_helper_tools/fake_driver.py)
halt_detector (cob_base_controller_utils/cob_halt_detector)
twist_mux (twist_mux/twist_mux)
twist_mux (twist_mux/twist_mux)
velocity_smoother (cob_base_velocity_smoother/velocity_smoother)
/bms/
power_state_aggregator (cob_bms_driver/power_state_aggregator.py)
/teleop/
teleop_node_lab_ubuntu_01_14813_5783477407708573307 (cob_teleop/cob_teleop)
```

auto-starting new master

process[master]: started with pid [14825]

ROS_MASTER_URI=http://localhost:11311

setting /run_id to 26344c56-5eb1-11f1-b340-0390f703237c

process[rosout-1]: started with pid [14835]

started core service [/rosout]

running rosparam delete /objects

ERROR: parameter [/objects] is not set

process[gazebo-2]: started with pid [14847]

process[gazebo_gui-3]: started with pid [14851]

process[spawn_urdf_world-4]: started with pid [14857]

process[bms-5]: started with pid [14858]

process[bms/power_state_aggregator-6]: started with pid [14859]

process[base/halt_detector-7]: started with pid [14860]

process[base/fake_driver-8]: started with pid [14861]

process[base/base_controller_spawner-9]: started with pid [14866]

[INFO] [1780425038.323804796]: waitForService: Service [/base/driver/halt] has not been advertised, waiting...

process[base/twist_mux-10]: started with pid [14868]

process[base/twist_marker-11]: started with pid [14869]

process[base/velocity_smoother-12]: started with pid [14874]

[INFO] [1780425038.342773935]: Topic handler 'collision_velocity_filter' subscribed to topic 'twist_mux/command_safe': timeout = 0.250000s, priority = 10
process[diagnostic_aggregator-13]: started with pid [14880]
[INFO] [1780425038.346130007]: Topic handler 'navigation' subscribed to topic 'twist_mux/command_navigation': timeout = 0.250000s, priority = 20
[INFO] [1780425038.348371654]: Topic handler 'syncmm' subscribed to topic 'twist_mux/command_syncmm': timeout = 0.500000s, priority = 60
[INFO] [1780425038.351035552]: Topic handler 'script_server' subscribed to topic 'twist_mux/command_script_server': timeout = 0.500000s, priority = 70
process[dashboard_aggregator-14]: started with pid [14883]
[INFO] [1780425038.354271408]: Topic handler 'teleop_keyboard' subscribed to topic 'twist_mux/command_teleop_keyboard': timeout = 0.500000s, priority = 80
[INFO] [1780425038.357797121]: Topic handler 'teleop_android' subscribed to topic 'twist_mux/command_teleop_android': timeout = 0.500000s, priority = 90
process[joy_node_lab_ubuntu_01_14813_2785014673887014938-15]: started with pid [14887]
[INFO] [1780425038.362443573]: Topic handler 'teleop_joystick' subscribed to topic 'twist_mux/command_teleop_joy': timeout = 0.250000s, priority = 100
process[teleop/teleop_node_lab_ubuntu_01_14813_5783477407708573307-16]: started with pid [14893]
process[script_server-17]: started with pid [14895]
[INFO] [1780425038.375393213]: Topic handler 'pause_navigation' subscribed to topic 'twist_mux/locks/pause_navigation': timeout = None, priority = 21
process[robot_state_publisher-18]: started with pid [14902]
[INFO] [1780425038.381441773]: Reconfigure request : 1.200000 1.000000 1.200000 0.500000 0.500000 0.500000 1.500000 2.500000
[INFO] [1780425038.383586676]: Topic handler 'pause_teleop' subscribed to topic 'twist_mux/locks/pause_teleop': timeout = None, priority = 101
process[joint_state_publisher-19]: started with pid [14909]
process[fake_diagnostics_lab_ubuntu_01_14813_5532009427712672313-20]: started with pid [14910]
[INFO] [1780425038.395044665]: Topic handler 'pause_all' subscribed to topic 'twist_mux/locks/pause_all': timeout = None, priority = 255
process[spawn_gazebo_model-21]: started with pid [14913]
[ERROR] [1780425038.445836998]: Couldn't open joystick /dev/joydev. Will retry every second.
[INFO] [1780425038.687628, 0.000000]: power state aggregator running
[INFO] [1780425038.709918, 0.000000]: fake_driver running
[INFO] [1780425038.711261, 0.000000]: Loading model XML from ros parameter world_description
[INFO] [1780425038.712241710]: waitForService: Service [/base/driver/halt] is now available.
[INFO] [1780425038.713096169]: waitForService: Service [/base/controller_manager/switch_controller] has not been advertised, waiting...
[INFO] [1780425038.713312, 0.000000]: Waiting for service /gazebo/spawn_urdf_model
[INFO] [1780425038.842263564]: Finished loading Gazebo ROS API Plugin.

[INFO] [1780425038.843227671]: waitForService: Service [/gazebo/set_physics_properties] has not been advertised, waiting...

[INFO] [1780425038.855156, 0.000000]: Waiting for model ipa-office before proceeding.

[INFO] [1780425038.918133955]: Finished loading Gazebo ROS API Plugin.

[INFO] [1780425038.918977686]: waitForService: Service [/gazebo_gui/set_physics_properties] has not been advertised, waiting...

[INFO] [1780425039.823214, 0.000000]: script_server is running

[INFO] [1780425039.863183685, 0.010000000]: waitForService: Service [/gazebo/set_physics_properties] is now available.

[INFO] [1780425039.878829441, 0.025000000]: Physics dynamic reconfigure ready.

[INFO] [1780425039.918822, 0.000000]: Calling service /gazebo/spawn_urdf_model

[INFO] [1780425040.058501, 0.000000]: Spawn status: SpawnModel: Successfully spawned entity

[INFO] [1780425040.087565093, 0.172000000]: Loading gazebo_ros_control plugin

[INFO] [1780425040.087648096, 0.172000000]: Starting gazebo_ros_control plugin in namespace: world

[INFO] [1780425040.088002600, 0.172000000]: gazebo_ros_control plugin is waiting for model URDF in parameter [/world_description] on the ROS param server.

[INFO] [1780425040.209455603, 0.172000000]: Loaded gazebo_ros_control.

[INFO] [1780425040.237143, 0.200000]: Loading model XML from ros parameter robot_description

[INFO] [1780425040.241780, 0.203000]: Waiting for service /gazebo/spawn_urdf_model

[INFO] [1780425040.243623, 0.206000]: Calling service /gazebo/spawn_urdf_model

[INFO] [1780425040.479325, 0.372000]: Spawn status: SpawnModel: Successfully spawned entity

[INFO] [1780425040.516624809, 0.372000000]: Loading cob_gazebo_ros_control plugin

[INFO] [1780425040.516694443, 0.372000000]: Enable joint-filtering. Using joint_names from parameter: "joint_names"

[INFO] [1780425040.516713753, 0.372000000]: Starting cob_gazebo_ros_control plugin in namespace: base

[INFO] [1780425040.618575476, 0.372000000]: JointFiltering is enabled! DoF: 6

[INFO] [1780425040.623000855, 0.372000000]: Loaded cob_gazebo_ros_control.

[INFO] [1780425040.636936414, 0.386000000]: waitForService: Service [/base/controller_manager/switch_controller] is now available.

[spawn_urdf_world-4] process has finished cleanly

log file:

/home/labuser/.ros/log/26344c56-5eb1-11f1-b340-0390f703237c/spawn_urdf_world-4*.log

Loaded 'joint_state_controller'

[INFO] [1780425040.893489917, 0.643000000]: configure all steers: s: 15.000000, d: 2.750000, m: 0.100000, v: 10.000000, a: 40.000000

[INFO] [1780425040.900854241, 0.650000000]: configure steer 0: s: 15.000000, d: 2.750000, m: 0.100000, v: 10.000000, a: 40.000000

[INFO] [1780425040.908211809, 0.657000000]: configure steer 1: s: 15.000000, d: 2.750000, m: 0.100000, v: 10.000000, a: 40.000000

[INFO] [1780425040.914336222, 0.663000000]: configure steer 2: s: 15.000000, d: 2.750000, m: 0.100000, v: 10.000000, a: 40.000000

Loaded 'twist_controller'

Loaded 'odometry_controller'

Started ['joint_state_controller'] successfully

Started ['twist_controller'] successfully

Started ['odometry_controller'] successfully

[spawn_gazebo_model-21] process has finished cleanly

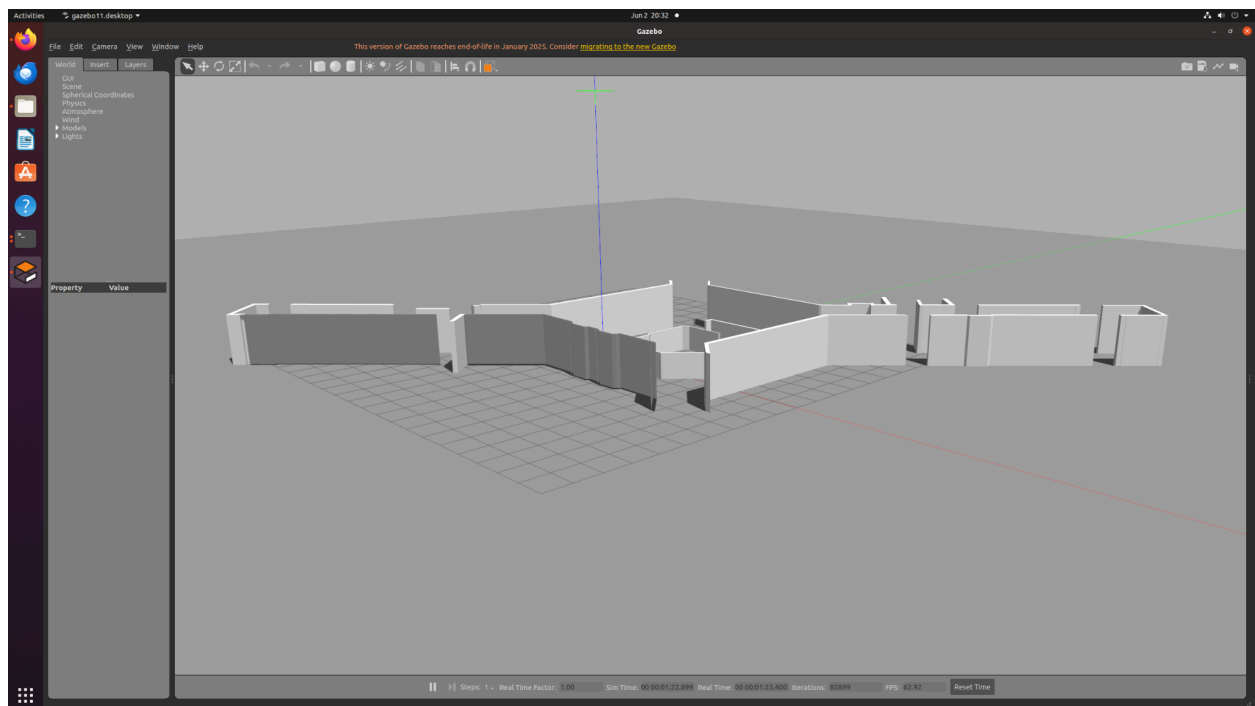
log file:

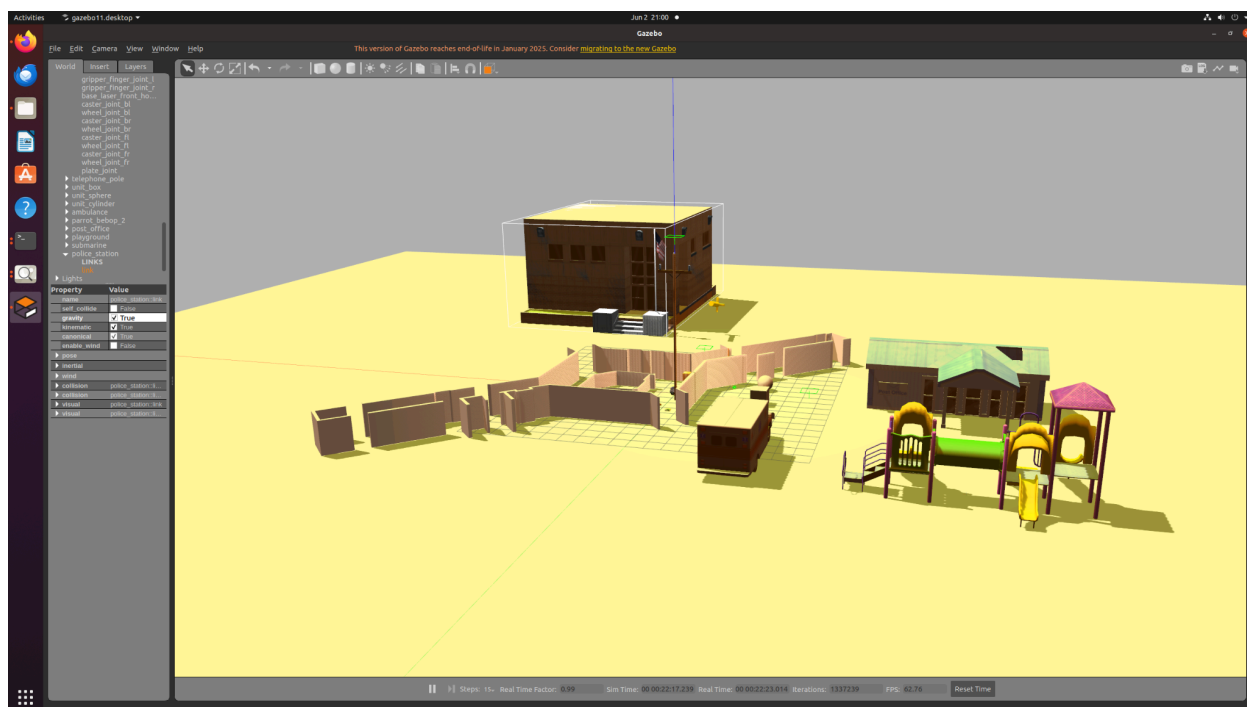
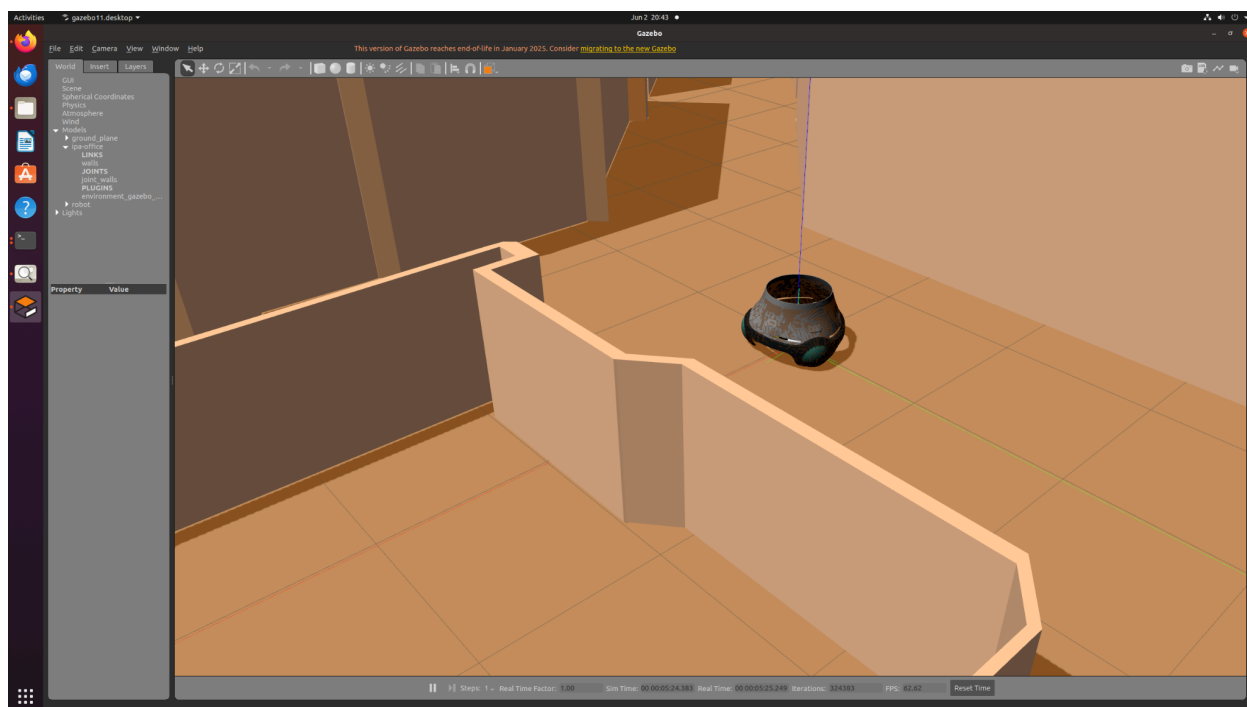
/home/labuser/.ros/log/26344c56-5eb1-11f1-b340-0390f703237c/spawn_gazebo_model-21*.log

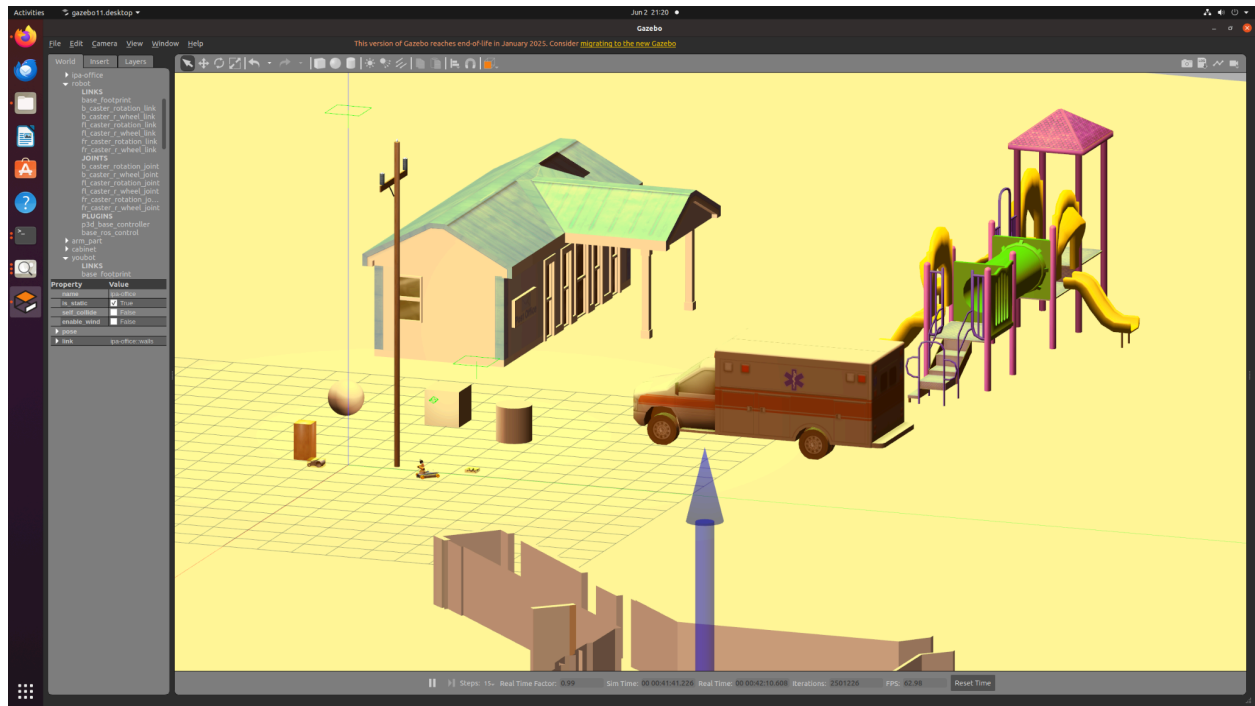
[base/base_controller_spawner-9] process has finished cleanly

log file:

/home/labuser/.ros/log/26344c56-5eb1-11f1-b340-0390f703237c/base-base_controller_spawner-9*.log







```
labuser@lab-ubuntu-01:~$ rostopic list
/base/collision_velocity_filter/command
/base/joint_states
/base/odometry_controller/odometry
/base/twist_controller/b_caster_rotation_joint/parameter_descriptions
/base/twist_controller/b_caster_rotation_joint/parameter_updates
/base/twist_controller/command
/base/twist_controller/default/steer_ctrl/parameter_descriptions
/base/twist_controller/default/steer_ctrl/parameter_updates
/base/twist_controller/fl_caster_rotation_joint/parameter_descriptions
/base/twist_controller/fl_caster_rotation_joint/parameter_updates
/base/twist_controller/fr_caster_rotation_joint/parameter_descriptions
/base/twist_controller/fr_caster_rotation_joint/parameter_updates
/base/twist_controller/wheel_commands
/base/twist_mux/command_navigation
/base/twist_mux/command_safe
/base/twist_mux/command_script_server
/base/twist_mux/command_syncmm
/base/twist_mux/command_teleop_android
/base/twist_mux/command_teleop_joy
/base/twist_mux/command_teleop_keyboard
/base/twist_mux/locks/pause_all
/base/twist_mux/locks/pause_navigation
```

/base/twist_mux/locks/pause_teleop
/base/twist_mux/marker
/base/velocity_smoother/command
/base/velocity_smoother/parameter_descriptions
/base/velocity_smoother/parameter_updates
/base_bumper
/base_pose_ground_truth
/bms/battery_state
/bms/current
/bms/full_charge_capacity
/bms/remaining_capacity
/bms/temperature
/bms/voltage
/clock
/dashboard_agg
/diagnostics
/diagnostics_agg
/diagnostics_toplevel_state
/docker_control/dock/cancel
/docker_control/dock/feedback
/docker_control/dock/goal
/docker_control/dock/result
/docker_control/dock/status
/docker_control/undock/cancel
/docker_control/undock/feedback
/docker_control/undock/goal
/docker_control/undock/result
/docker_control/undock/status
/emergency_stop_state
/gazebo/link_states
/gazebo/model_states
/gazebo/parameter_descriptions
/gazebo/parameter_updates
/gazebo/performance_metrics
/gazebo/set_link_state
/gazebo/set_model_state
/joint_states
/joy
/joy/set_feedback
/power_state
/rosout
/rosout_agg
/script_server/cancel
/script_server/feedback


```
labuser@lab-ubuntu-01:~$ echo $ROBOT
```

```
labuser@lab-ubuntu-01:~$ export ROBOT=cob4-4
```

```
labuser@lab-ubuntu-01:~$ export ROBOT_ENV=ipa-office
```

```
labuser@lab-ubuntu-01:~$
```

```
labuser@lab-ubuntu-01:~$ echo $ROBOT
```

```
cob4-4
```

```
labuser@lab-ubuntu-01:~$ rosparam get /robot_description > robot.urdf
```

```
labuser@lab-ubuntu-01:~$ grep -i "sensor" robot.urdf
```

```
\    <self_collide>false</self_collide>\n  <sensor name=\"base_contact_sensor\"\\n  </plugin>\n  </sensor>\n  </gazebo>\n  <gazebo>\n    <plugin filename=\"
```

```
labuser@lab-ubuntu-01:~$ grep -i "laser" robot.urdf
```

```
labuser@lab-ubuntu-01:~$ grep -i "hokuyo" robot.urdf
```

```
labuser@lab-ubuntu-01:~$ grep -i "camera" robot.urdf
```

```
labuser@lab-ubuntu-01:~$ grep -i "kinect" robot.urdf
```

```
labuser@lab-ubuntu-01:~$ grep -i "realsense" robot.urdf
```

```
labuser@lab-ubuntu-01:~$ find /home/labuser/catkin_ws/src -name "*.urdf" | grep cob4
```

```
labuser@lab-ubuntu-01:~$ find /home/labuser/catkin_ws/src -name "*.xacro" | grep cob4
```

```
labuser@lab-ubuntu-01:~$ find /home/labuser/catkin_ws/src -name "*laser*"
```

```
labuser@lab-ubuntu-01:~$ find /home/labuser/catkin_ws/src -name "*camera*"
```

```
/home/labuser/catkin_ws/src/cob_demo/scripts/camera_demo.py
```

```
labuser@lab-ubuntu-01:~$ rostopic echo /robot_description | head
```

```
WARNING: topic [/robot_description] does not appear to be published yet
```

```
^C
```

```
labuser@lab-ubuntu-01:~$ rosparam list | grep robot
```

```
/base/velocity_smoother/robot_feedback
```

```
/robot_description
```

```
labuser@lab-ubuntu-01:~$
```

```
labuser@lab-ubuntu-01:~$
```

```
*****
```

```
labuser@lab-ubuntu-01:~$ rostopic list
```

```
/base/collision_velocity_filter/command
```

```
/base/joint_states
```

```
/base/odometry_controller/odometry
```

```
/base/twist_controller/b_caster_rotation_joint/parameter_descriptions
```

```
/base/twist_controller/b_caster_rotation_joint/parameter_updates
```

```
/base/twist_controller/command
```

```
/base/twist_controller/default/steer_ctrl/parameter_descriptions
```

```
/base/twist_controller/default/steer_ctrl/parameter_updates
```

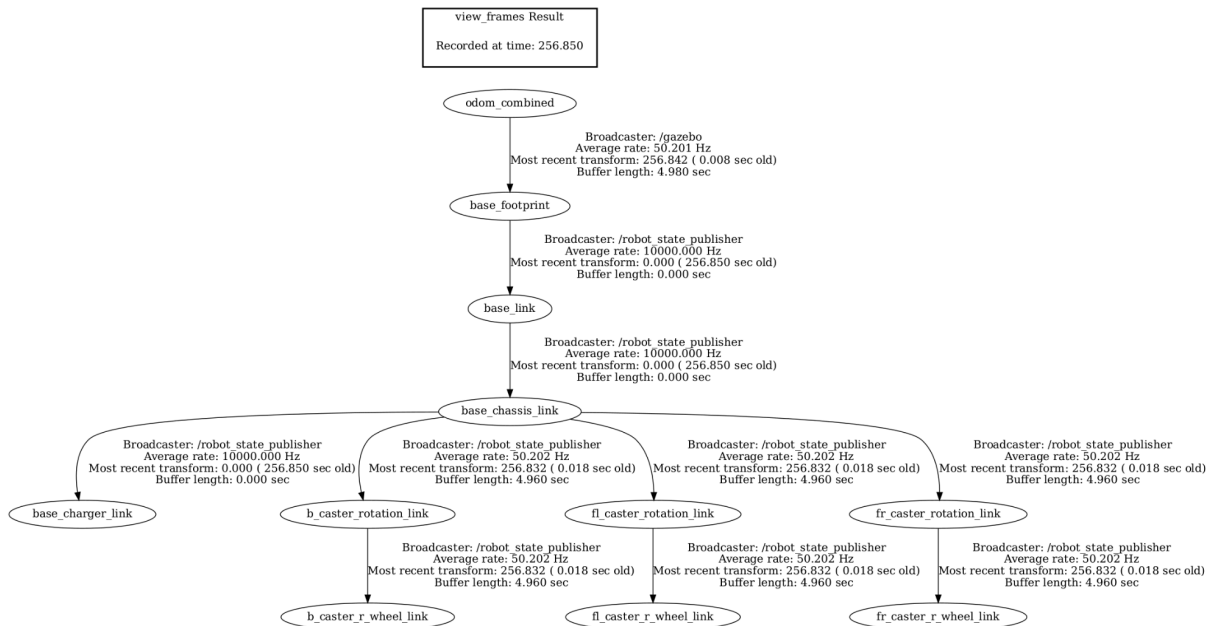
```
/base/twist_controller/fl_caster_rotation_joint/parameter_descriptions
```

```
/base/twist_controller/fl_caster_rotation_joint/parameter_updates
```

/base/twist_controller/fr_caster_rotation_joint/parameter_descriptions
/base/twist_controller/fr_caster_rotation_joint/parameter_updates
/base/twist_controller/wheel_commands
/base/twist_mux/command_navigation
/base/twist_mux/command_safe
/base/twist_mux/command_script_server
/base/twist_mux/command_syncmm
/base/twist_mux/command_teleop_android
/base/twist_mux/command_teleop_joy
/base/twist_mux/command_teleop_keyboard
/base/twist_mux/locks/pause_all
/base/twist_mux/locks/pause_navigation
/base/twist_mux/locks/pause_teleop
/base/twist_mux/marker
/base/velocity_smoother/command
/base/velocity_smoother/parameter_descriptions
/base/velocity_smoother/parameter_updates
/base_bumper
/base_pose_ground_truth
/bms/battery_state
/bms/current
/bms/full_charge_capacity
/bms/remaining_capacity
/bms/temperature
/bms/voltage
/clock
/dashboard_agg
/diagnostics
/diagnostics_agg
/diagnostics_toplevel_state
/docker_control/dock/cancel
/docker_control/dock/feedback
/docker_control/dock/goal
/docker_control/dock/result
/docker_control/dock/status
/docker_control/undock/cancel
/docker_control/undock/feedback
/docker_control/undock/goal
/docker_control/undock/result
/docker_control/undock/status
/emergency_stop_state
/gazebo/link_states
/gazebo/model_states
/gazebo/parameter_descriptions

```
/gazebo/parameter_updates
/gazebo/performance_metrics
/gazebo/set_link_state
/gazebo/set_model_state
/joint_states
/joy
/joy/set_feedback
/power_state
/rosout
/rosout_agg
/script_server/cancel
/script_server/feedback
/script_server/goal
/script_server/result
/script_server/status
/sound/say/cancel
/sound/say/feedback
/sound/say/goal
/sound/say/result
/sound/say/status
/teleop/set_light/cancel
/teleop/set_light/feedback
/teleop/set_light/goal
/teleop/set_light/result
/teleop/set_light/status
/tf
/tf_static
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ /scan
bash: /scan: No such file or directory
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ rostopic echo /scan
WARNING: topic [/scan] does not appear to be published yet
rostopic list | grep image
^Clabuser@lab-ubuntu-01:~$ rostopic list | grep image
labuser@lab-ubuntu-01:~$ rqt_image_view
labuser@lab-ubuntu-01:~$ rosrn rqt_tf_tree rqt_tf_tree
labuser@lab-ubuntu-01:~$ rosrn tf view_frames
WARNING: tf view_frames is deprecated. Use tf2_tools view_frames.py instead.
Listening to /tf for 5.0 seconds
Done Listening
frames.pdf generated
labuser@lab-ubuntu-01:~$ ls
```


catkin_ws Desktop Documents Downloads frames.gv frames.pdf Music Pictures Public
 robot.urdf snap Templates Videos
 labuser@lab-ubuntu-01:~\$



labuser@lab-ubuntu-01:~\$ rostopic echo -n 1 /gazebo/model_states

name:

- ground_plane
- ipa-office
- robot

pose:

-

position:

x: 0.0

y: 0.0

z: 0.0

orientation:

x: 0.0

y: 0.0

z: 0.0

w: 1.0

-

position:

x: 0.0

y: 0.0

z: 0.01

orientation:

x: 0.0

y: 0.0

z: 0.0

w: 1.0

-

position:

x: 0.030001344028995753

y: 0.026787698656285257

z: -0.00018629203649922195

orientation:

x: -4.948477951244622e-06

y: -2.426316206214485e-08

z: 0.08027652852562189

w: 0.9967726315180344

twist:

-

linear:

x: 0.0

y: 0.0

z: 0.0

angular:

x: 0.0

y: 0.0

z: 0.0

-

linear:

x: 0.0

y: 0.0

z: 0.0

angular:

x: 0.0

y: 0.0

z: 0.0

-

linear:

x: 6.335243966470061e-05

y: 5.250681798531267e-05

z: 0.0024339022519819396

angular:

```
labuser@lab-ubuntu-01:~$ rosparam list | grep robot  
/base/velocity_smoother/robot_feedback  
/robot_description  
labuser@lab-ubuntu-01:~$ rosparam get robot_description | head  
"<?xml version='\"1.0\"' ?>\n<!--  
=====\\  
=====  
 \ -->\n<!-- | This document was autogenerated by xacro from  
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-4/urdf/cob4-4.urdf.xacro\  
 \ | -->\n<!-- | EDITING THIS FILE BY HAND IS NOT RECOMMENDED      \  
 \           | -->\n<!--  
=====\\  
=====  
 \ -->\n<robot name= \"cob4-4\">\n <material name= \"IPA/LightGrey\">\n <color\  
 \ rgba= \"0.7 0.7 0.7 1.0\"/>\n </material>\n <material name= \"IPA/DarkGrey\">\n \  
 \   <color rgba= \"0.4 0.4 0.4 1.0\"/>\n </material>\n <material name= \"IPA/Black\"\  
 >\n <color rgba= \"0.0 0.0 0.0 1.0\"/>\n </material>\n <material name= \"IPA/Metall\"\  
 >\n <color rgba= \"0.7 0.7 0.7 1.0\"/>\n </material>\n <material name= \"IPA/YellowRubber\"\  
 >\n <color rgba= \"0.4 0.27 0.16 1.0\"/>\n </material>\n <material name= \"IPA/Orange\"\  
labuser@lab-ubuntu-01:~$ xdg-open frames.pdf  
labuser@lab-ubuntu-01:~$ rosnode list  
/base/fake_driver  
/base/halt_detector  
/base/twist_marker  
/base/twist_mux  
/base/velocity_smoother  
/bms  
/bms/power_state_aggregator  
/dashboard_aggregator  
/diagnostic_aggregator  
/fake_diagnostics_lab_ubuntu_01_3677_5689824313652294819  
/gazebo  
/gazebo_gui  
/joint_state_publisher  
/joy_node_lab_ubuntu_01_3677_7471172081649935816  
/robot_state_publisher  
/rosout  
/script_server  
/teleop/teleop_node_lab_ubuntu_01_3677_5349299951740356536  
labuser@lab-ubuntu-01:~$
```

```
labuser@lab-ubuntu-01:~$ rosparam get robot_description > robot.urdf
labuser@lab-ubuntu-01:~$ grep -i camera robot.urdf
labuser@lab-ubuntu-01:~$ grep -i laser robot.urdf
labuser@lab-ubuntu-01:~$ grep -i hokuyo robot.urdf
labuser@lab-ubuntu-01:~$ grep -i kinect robot.urdf
labuser@lab-ubuntu-01:~$ grep -i sensor robot.urdf
\      <self_collide>false</self_collide>\n  <sensor name=\"base_contact_sensor\"\\
\      </plugin>\n    </sensor>\n </gazebo>\n <gazebo>\n      <plugin filename=\"\n
labuser@lab-ubuntu-01:~$
```

```
labuser@lab-ubuntu-01:~$ roslaunch --files cob_bringup_sim robot.launch
xacro: in-order processing became default in ROS Melodic. You can drop the option.
xacro: in-order processing became default in ROS Melodic. You can drop the option.
/opt/ros/noetic/share/cob_default_robot_config/robots/cob4-4/upload_param.xml
/opt/ros/noetic/share/cob_bringup/tools/velocity_smoother.launch
/opt/ros/noetic/share/cob_bringup/components/cob4_base.launch
/opt/ros/noetic/share/cob_bringup/tools/diagnostics_aggregator.launch
/opt/ros/noetic/share/cob_script_server/launch/script_server.launch
/opt/ros/noetic/share/cob_default_env_config/upload_object_locations.launch
/opt/ros/noetic/share/cob_bringup/tools/teleop.launch
/opt/ros/noetic/share/cob_bringup/tools/robot_state_publisher.launch
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo/launch/robot.launch
/opt/ros/noetic/share/cob_bringup/components/ros_control_base.launch
/opt/ros/noetic/share/cob_bringup/tools/fake_diagnostics.launch
/home/labuser/catkin_ws/src/cob_simulation/cob_bringup_sim/launch/robot.launch
/opt/ros/noetic/share/cob_bringup/drivers/joy.launch
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/ipa-office.xml
/opt/ros/noetic/share/cob_bringup/robots/cob4-4.xml
/opt/ros/noetic/share/cob_default_env_config/upload_navigation_goals.launch
/opt/ros/noetic/share/gazebo_ros/launch/empty_world.launch
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/world.launch
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/world_urdf.xml
/opt/ros/noetic/share/cob_hardware_config/upload_robot.launch
/opt/ros/noetic/share/cob_default_robot_config/upload_param.launch
/opt/ros/noetic/share/cob_bringup/tools/twist_mux.launch
/opt/ros/noetic/share/cob_bringup/drivers/bms.launch
labuser@lab-ubuntu-01:~$
```

```
labuser@lab-ubuntu-01:~$ ls /opt/ros/noetic/share/cob_hardware_config/robots/
cob4 cob4-10 cob4-11 cob4-13 cob4-16 cob4-18 cob4-20 cob4-23 cob4-24 cob4-25
cob4-29 cob4-3 cob4-30 cob4-4 cob4-5 cob4-6 cob4-7 cob4-8 cob4-9 common raw3
raw3-1 raw3-3 raw3-5 raw-mini
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ ls /opt/ros/noetic/share/cob_default_robot_config/robots/
cob4 cob4-10 cob4-11 cob4-13 cob4-16 cob4-18 cob4-20 cob4-23 cob4-24 cob4-25
cob4-29 cob4-3 cob4-30 cob4-4 cob4-5 cob4-6 cob4-7 cob4-8 cob4-9 raw3 raw3-1
raw3-3 raw3-5
labuser@lab-ubuntu-01:~$
```

```
labuser@lab-ubuntu-01:~$ grep -R "camera\\|laser\\|hokuyo\\|kinect\\|head_cam"
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-* 2>/dev/null | head -100
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:    <!--
laser front mount positions | relative to base_link -->
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="laser_front_x" value="${def_laser_front_x+offset_laser_front_x}"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="laser_front_y" value="${def_laser_front_y+offset_laser_front_y}"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="laser_front_z" value="${def_laser_front_z+offset_laser_front_z}"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="laser_front_roll" value="${def_laser_front_roll+offset_laser_front_roll}"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="laser_front_pitch"
value="${def_laser_front_pitch+offset_laser_front_pitch}"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="laser_front_yaw"
value="${def_laser_front_yaw+offset_laser_front_yaw}"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:    <!--
laser rear mount positions | relative to base_link -->
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="laser_left_x" value="${def_laser_left_x+offset_laser_left_x}"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="laser_left_y" value="${def_laser_left_y+offset_laser_left_y}"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="laser_left_z" value="${def_laser_left_z+offset_laser_left_z}"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="laser_left_roll" value="${def_laser_left_roll+offset_laser_left_roll}"/>
```

```

/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="laser_left_pitch"
value="${def_laser_left_pitch+offset_laser_left_pitch}"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="laser_left_yaw" value="${def_laser_left_yaw+offset_laser_left_yaw}"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:    <!--
laser top mount positions | relative to base_link -->
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="laser_right_x" value="${def_laser_right_x+offset_laser_right_x}"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="laser_right_y" value="${def_laser_right_y+offset_laser_right_y}"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="laser_right_z" value="${def_laser_right_z+offset_laser_right_z}"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="laser_right_roll" value="${def_laser_right_roll+offset_laser_right_roll}"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="laser_right_pitch"
value="${def_laser_right_pitch+offset_laser_right_pitch}"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="laser_right_yaw"
value="${def_laser_right_yaw+offset_laser_right_yaw}"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="head_cam_x" value="${def_head_cam_x+offset_head_cam_x}"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="head_cam_y" value="${def_head_cam_y+offset_head_cam_y}"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="head_cam_z" value="${def_head_cam_z+offset_head_cam_z}"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="head_cam_roll" value="${def_head_cam_roll+offset_head_cam_roll}"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="head_cam_pitch"
value="${def_head_cam_pitch+offset_head_cam_pitch}"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/properties.urdf.xacro:
<xacro:property name="head_cam_yaw"
value="${def_head_cam_yaw+offset_head_cam_yaw}"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/cob4-10.urdf.xacro:
<xacro:include filename="$(find cob_description)/urdf/sensors/kinect.urdf.xacro" />
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/cob4-10.urdf.xacro:
<xacro:include filename="$(find cob_description)/urdf/sensors/sick_s300_laser.urdf.xacro"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/cob4-10.urdf.xacro:    <!-- laser
scanners -->
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/cob4-10.urdf.xacro:
<xacro:sick_s300_laser_v0 name="base_laser_front" parent="base">

```

```

/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/cob4-10.urdf.xacro:      <origin
xyz="{laser_front_x} {laser_front_y} {laser_front_z}" rpy="{laser_front_roll}
{laser_front_pitch} {laser_front_yaw}" />
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/cob4-10.urdf.xacro:
</xacro:sick_s300_laser_v0>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/cob4-10.urdf.xacro:
<xacro:sick_s300_laser_v0 name="base_laser_left" parent="base">
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/cob4-10.urdf.xacro:      <origin
xyz="{laser_left_x} {laser_left_y} {laser_left_z}" rpy="{laser_left_roll} {laser_left_pitch}
{laser_left_yaw}" />
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/cob4-10.urdf.xacro:
</xacro:sick_s300_laser_v0>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/cob4-10.urdf.xacro:
<xacro:sick_s300_laser_v0 name="base_laser_right" parent="base">
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/cob4-10.urdf.xacro:      <origin
xyz="{laser_right_x} {laser_right_y} {laser_right_z}" rpy="{laser_right_roll}
{laser_right_pitch} {laser_right_yaw}" />
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/cob4-10.urdf.xacro:
</xacro:sick_s300_laser_v0>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/cob4-10.urdf.xacro: <!-- left
camera -->
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/cob4-10.urdf.xacro:
<xacro:cob_kinect_v0 name="torso_cam3d_left" ros_topic="torso_cam3d_left"
parent="torso_3_link" update_rate="30" data_skip="9">
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/cob4-10.urdf.xacro:
</xacro:cob_kinect_v0>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/cob4-10.urdf.xacro: <!-- right
camera -->
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/cob4-10.urdf.xacro:
<xacro:cob_kinect_v0 name="torso_cam3d_right" ros_topic="torso_cam3d_right"
parent="torso_3_link" update_rate="30" data_skip="9">
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/cob4-10.urdf.xacro:
</xacro:cob_kinect_v0>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/cob4-10.urdf.xacro: <!-- down
camera -->
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/cob4-10.urdf.xacro:
<xacro:usb_cam name="head_cam" parent="head_3_link" ros_topic="head_cam"
update_rate="3.5" height="1536" width="2048" > <!-- get this from config/head_cam.yaml -->
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/cob4-10.urdf.xacro:      <origin
xyz="{head_cam_x} {head_cam_y} {head_cam_z}" rpy="{head_cam_roll}
{head_cam_pitch} {head_cam_yaw}" />
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/cob4-10.urdf.xacro: <!--
cameras -->

```

```

/opt/ros/noetic/share/cobHardwareConfig/robots/cob4-10/urdf/cob4-10.urdf.xacro:
<xacro:cob_kinect_v0 name="sensorring_cam3d_back" ros_topic="sensorring_cam3d_back"
parent="sensorring_link" update_rate="30" data_skip="9">
/opt/ros/noetic/share/cobHardwareConfig/robots/cob4-10/urdf/cob4-10.urdf.xacro:
</xacro:cob_kinect_v0>
/opt/ros/noetic/share/cobHardwareConfig/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<!-- laser front mount positions | relative to base_link -->
/opt/ros/noetic/share/cobHardwareConfig/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_laser_front_x" value="0.244"/>
/opt/ros/noetic/share/cobHardwareConfig/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_laser_front_y" value="0.0"/>
/opt/ros/noetic/share/cobHardwareConfig/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_laser_front_z" value="0.20858"/>
/opt/ros/noetic/share/cobHardwareConfig/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_laser_front_roll" value="0.0"/>
/opt/ros/noetic/share/cobHardwareConfig/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_laser_front_pitch" value="0.0"/>
/opt/ros/noetic/share/cobHardwareConfig/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_laser_front_yaw" value="0.0"/>
/opt/ros/noetic/share/cobHardwareConfig/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<!-- laser rear mount positions | relative to base_link -->
/opt/ros/noetic/share/cobHardwareConfig/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_laser_left_x" value="-0.122"/>
/opt/ros/noetic/share/cobHardwareConfig/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_laser_left_y" value="0.21131"/>
/opt/ros/noetic/share/cobHardwareConfig/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_laser_left_z" value="0.20858"/>
/opt/ros/noetic/share/cobHardwareConfig/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_laser_left_roll" value="0.0"/>
/opt/ros/noetic/share/cobHardwareConfig/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_laser_left_pitch" value="0.0"/>
/opt/ros/noetic/share/cobHardwareConfig/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_laser_left_yaw" value="2.094"/>
/opt/ros/noetic/share/cobHardwareConfig/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<!-- laser top mount positions | relative to base_link -->
/opt/ros/noetic/share/cobHardwareConfig/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_laser_right_x" value="-0.122"/>
/opt/ros/noetic/share/cobHardwareConfig/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_laser_right_y" value="-0.21131"/>
/opt/ros/noetic/share/cobHardwareConfig/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_laser_right_z" value="0.20858"/>
/opt/ros/noetic/share/cobHardwareConfig/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_laser_right_roll" value="0.0"/>

```



```

/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_laser_right_pitch" value="0.0"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_laser_right_yaw" value="-2.094"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_head_cam_x" value="-0.181"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_head_cam_y" value="0.0"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_head_cam_z" value="0.325"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_head_cam_roll" value="-1.57079"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_head_cam_pitch" value="-0.73"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/urdf/calibration_default.urdf.xacro:
<xacro:property name="def_head_cam_yaw" value="0.0"/>
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/config/footprint_observer_params.yaml:
frames_to_check: base_laser_front_link base_laser_left_link base_laser_right_link
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/config/hz_monitor_sensoring_cam3d_front.yaml:
topics: [/sensoring_cam3d_front/depth/camera_info]
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/config/base_laser_front.yaml:
frame_id: base_laser_front_link
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/config/base_laser_right.yaml:
frame_id: base_laser_right_link
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/config/base_laser_left.yaml:
frame_id: base_laser_left_link
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/config/hz_monitor_head_cam.yaml:
diagnostics_name: head_cam
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/config/hz_monitor_head_cam.yaml:
topics: [/head_cam/camera_info]
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/config/hz_monitor_torso_cam3d_down.yaml:
topics: [/torso_cam3d_down/rgb/camera_info, /torso_cam3d_down/depth/camera_info]
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/config/hz_monitor_sensoring_cam3d_back.yaml:
topics: [/sensoring_cam3d_back/rgb/camera_info]
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/config/hz_monitor_torso_cam3d_right.yaml:
topics: [/torso_cam3d_right/rgb/camera_info, /torso_cam3d_right/depth/camera_info]
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/config/docking_laser_filter.yaml:
type: laser_filters/LaserScanRangeFilter
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/config/docking_laser_filter.yaml:
type: laser_filters/LaserScanIntensityFilter
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/config/local_costmap_params.yaml:
inf_is_valid: false          # Allows for Inf values in "LaserScan" observation messages. The
                              Inf values are converted to the laser maximum range. default: false

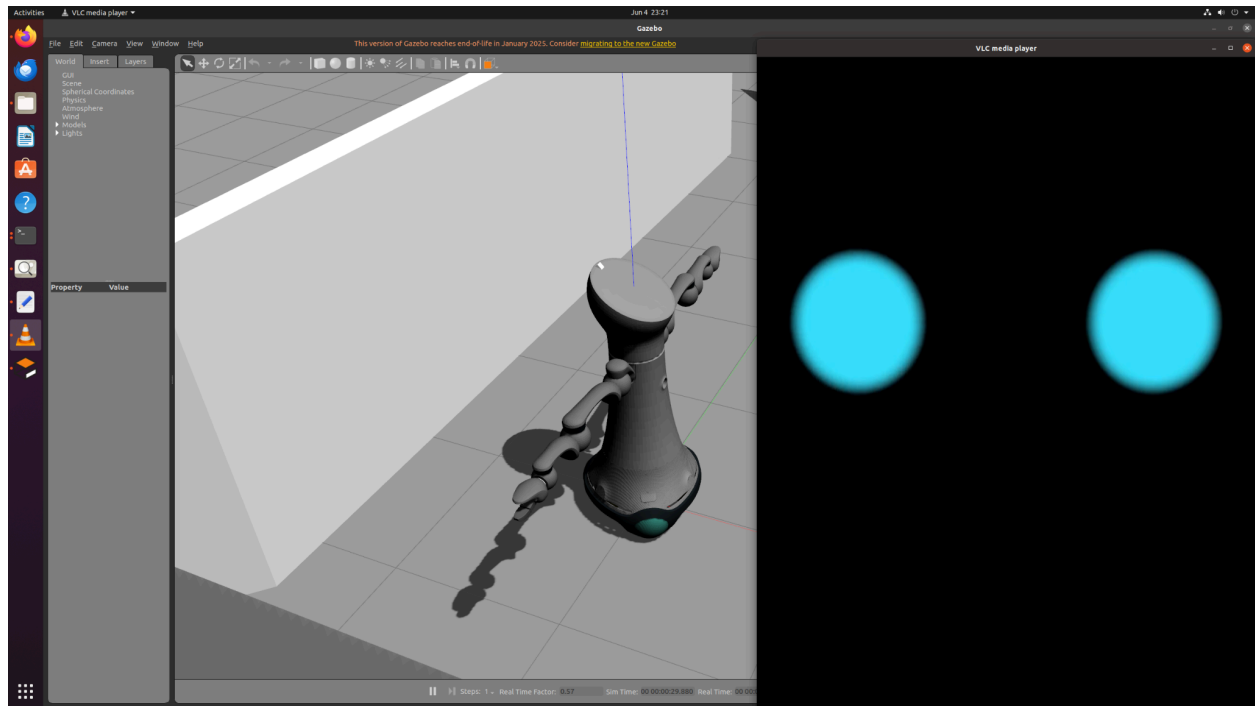
```

```
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/config/hz_monitor_torso_cam3d_left.
yaml:topics: [/torso_cam3d_left/rgb/camera_info, /torso_cam3d_left/depth/camera_info]
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/config/scan_unifier_filter.yaml:
type: laser_filters/LaserScanRangeFilter
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/cob4-10.rviz:
base_laser_front_link:
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/cob4-10.rviz:
base_laser_left_link:
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/cob4-10.rviz:
base_laser_right_link:
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/cob4-10.rviz:
gripper_left_camera_link:
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/cob4-10.rviz:
gripper_right_camera_link:
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/cob4-10.rviz: Topic:
/base_laser_front/scan
/opt/ros/noetic/share/cob_hardware_config/robots/cob4-10/cob4-10.rviz: Topic:
/base_laser_left/scan
labuser@lab-ubuntu-01:~$
```

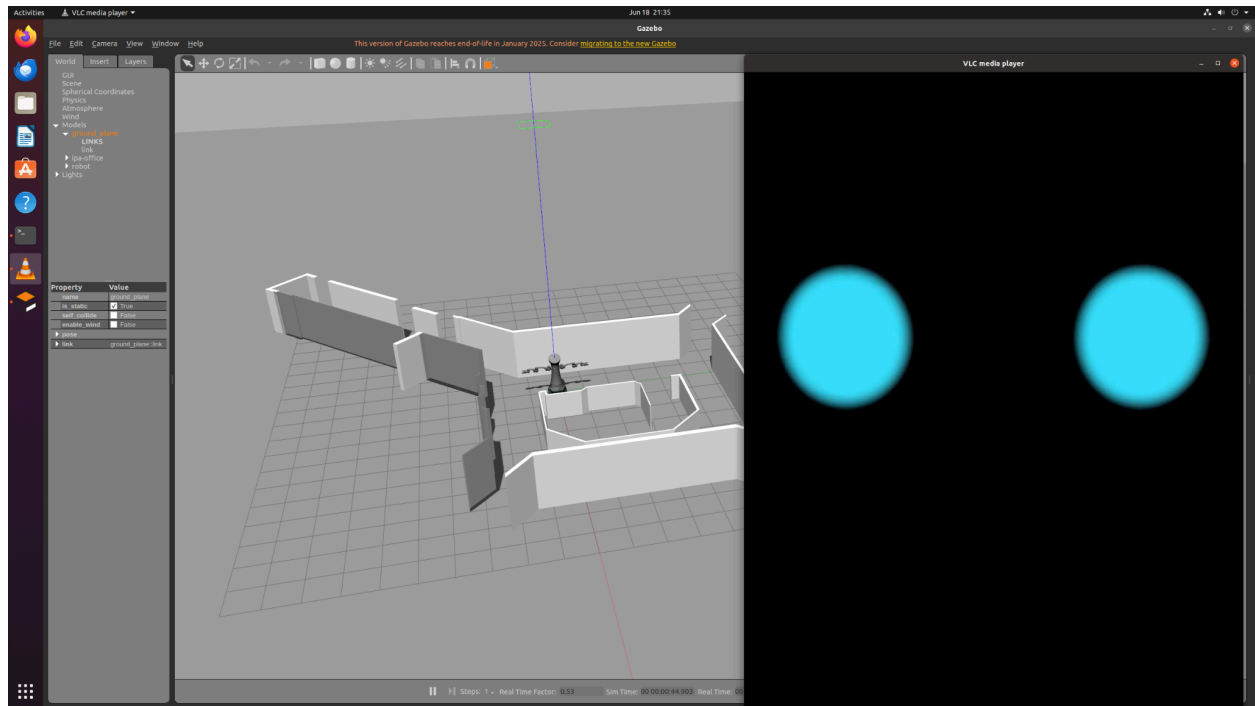
```
source /opt/ros/noetic/setup.bash
source ~/catkin_ws/devel/setup.bash

export ROBOT=cob4-10
export ROBOT_ENV=ipa-office

roslaunch cob_bringup_sim robot.launch
```



```
labuser@lab-ubuntu-01:~$ source /opt/ros/noetic/setup.bash
labuser@lab-ubuntu-01:~$ source ~/catkin_ws/devel/setup.bash
labuser@lab-ubuntu-01:~$ export ROBOT=cob4-5
labuser@lab-ubuntu-01:~$ export ROBOT_ENV=ipa-office
labuser@lab-ubuntu-01:~$ roslaunch cob_bringup_sim robot.launch
```



```
labuser@lab-ubuntu-01:~$ source /opt/ros/noetic/setup.bash
labuser@lab-ubuntu-01:~$ source ~/catkin_ws/devel/setup.bash
labuser@lab-ubuntu-01:~$ rostopic list | grep image
/head_cam/image_color
/head_cam/image_color/compressed
/head_cam/image_color/compressed/parameter_descriptions
/head_cam/image_color/compressed/parameter_updates
/head_cam/image_color/compressedDepth
/head_cam/image_color/compressedDepth/parameter_descriptions
/head_cam/image_color/compressedDepth/parameter_updates
/head_cam/image_color/theora
/head_cam/image_color/theora/parameter_descriptions
/head_cam/image_color/theora/parameter_updates
/head_cam/image_flip_head_cam/colorimage_inplane_transform
/head_cam/image_mono
/head_cam/image_mono/compressed
/head_cam/image_mono/compressed/parameter_descriptions
/head_cam/image_mono/compressed/parameter_updates
/head_cam/image_mono/compressedDepth
/head_cam/image_mono/compressedDepth/parameter_descriptions
/head_cam/image_mono/compressedDepth/parameter_updates
/head_cam/image_mono/theora
/head_cam/image_mono/theora/parameter_descriptions
```

/head_cam/image_mono/theora/parameter_updates
/head_cam/image_proc_debayer/parameter_descriptions
/head_cam/image_proc_debayer/parameter_updates
/head_cam/image_proc_rectify_color/parameter_descriptions
/head_cam/image_proc_rectify_color/parameter_updates
/head_cam/image_proc_rectify_mono/parameter_descriptions
/head_cam/image_proc_rectify_mono/parameter_updates
/head_cam/image_raw
/head_cam/image_raw/compressed
/head_cam/image_raw/compressed/parameter_descriptions
/head_cam/image_raw/compressed/parameter_updates
/head_cam/image_raw/compressedDepth
/head_cam/image_raw/compressedDepth/parameter_descriptions
/head_cam/image_raw/compressedDepth/parameter_updates
/head_cam/image_raw/theora
/head_cam/image_raw/theora/parameter_descriptions
/head_cam/image_raw/theora/parameter_updates
/head_cam/image_rect
/head_cam/image_rect/compressed
/head_cam/image_rect/compressed/parameter_descriptions
/head_cam/image_rect/compressed/parameter_updates
/head_cam/image_rect/compressedDepth
/head_cam/image_rect/compressedDepth/parameter_descriptions
/head_cam/image_rect/compressedDepth/parameter_updates
/head_cam/image_rect/theora
/head_cam/image_rect/theora/parameter_descriptions
/head_cam/image_rect/theora/parameter_updates
/head_cam/image_rect_color
/head_cam/image_rect_color/compressed
/head_cam/image_rect_color/compressed/parameter_descriptions
/head_cam/image_rect_color/compressed/parameter_updates
/head_cam/image_rect_color/compressedDepth
/head_cam/image_rect_color/compressedDepth/parameter_descriptions
/head_cam/image_rect_color/compressedDepth/parameter_updates
/head_cam/image_rect_color/theora
/head_cam/image_rect_color/theora/parameter_descriptions
/head_cam/image_rect_color/theora/parameter_updates
/head_cam_upright/image_rect_color
/head_cam_upright/image_rect_color/compressed
/head_cam_upright/image_rect_color/compressed/parameter_descriptions
/head_cam_upright/image_rect_color/compressed/parameter_updates
/head_cam_upright/image_rect_color/compressedDepth
/head_cam_upright/image_rect_color/compressedDepth/parameter_descriptions
/head_cam_upright/image_rect_color/compressedDepth/parameter_updates

/head_cam_upright/image_rect_color/theora
/head_cam_upright/image_rect_color/theora/parameter_descriptions
/head_cam_upright/image_rect_color/theora/parameter_updates
/sensoring_cam3d_back/depth/image_raw
/sensoring_cam3d_back/depth/registered/image_raw
/sensoring_cam3d_back/image_flip_nodelet/colorimage_inplane_transform
/sensoring_cam3d_back/image_flip_nodelet/pointcloud_inplane_transform
/sensoring_cam3d_back/rgb/image_raw
/sensoring_cam3d_back/rgb/image_raw/compressed
/sensoring_cam3d_back/rgb/image_raw/compressed/parameter_descriptions
/sensoring_cam3d_back/rgb/image_raw/compressed/parameter_updates
/sensoring_cam3d_back/rgb/image_raw/compressedDepth
/sensoring_cam3d_back/rgb/image_raw/compressedDepth/parameter_descriptions
/sensoring_cam3d_back/rgb/image_raw/compressedDepth/parameter_updates
/sensoring_cam3d_back/rgb/image_raw/theora
/sensoring_cam3d_back/rgb/image_raw/theora/parameter_descriptions
/sensoring_cam3d_back/rgb/image_raw/theora/parameter_updates
/sensoring_cam3d_back_upright/rgb/image_raw
/sensoring_cam3d_back_upright/rgb/image_raw/compressed
/sensoring_cam3d_back_upright/rgb/image_raw/compressed/parameter_descriptions
/sensoring_cam3d_back_upright/rgb/image_raw/compressed/parameter_updates
/sensoring_cam3d_back_upright/rgb/image_raw/compressedDepth
/sensoring_cam3d_back_upright/rgb/image_raw/compressedDepth/parameter_descriptions
/sensoring_cam3d_back_upright/rgb/image_raw/compressedDepth/parameter_updates
/sensoring_cam3d_back_upright/rgb/image_raw/theora
/sensoring_cam3d_back_upright/rgb/image_raw/theora/parameter_descriptions
/sensoring_cam3d_back_upright/rgb/image_raw/theora/parameter_updates
/sensoring_cam3d_front/depth/image
/sensoring_cam3d_front/image_flip_sensoring_cam3d_front/colorimage_inplane_transform
/sensoring_cam3d_front/image_flip_sensoring_cam3d_front/pointcloud_inplane_transform
/sensoring_cam3d_front/image_raw
/sensoring_cam3d_front/image_raw/compressed
/sensoring_cam3d_front/image_raw/compressed/parameter_descriptions
/sensoring_cam3d_front/image_raw/compressed/parameter_updates
/sensoring_cam3d_front/image_raw/compressedDepth
/sensoring_cam3d_front/image_raw/compressedDepth/parameter_descriptions
/sensoring_cam3d_front/image_raw/compressedDepth/parameter_updates
/sensoring_cam3d_front/image_raw/theora
/sensoring_cam3d_front/image_raw/theora/parameter_descriptions
/sensoring_cam3d_front/image_raw/theora/parameter_updates
/sensoring_cam3d_front_upright/rgb/image_raw
/sensoring_cam3d_front_upright/rgb/image_raw/compressed
/sensoring_cam3d_front_upright/rgb/image_raw/compressed/parameter_descriptions
/sensoring_cam3d_front_upright/rgb/image_raw/compressed/parameter_updates

/sensoring_cam3d_front_upright/rgb/image_raw/compressedDepth
/sensoring_cam3d_front_upright/rgb/image_raw/compressedDepth/parameter_descriptions
/sensoring_cam3d_front_upright/rgb/image_raw/compressedDepth/parameter_updates
/sensoring_cam3d_front_upright/rgb/image_raw/theora
/sensoring_cam3d_front_upright/rgb/image_raw/theora/parameter_descriptions
/sensoring_cam3d_front_upright/rgb/image_raw/theora/parameter_updates
/torso_cam3d_down/depth/image_raw
/torso_cam3d_down/rgb/image_raw
/torso_cam3d_down/rgb/image_raw/compressed
/torso_cam3d_down/rgb/image_raw/compressed/parameter_descriptions
/torso_cam3d_down/rgb/image_raw/compressed/parameter_updates
/torso_cam3d_down/rgb/image_raw/compressedDepth
/torso_cam3d_down/rgb/image_raw/compressedDepth/parameter_descriptions
/torso_cam3d_down/rgb/image_raw/compressedDepth/parameter_updates
/torso_cam3d_down/rgb/image_raw/theora
/torso_cam3d_down/rgb/image_raw/theora/parameter_descriptions
/torso_cam3d_down/rgb/image_raw/theora/parameter_updates
/torso_cam3d_left/depth/image_raw
/torso_cam3d_left/depth_registered/image_raw
/torso_cam3d_left/image_flip_nodelet/colorimage_inplane_transform
/torso_cam3d_left/image_flip_nodelet/pointcloud_inplane_transform
/torso_cam3d_left/rgb/image_raw
/torso_cam3d_left/rgb/image_raw/compressed
/torso_cam3d_left/rgb/image_raw/compressed/parameter_descriptions
/torso_cam3d_left/rgb/image_raw/compressed/parameter_updates
/torso_cam3d_left/rgb/image_raw/compressedDepth
/torso_cam3d_left/rgb/image_raw/compressedDepth/parameter_descriptions
/torso_cam3d_left/rgb/image_raw/compressedDepth/parameter_updates
/torso_cam3d_left/rgb/image_raw/theora
/torso_cam3d_left/rgb/image_raw/theora/parameter_descriptions
/torso_cam3d_left/rgb/image_raw/theora/parameter_updates
/torso_cam3d_left_upright/rgb/image_raw
/torso_cam3d_left_upright/rgb/image_raw/compressed
/torso_cam3d_left_upright/rgb/image_raw/compressed/parameter_descriptions
/torso_cam3d_left_upright/rgb/image_raw/compressed/parameter_updates
/torso_cam3d_left_upright/rgb/image_raw/compressedDepth
/torso_cam3d_left_upright/rgb/image_raw/compressedDepth/parameter_descriptions
/torso_cam3d_left_upright/rgb/image_raw/compressedDepth/parameter_updates
/torso_cam3d_left_upright/rgb/image_raw/theora
/torso_cam3d_left_upright/rgb/image_raw/theora/parameter_descriptions
/torso_cam3d_left_upright/rgb/image_raw/theora/parameter_updates
/torso_cam3d_right/depth/image_raw
/torso_cam3d_right/depth_registered/image_raw
/torso_cam3d_right/image_flip_nodelet/colorimage_inplane_transform

```
/torso_cam3d_right/image_flip_nodelet/pointcloud_inplane_transform
/torso_cam3d_right/rgb/image_raw
/torso_cam3d_right/rgb/image_raw/compressed
/torso_cam3d_right/rgb/image_raw/compressed/parameter_descriptions
/torso_cam3d_right/rgb/image_raw/compressed/parameter_updates
/torso_cam3d_right/rgb/image_raw/compressedDepth
/torso_cam3d_right/rgb/image_raw/compressedDepth/parameter_descriptions
/torso_cam3d_right/rgb/image_raw/compressedDepth/parameter_updates
/torso_cam3d_right/rgb/image_raw/theora
/torso_cam3d_right/rgb/image_raw/theora/parameter_descriptions
/torso_cam3d_right/rgb/image_raw/theora/parameter_updates
/torso_cam3d_right_upright/rgb/image_raw
/torso_cam3d_right_upright/rgb/image_raw/compressed
/torso_cam3d_right_upright/rgb/image_raw/compressed/parameter_descriptions
/torso_cam3d_right_upright/rgb/image_raw/compressed/parameter_updates
/torso_cam3d_right_upright/rgb/image_raw/compressedDepth
/torso_cam3d_right_upright/rgb/image_raw/compressedDepth/parameter_descriptions
/torso_cam3d_right_upright/rgb/image_raw/compressedDepth/parameter_updates
/torso_cam3d_right_upright/rgb/image_raw/theora
/torso_cam3d_right_upright/rgb/image_raw/theora/parameter_descriptions
/torso_cam3d_right_upright/rgb/image_raw/theora/parameter_updates
labuser@lab-ubuntu-01:~$ rostopic list | grep camera
/head_cam/camera_info
/sensorring_cam3d_back/depth/camera_info
/sensorring_cam3d_back/depth_registered/camera_info
/sensorring_cam3d_back/rgb/camera_info
/sensorring_cam3d_front/camera_info
/sensorring_cam3d_front/depth/camera_info
/torso_cam3d_down/depth/camera_info
/torso_cam3d_down/rgb/camera_info
/torso_cam3d_left/depth/camera_info
/torso_cam3d_left/depth_registered/camera_info
/torso_cam3d_left/rgb/camera_info
/torso_cam3d_right/depth/camera_info
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/torso_cam3d_left_upright/rgb/image_raw/compressedDepth/parameter_descriptions
/torso_cam3d_left_upright/rgb/image_raw/compressedDepth/parameter_updates
/torso_cam3d_left_upright/rgb/image_raw/theora
/torso_cam3d_left_upright/rgb/image_raw/theora/parameter_descriptions
/torso_cam3d_left_upright/rgb/image_raw/theora/parameter_updates
/torso_cam3d_right/depth/camera_info
/torso_cam3d_right/depth/image_raw
/torso_cam3d_right/depth/points
/torso_cam3d_right/depth_registered/camera_info
/torso_cam3d_right/depth_registered/image_raw
/torso_cam3d_right/depth_registered/points
/torso_cam3d_right/image_flip_nodelet/colorimage_inplane_transform
/torso_cam3d_right/image_flip_nodelet/pointcloud_inplane_transform
/torso_cam3d_right/parameter_descriptions
/torso_cam3d_right/parameter_updates
/torso_cam3d_right/rgb/camera_info
/torso_cam3d_right/rgb/image_raw
/torso_cam3d_right/rgb/image_raw/compressed

```
/torso_cam3d_right/rgb/image_raw/compressed/parameter_descriptions
/torso_cam3d_right/rgb/image_raw/compressed/parameter_updates
/torso_cam3d_right/rgb/image_raw/compressedDepth
/torso_cam3d_right/rgb/image_raw/compressedDepth/parameter_descriptions
/torso_cam3d_right/rgb/image_raw/compressedDepth/parameter_updates
/torso_cam3d_right/rgb/image_raw/theora
/torso_cam3d_right/rgb/image_raw/theora/parameter_descriptions
/torso_cam3d_right/rgb/image_raw/theora/parameter_updates
/torso_cam3d_right/torso_cam3d_right_nodelet_manager/bond
/torso_cam3d_right_upright/depth_registered/points
/torso_cam3d_right_upright/rgb/image_raw
/torso_cam3d_right_upright/rgb/image_raw/compressed
/torso_cam3d_right_upright/rgb/image_raw/compressed/parameter_descriptions
/torso_cam3d_right_upright/rgb/image_raw/compressed/parameter_updates
/torso_cam3d_right_upright/rgb/image_raw/compressedDepth
/torso_cam3d_right_upright/rgb/image_raw/compressedDepth/parameter_descriptions
/torso_cam3d_right_upright/rgb/image_raw/compressedDepth/parameter_updates
/torso_cam3d_right_upright/rgb/image_raw/theora
/torso_cam3d_right_upright/rgb/image_raw/theora/parameter_descriptions
/torso_cam3d_right_upright/rgb/image_raw/theora/parameter_updates
labuser@lab-ubuntu-01:~$ rostopic echo /scan
WARNING: topic [/scan] does not appear to be published yet
^Clabuser@lab-ubuntu-01:~$ rostopic list | grep scan
/base_laser_front/scan
/base_laser_front/scan_raw
/base_laser_left/scan
/base_laser_left/scan_raw
/base_laser_right/scan
/base_laser_right/scan_raw
/scan_unified
/scan_unified_filtered
/scan_unified_throttled
/scan_unifier_filter/range/parameter_descriptions
/scan_unifier_filter/range/parameter_updates
/station_detector/scan
labuser@lab-ubuntu-01:~$ rosrn tf view_frames
WARNING: tf view_frames is deprecated. Use tf2_tools view_frames.py instead.
Listening to /tf for 5.0 seconds
Done Listening
frames.pdf generated
labuser@lab-ubuntu-01:~$ rviz
[INFO] [1781812152.400845394]: rviz version 1.14.26
[INFO] [1781812152.400909295]: compiled against Qt version 5.12.8
[INFO] [1781812152.400920959]: compiled against OGRE version 1.9.0 (Ghadamon)
```

```
[INFO] [1781812152.412656658]: Forcing OpenGL version 0.
[INFO] [1781812153.297852387, 568.139000000]: Stereo is NOT SUPPORTED
[INFO] [1781812153.297918833, 568.139000000]: OpenGL device: NVIDIA GeForce GTX 1660
SUPER/PCIe/SSE2
[INFO] [1781812153.297947340, 568.139000000]: OpenGL version: 4,6 (GLSL 4,6).
labuser@lab-ubuntu-01:~$ rostopic list | grep cmd_vel
labuser@lab-ubuntu-01:~$
```

```
labuser@lab-ubuntu-01:~$ rostopic hz /scan_unified
subscribed to [/scan_unified]
WARNING: may be using simulated time
average rate: 11.945
      min: 0.083s max: 0.086s std dev: 0.00103s window: 8
average rate: 11.925
      min: 0.082s max: 0.086s std dev: 0.00119s window: 15
average rate: 11.905
      min: 0.082s max: 0.086s std dev: 0.00121s window: 23
average rate: 11.905
      min: 0.082s max: 0.086s std dev: 0.00114s window: 30
average rate: 11.912
      min: 0.082s max: 0.086s std dev: 0.00116s window: 38
average rate: 11.908
      min: 0.082s max: 0.086s std dev: 0.00114s window: 45
^Cno new messages
labuser@lab-ubuntu-01:~$ rostopic list | grep map
/collision_velocity_filter/anti_collision_costmap/costmap
/collision_velocity_filter/anti_collision_costmap/costmap_updates
/collision_velocity_filter/anti_collision_costmap/footprint
/collision_velocity_filter/anti_collision_costmap/obstacle_layer_0/parameter_descriptions
/collision_velocity_filter/anti_collision_costmap/obstacle_layer_0/parameter_updates
/collision_velocity_filter/anti_collision_costmap/parameter_descriptions
/collision_velocity_filter/anti_collision_costmap/parameter_updates
labuser@lab-ubuntu-01:~$ rospack find gmapping
[rospack] Error: package 'gmapping' not found
labuser@lab-ubuntu-01:~$ rostopic echo /head_cam/camera_info
header:
  seq: 1768
  stamp:
    secs: 719
    nsecs: 547000000
```

```
frame_id: "head_cam_optical_frame"
height: 1536
width: 2048
distortion_model: "plumb_bob"
D: [0.0, 0.0, 0.0, 0.0, 0.0]
K: [89.58825132197838, 0.0, 1024.5, 0.0, 89.58825132197838, 768.5, 0.0, 0.0, 1.0]
R: [1.0, 0.0, 0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 1.0]
P: [89.58825132197838, 0.0, 1024.5, -6.2711775925384865, 0.0, 89.58825132197838, 768.5,
0.0, 0.0, 0.0, 1.0, 0.0]
binning_x: 0
binning_y: 0
roi:
  x_offset: 0
  y_offset: 0
  height: 0
  width: 0
  do_rectify: False
---
header:
  seq: 1769
  stamp:
    secs: 719
    nsecs: 833000000
  frame_id: "head_cam_optical_frame"
height: 1536
width: 2048
distortion_model: "plumb_bob"
D: [0.0, 0.0, 0.0, 0.0, 0.0]
K: [89.58825132197838, 0.0, 1024.5, 0.0, 89.58825132197838, 768.5, 0.0, 0.0, 1.0]
R: [1.0, 0.0, 0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 1.0]
P: [89.58825132197838, 0.0, 1024.5, -6.2711775925384865, 0.0, 89.58825132197838, 768.5,
0.0, 0.0, 0.0, 1.0, 0.0]
binning_x: 0
binning_y: 0
roi:
  x_offset: 0
  y_offset: 0
  height: 0
  width: 0
  do_rectify: False
---
```

```
labuser@lab-ubuntu-01:~$ rospack list | grep -Ei
 slam|mapping|gmapping|hector|cartographer|karto"
labuser@lab-ubuntu-01:~$ rospack list | grep move_base
move_base_msgs /opt/ros/noetic/share/move_base_msgs
labuser@lab-ubuntu-01:~$ rospack list | grep map_server
labuser@lab-ubuntu-01:~$ rospack list | grep amcl
labuser@lab-ubuntu-01:~$ find ~/catkin_ws/src -iname "**slam*.launch"
labuser@lab-ubuntu-01:~$ find ~/catkin_ws/src -iname "**mapping*.launch"
labuser@lab-ubuntu-01:~$
```

```
labuser@lab-ubuntu-01:~$ rospack list | grep cob_
cob_actions /opt/ros/noetic/share/cob_actions
cob_android_msgs /opt/ros/noetic/share/cob_android_msgs
cob_android_script_server /opt/ros/noetic/share/cob_android_script_server
cob_base_controller_utils /opt/ros/noetic/share/cob_base_controller_utils
cob_base_velocity_smoother /opt/ros/noetic/share/cob_base_velocity_smoother
cob_bms_driver /opt/ros/noetic/share/cob_bms_driver
cob_bringup /opt/ros/noetic/share/cob_bringup
cob_bringup_sim /home/labuser/catkin_ws/src/cob_simulation/cob_bringup_sim
cob_calibration_data /opt/ros/noetic/share/cob_calibration_data
cob_cam3d_throttle /opt/ros/noetic/share/cob_cam3d_throttle
cob_collision_monitor /opt/ros/noetic/share/cob_collision_monitor
cob_collision_velocity_filter /opt/ros/noetic/share/cob_collision_velocity_filter
cob_command_gui /opt/ros/noetic/share/cob_command_gui
cob_control_mode_adapter /opt/ros/noetic/share/cob_control_mode_adapter
cob_control_msgs /opt/ros/noetic/share/cob_control_msgs
cob_dashboard /opt/ros/noetic/share/cob_dashboard
cob_default_env_config /opt/ros/noetic/share/cob_default_env_config
cob_default_robot_behavior /opt/ros/noetic/share/cob_default_robot_behavior
cob_default_robot_config /opt/ros/noetic/share/cob_default_robot_config
cob_demo /home/labuser/catkin_ws/src/cob_demo
cob_description /opt/ros/noetic/share/cob_description
cob_docker_control /opt/ros/noetic/share/cob_docker_control
cob_footprint_observer /opt/ros/noetic/share/cob_footprint_observer
cob_frame_tracker /opt/ros/noetic/share/cob_frame_tracker
cob_gazebo /home/labuser/catkin_ws/src/cob_simulation/cob_gazebo
cob_gazebo_objects /home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_objects
cob_gazebo_ros_control /opt/ros/noetic/share/cob_gazebo_ros_control
cob_gazebo_tools /home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_tools
```

```
cob_gazebo_worlds /home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds
cob_hand_bridge /opt/ros/noetic/share/cob_hand_bridge
cob_hardware_config /opt/ros/noetic/share/cob_hardware_config
cob_helper_tools /opt/ros/noetic/share/cob_helper_tools
cob_image_flip /opt/ros/noetic/share/cob_image_flip
cob_light /opt/ros/noetic/share/cob_light
cob_linear_nav /opt/ros/noetic/share/cob_linear_nav
cob_mecanum_controller /opt/ros/noetic/share/cob_mecanum_controller
cob_mimic /opt/ros/noetic/share/cob_mimic
cob_monitoring /opt/ros/noetic/share/cob_monitoring
cob_moveit_config /opt/ros/noetic/share/cob_moveit_config
cob_msgs /opt/ros/noetic/share/cob_msgs
cob_obstacle_distance /opt/ros/noetic/share/cob_obstacle_distance
cob_omni_drive_controller /opt/ros/noetic/share/cob_omni_drive_controller
cob_perception_msgs /opt/ros/noetic/share/cob_perception_msgs
cob_phidget_em_state /opt/ros/noetic/share/cob_phidget_em_state
cob_phidget_power_state /opt/ros/noetic/share/cob_phidget_power_state
cob_phidgets /opt/ros/noetic/share/cob_phidgets
cob_reflector_referencing /opt/ros/noetic/share/cob_reflector_referencing
cob_safety_controller /opt/ros/noetic/share/cob_safety_controller
cob_scan_unifier /opt/ros/noetic/share/cob_scan_unifier
cob_script_server /opt/ros/noetic/share/cob_script_server
cob_sick_lms1xx /opt/ros/noetic/share/cob_sick_lms1xx
cob_sick_s300 /opt/ros/noetic/share/cob_sick_s300
cob_sound /opt/ros/noetic/share/cob_sound
cob_srvs /opt/ros/noetic/share/cob_srvs
cob_supported_robots /home/labuser/catkin_ws/src/cob_supported_robots
cob_teleop /opt/ros/noetic/share/cob_teleop
cob_twist_controller /opt/ros/noetic/share/cob_twist_controller
cob_voltage_control /opt/ros/noetic/share/cob_voltage_control
labuser@lab-ubuntu-01:~$
labuser@lab-ubuntu-01:~$ find ~/catkin_ws/src -name "*.launch" | grep cob
/home/labuser/catkin_ws/src/cob_simulation/cob_bringup_sim/launch/robot.launch
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/world.launch
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/world_joint_state_publisher.launch
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/launch/display_world.launch
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_objects/launch/spawn_object_urdf.launch
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo/launch/robot.launch
labuser@lab-ubuntu-01:~$
```


*****+

```
labuser@lab-ubuntu-01:~$ rostopic info /base/twist_mux/command_navigation
Type: geometry_msgs/Twist
```

Publishers:

* /docker_control/move_base_linear (http://lab-ubuntu-01:41591/)

Subscribers:

* /base/twist_mux (http://lab-ubuntu-01:43679/)

```
labuser@lab-ubuntu-01:~$
```

```
labuser@lab-ubuntu-01:~$ rospack find teleop_twist_keyboard
```

```
/opt/ros/noetic/share/teleop_twist_keyboard
```

```
labuser@lab-ubuntu-01:~$
```

```
labuser@lab-ubuntu-01:~$ source /opt/ros/noetic/setup.bash
```

```
labuser@lab-ubuntu-01:~$ source ~/catkin_ws/devel/setup.bash
```

```
labuser@lab-ubuntu-01:~$ rosrn teleop_twist_keyboard teleop_twist_keyboard.py
```

```
cmd_vel:=/base/twist_mux/command_navigation
```

Reading from the keyboard and Publishing to Twist!

Moving around:

u	i	o
j	k	l
m	,	.

For Holonomic mode (strafting), hold down the shift key:

U	I	O
J	K	L
M	<	>

t : up (+z)

b : down (-z)

anything else : stop

q/z : increase/decrease max speeds by 10%

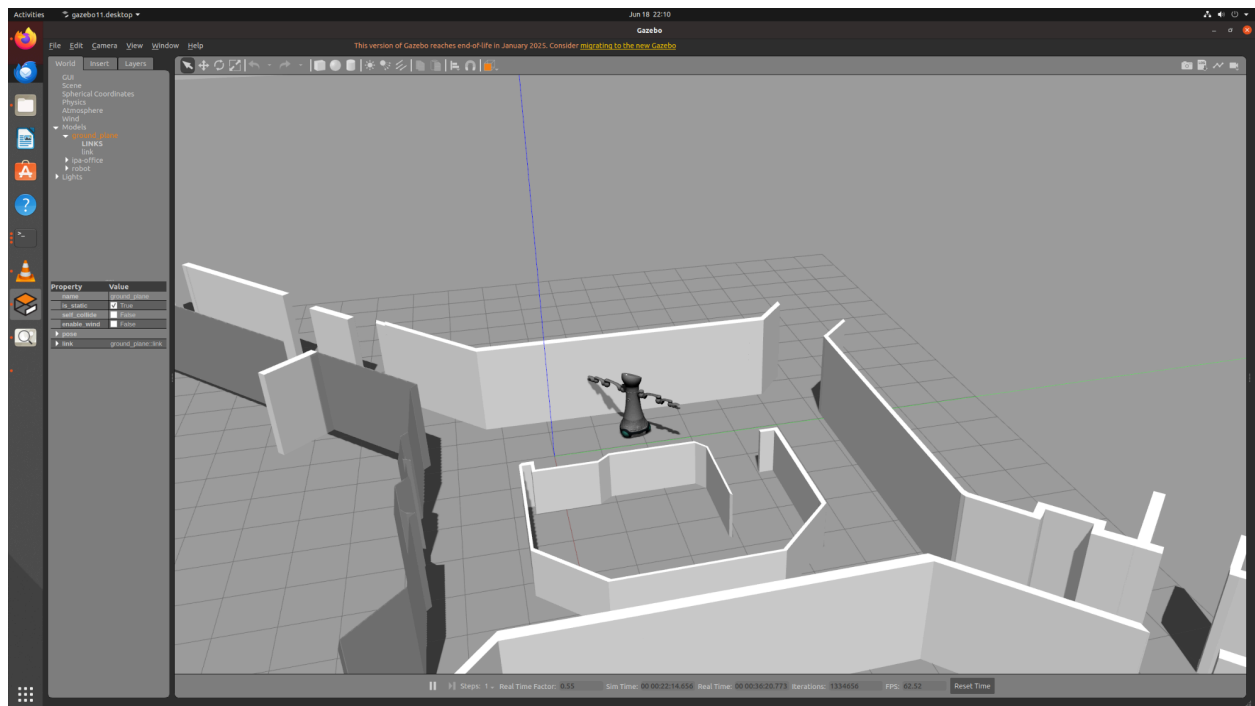
w/x : increase/decrease only linear speed by 10%

e/c : increase/decrease only angular speed by 10%

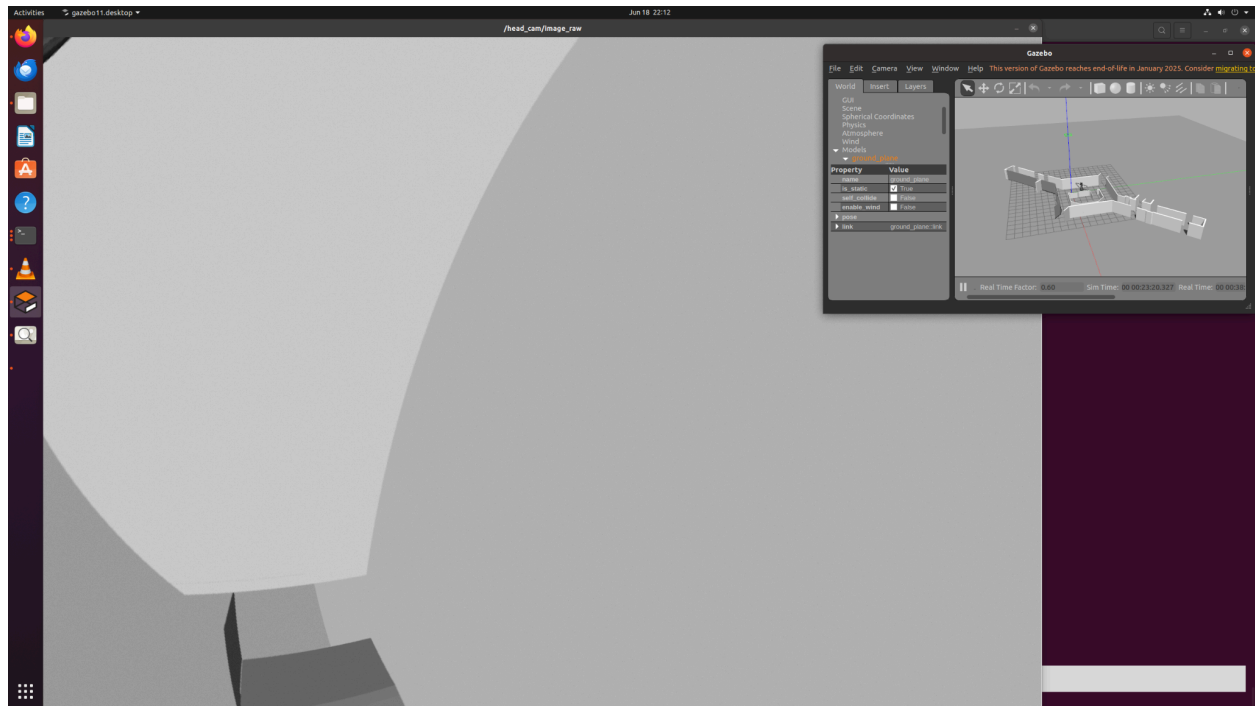
CTRL-C to quit

currently: speed 0.5 turn 1.0

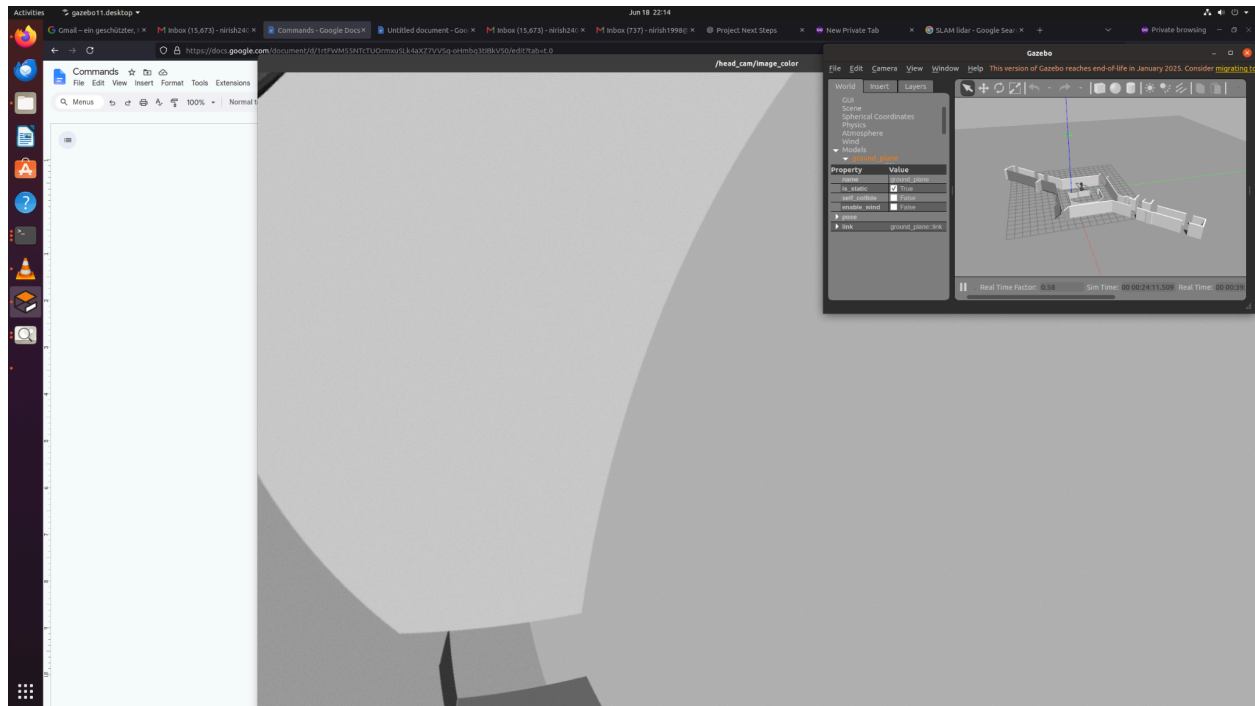
The Robot is moving in Gazebo



```
labuser@lab-ubuntu-01:~$ rosrn image_view image_view image:=/head_cam/image_raw
[INFO] [1781813355.678286751]: Initializing nodelet with 16 worker threads.
[INFO] [1781813355.778873141]: Using transport "raw"
```



```
labuser@lab-ubuntu-01:~$ roslaunch image_view image_view image:=/head_cam/image_color
[INFO] [1781813596.962098816]: Initializing nodelet with 16 worker threads.
[INFO] [1781813597.065812300]: Using transport "raw"
```



```
labuser@lab-ubuntu-01:~$ rostopic hz /head_cam/image_raw
subscribed to [/head_cam/image_raw]
WARNING: may be using simulated time
average rate: 4.115
    min: 0.243s max: 0.243s std dev: 0.00000s window: 2
average rate: 2.725
    min: 0.243s max: 0.585s std dev: 0.15464s window: 4
average rate: 2.880
    min: 0.243s max: 0.585s std dev: 0.13822s window: 5
average rate: 2.675
    min: 0.243s max: 0.585s std dev: 0.14480s window: 7
average rate: 2.767
    min: 0.243s max: 0.585s std dev: 0.13746s window: 8
^Caverage rate: 2.582
    min: 0.243s max: 0.585s std dev: 0.14560s window: 9
labuser@lab-ubuntu-01:~$
```

```
labuser@lab-ubuntu-01:~$ cd ~/catkin_ws/src
```

```
labuser@lab-ubuntu-01:~/catkin_ws/src$ catkin_create_pkg privacy_environment_map rospy
sensor_msgs cv_bridge image_transport std_msgs
```

```
Created file privacy_environment_map/package.xml
```

```
Created file privacy_environment_map/CMakeLists.txt
```

```
Created folder privacy_environment_map/src
```

```
Successfully created files in /home/labuser/catkin_ws/src/privacy_environment_map. Please
adjust the values in package.xml.
```

```
labuser@lab-ubuntu-01:~/catkin_ws/src$ cd ~/catkin_ws
```

```
labuser@lab-ubuntu-01:~/catkin_ws$ catkin_make
```

```
Base path: /home/labuser/catkin_ws
```

```
Source space: /home/labuser/catkin_ws/src
```

```
Build space: /home/labuser/catkin_ws/build
```

```
Devel space: /home/labuser/catkin_ws/devel
```

```
Install space: /home/labuser/catkin_ws/install
```

```
####
```

```
#### Running command: "cmake /home/labuser/catkin_ws/src
```

```
-DCATKIN_DEVEL_PREFIX=/home/labuser/catkin_ws/devel
```

```
-DCMAKE_INSTALL_PREFIX=/home/labuser/catkin_ws/install -G Unix Makefiles" in
"/home/labuser/catkin_ws/build"
```

```
####
```

```
-- Using CATKIN_DEVEL_PREFIX: /home/labuser/catkin_ws/devel
```

```
-- Using CMAKE_PREFIX_PATH: /home/labuser/catkin_ws/devel;/opt/ros/noetic
```

```
-- This workspace overlays: /home/labuser/catkin_ws/devel;/opt/ros/noetic
```

```
-- Found PythonInterp: /usr/bin/python3 (found suitable version "3.8.10", minimum required is
"3")
```

```
-- Using PYTHON_EXECUTABLE: /usr/bin/python3
```

```
-- Using Debian Python package layout
```

```
-- Using emp: /usr/lib/python3/dist-packages/em.py
```

```
-- Using CATKIN_ENABLE_TESTING: ON
```

```
-- Call enable_testing()
```

```
-- Using CATKIN_TEST_RESULTS_DIR: /home/labuser/catkin_ws/build/test_results
```

```
-- Forcing gtest/gmock from source, though one was otherwise available.
```

```
-- Found gtest sources under '/usr/src/gtest': gtests will be built
```

```
-- Found gmock sources under '/usr/src/gtest': gmock will be built
```

```
-- Found PythonInterp: /usr/bin/python3 (found version "3.8.10")
```

```
-- Using Python nosetests: /usr/bin/nosetests3
```

```
-- catkin 0.8.12
```

```
-- BUILD_SHARED_LIBS is on
```

```
-- BUILD_SHARED_LIBS is on
```

```
-- ~~~~~
```

```
-- ~~ traversing 9 packages in topological order:
```

```
-- ~~ - cob_gazebo_objects
```

```
-- ~~ - cob_gazebo_tools
```

```
-- ~~ - cob_simulation (metapackage)
```

```

-- ~~ - cob_supported_robots
-- ~~ - cob_demo
-- ~~ - cob_bringup_sim
-- ~~ - cob_gazebo
-- ~~ - cob_gazebo_worlds
-- ~~ - privacy_environment_map
-- ~~~~~~
-- +++ processing catkin package: 'cob_gazebo_objects'
-- ==> add_subdirectory(cob_simulation/cob_gazebo_objects)
-- +++ processing catkin package: 'cob_gazebo_tools'
-- ==> add_subdirectory(cob_simulation/cob_gazebo_tools)
-- Installing devel-space wrapper
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_tools/scripts/move_object.py to
/home/labuser/catkin_ws/devel/lib/cob_gazebo_tools
-- Installing devel-space wrapper
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_tools/scripts/remove_object.py to
/home/labuser/catkin_ws/devel/lib/cob_gazebo_tools
-- Installing devel-space wrapper
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_tools/scripts/spawn_object.py to
/home/labuser/catkin_ws/devel/lib/cob_gazebo_tools
-- +++ processing catkin metapackage: 'cob_simulation'
-- ==> add_subdirectory(cob_simulation/cob_simulation)
-- +++ processing catkin package: 'cob_supported_robots'
-- ==> add_subdirectory(cob_supported_robots)
-- +++ processing catkin package: 'cob_demo'
-- ==> add_subdirectory(cob_demo)
-- Using these message generators: gencpp;geneus;genlisp;gennodejs;genpy
-- +++ processing catkin package: 'cob_bringup_sim'
-- ==> add_subdirectory(cob_simulation/cob_bringup_sim)
testing for robot: cob4-3 in env: empty
testing for robot: cob4-3 in env: ipa-apartment
testing for robot: cob4-3 in env: ipa-kitchen
testing for robot: cob4-3 in env: ipa-office
testing for robot: cob4-3 in env: ipa-production-plant
testing for robot: cob4-3 in env: iza-campus
testing for robot: cob4-3 in env: iza-carpark-top
testing for robot: cob4-4 in env: empty
testing for robot: cob4-4 in env: ipa-apartment
testing for robot: cob4-4 in env: ipa-kitchen
testing for robot: cob4-4 in env: ipa-office
testing for robot: cob4-4 in env: ipa-production-plant
testing for robot: cob4-4 in env: iza-campus
testing for robot: cob4-4 in env: iza-carpark-top
testing for robot: cob4-5 in env: empty

```

testing for robot: cob4-5 in env: ipa-apartment
testing for robot: cob4-5 in env: ipa-kitchen
testing for robot: cob4-5 in env: ipa-office
testing for robot: cob4-5 in env: ipa-production-plant
testing for robot: cob4-5 in env: iza-campus
testing for robot: cob4-5 in env: iza-carpark-top
testing for robot: cob4-6 in env: empty
testing for robot: cob4-6 in env: ipa-apartment
testing for robot: cob4-6 in env: ipa-kitchen
testing for robot: cob4-6 in env: ipa-office
testing for robot: cob4-6 in env: ipa-production-plant
testing for robot: cob4-6 in env: iza-campus
testing for robot: cob4-6 in env: iza-carpark-top
testing for robot: cob4-7 in env: empty
testing for robot: cob4-7 in env: ipa-apartment
testing for robot: cob4-7 in env: ipa-kitchen
testing for robot: cob4-7 in env: ipa-office
testing for robot: cob4-7 in env: ipa-production-plant
testing for robot: cob4-7 in env: iza-campus
testing for robot: cob4-7 in env: iza-carpark-top
testing for robot: cob4-8 in env: empty
testing for robot: cob4-8 in env: ipa-apartment
testing for robot: cob4-8 in env: ipa-kitchen
testing for robot: cob4-8 in env: ipa-office
testing for robot: cob4-8 in env: ipa-production-plant
testing for robot: cob4-8 in env: iza-campus
testing for robot: cob4-8 in env: iza-carpark-top
testing for robot: cob4-9 in env: empty
testing for robot: cob4-9 in env: ipa-apartment
testing for robot: cob4-9 in env: ipa-kitchen
testing for robot: cob4-9 in env: ipa-office
testing for robot: cob4-9 in env: ipa-production-plant
testing for robot: cob4-9 in env: iza-campus
testing for robot: cob4-9 in env: iza-carpark-top
testing for robot: cob4-10 in env: empty
testing for robot: cob4-10 in env: ipa-apartment
testing for robot: cob4-10 in env: ipa-kitchen
testing for robot: cob4-10 in env: ipa-office
testing for robot: cob4-10 in env: ipa-production-plant
testing for robot: cob4-10 in env: iza-campus
testing for robot: cob4-10 in env: iza-carpark-top
testing for robot: cob4-11 in env: empty
testing for robot: cob4-11 in env: ipa-apartment
testing for robot: cob4-11 in env: ipa-kitchen

testing for robot: cob4-11 in env: ipa-office
testing for robot: cob4-11 in env: ipa-production-plant
testing for robot: cob4-11 in env: iza-campus
testing for robot: cob4-11 in env: iza-carpark-top
testing for robot: cob4-13 in env: empty
testing for robot: cob4-13 in env: ipa-apartment
testing for robot: cob4-13 in env: ipa-kitchen
testing for robot: cob4-13 in env: ipa-office
testing for robot: cob4-13 in env: ipa-production-plant
testing for robot: cob4-13 in env: iza-campus
testing for robot: cob4-13 in env: iza-carpark-top
testing for robot: cob4-16 in env: empty
testing for robot: cob4-16 in env: ipa-apartment
testing for robot: cob4-16 in env: ipa-kitchen
testing for robot: cob4-16 in env: ipa-office
testing for robot: cob4-16 in env: ipa-production-plant
testing for robot: cob4-16 in env: iza-campus
testing for robot: cob4-16 in env: iza-carpark-top
testing for robot: cob4-18 in env: empty
testing for robot: cob4-18 in env: ipa-apartment
testing for robot: cob4-18 in env: ipa-kitchen
testing for robot: cob4-18 in env: ipa-office
testing for robot: cob4-18 in env: ipa-production-plant
testing for robot: cob4-18 in env: iza-campus
testing for robot: cob4-18 in env: iza-carpark-top
testing for robot: cob4-20 in env: empty
testing for robot: cob4-20 in env: ipa-apartment
testing for robot: cob4-20 in env: ipa-kitchen
testing for robot: cob4-20 in env: ipa-office
testing for robot: cob4-20 in env: ipa-production-plant
testing for robot: cob4-20 in env: iza-campus
testing for robot: cob4-20 in env: iza-carpark-top
testing for robot: cob4-23 in env: empty
testing for robot: cob4-23 in env: ipa-apartment
testing for robot: cob4-23 in env: ipa-kitchen
testing for robot: cob4-23 in env: ipa-office
testing for robot: cob4-23 in env: ipa-production-plant
testing for robot: cob4-23 in env: iza-campus
testing for robot: cob4-23 in env: iza-carpark-top
testing for robot: cob4-24 in env: empty
testing for robot: cob4-24 in env: ipa-apartment
testing for robot: cob4-24 in env: ipa-kitchen
testing for robot: cob4-24 in env: ipa-office
testing for robot: cob4-24 in env: ipa-production-plant

testing for robot: cob4-24 in env: iza-campus
testing for robot: cob4-24 in env: iza-carpark-top
testing for robot: cob4-25 in env: empty
testing for robot: cob4-25 in env: ipa-apartment
testing for robot: cob4-25 in env: ipa-kitchen
testing for robot: cob4-25 in env: ipa-office
testing for robot: cob4-25 in env: ipa-production-plant
testing for robot: cob4-25 in env: iza-campus
testing for robot: cob4-25 in env: iza-carpark-top
testing for robot: cob4-29 in env: empty
testing for robot: cob4-29 in env: ipa-apartment
testing for robot: cob4-29 in env: ipa-kitchen
testing for robot: cob4-29 in env: ipa-office
testing for robot: cob4-29 in env: ipa-production-plant
testing for robot: cob4-29 in env: iza-campus
testing for robot: cob4-29 in env: iza-carpark-top
testing for robot: cob4-30 in env: empty
testing for robot: cob4-30 in env: ipa-apartment
testing for robot: cob4-30 in env: ipa-kitchen
testing for robot: cob4-30 in env: ipa-office
testing for robot: cob4-30 in env: ipa-production-plant
testing for robot: cob4-30 in env: iza-campus
testing for robot: cob4-30 in env: iza-carpark-top
testing for robot: raw3-1 in env: empty
testing for robot: raw3-1 in env: ipa-apartment
testing for robot: raw3-1 in env: ipa-kitchen
testing for robot: raw3-1 in env: ipa-office
testing for robot: raw3-1 in env: ipa-production-plant
testing for robot: raw3-1 in env: iza-campus
testing for robot: raw3-1 in env: iza-carpark-top
testing for robot: raw3-3 in env: empty
testing for robot: raw3-3 in env: ipa-apartment
testing for robot: raw3-3 in env: ipa-kitchen
testing for robot: raw3-3 in env: ipa-office
testing for robot: raw3-3 in env: ipa-production-plant
testing for robot: raw3-3 in env: iza-campus
testing for robot: raw3-3 in env: iza-carpark-top
testing for robot: raw3-5 in env: empty
testing for robot: raw3-5 in env: ipa-apartment
testing for robot: raw3-5 in env: ipa-kitchen
testing for robot: raw3-5 in env: ipa-office
testing for robot: raw3-5 in env: ipa-production-plant
testing for robot: raw3-5 in env: iza-campus
testing for robot: raw3-5 in env: iza-carpark-top

```
-- +++ processing catkin package: 'cob_gazebo'
-- ==> add_subdirectory(cob_simulation/cob_gazebo)
testing for robot: cob4-3
testing for robot: cob4-4
testing for robot: cob4-5
testing for robot: cob4-6
testing for robot: cob4-7
testing for robot: cob4-8
testing for robot: cob4-9
testing for robot: cob4-10
testing for robot: cob4-11
testing for robot: cob4-13
testing for robot: cob4-16
testing for robot: cob4-18
testing for robot: cob4-20
testing for robot: cob4-23
testing for robot: cob4-24
testing for robot: cob4-25
testing for robot: cob4-29
testing for robot: cob4-30
testing for robot: raw3-1
testing for robot: raw3-3
testing for robot: raw3-5
-- Installing devel-space wrapper
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo/ros/test/component_test.py to
/home/labuser/catkin_ws/devel/lib/cob_gazebo
-- +++ processing catkin package: 'cob_gazebo_worlds'
-- ==> add_subdirectory(cob_simulation/cob_gazebo_worlds)
testing for env: empty
testing for env: ipa-apartment
testing for env: ipa-kitchen
testing for env: ipa-office
testing for env: ipa-production-plant
testing for env: iza-campus
testing for env: iza-carpark-top
-- Installing devel-space wrapper
/home/labuser/catkin_ws/src/cob_simulation/cob_gazebo_worlds/scripts/elevator.py to
/home/labuser/catkin_ws/devel/lib/cob_gazebo_worlds
-- +++ processing catkin package: 'privacy_environment_map'
-- ==> add_subdirectory(privacy_environment_map)
-- Configuring done
-- Generating done
-- Build files have been written to: /home/labuser/catkin_ws/build
####
```

Running command: "make -j16 -l16" in "/home/labuser/catkin_ws/build"

####

labuser@lab-ubuntu-01:~/catkin_ws\$ source devel/setup.bash

labuser@lab-ubuntu-01:~/catkin_ws\$

~/catkin_ws/src/privacy_environment_map/scripts/camera_view.py

bash: /home/labuser/catkin_ws/src/privacy_environment_map/scripts/camera_view.py: No such file or directory

labuser@lab-ubuntu-01:~/catkin_ws\$ chmod +x camera_view.py

chmod: cannot access 'camera_view.py': No such file or directory

labuser@lab-ubuntu-01:~/catkin_ws\$

labuser@lab-ubuntu-01:~/catkin_ws\$ mkdir -p

~/catkin_ws/src/privacy_environment_map/scripts

labuser@lab-ubuntu-01:~/catkin_ws\$ touch

~/catkin_ws/src/privacy_environment_map/scripts/camera_view.py

labuser@lab-ubuntu-01:~/catkin_ws\$ chmod +x camera_view.py

chmod: cannot access 'camera_view.py': No such file or directory

labuser@lab-ubuntu-01:~/catkin_ws\$ chmod +x

~/catkin_ws/src/privacy_environment_map/scripts/camera_view.py

labuser@lab-ubuntu-01:~/catkin_ws\$

labuser@lab-ubuntu-01:~/catkin_ws\$

#!/usr/bin/env python3

import rospy

from sensor_msgs.msg import Image

from cv_bridge import CvBridge

import cv2

bridge = CvBridge()

def callback(msg):

 frame = bridge.imgmsg_to_cv2(msg, desired_encoding="bgr8")

 cv2.imshow("Head Camera", frame)

 cv2.waitKey(1)

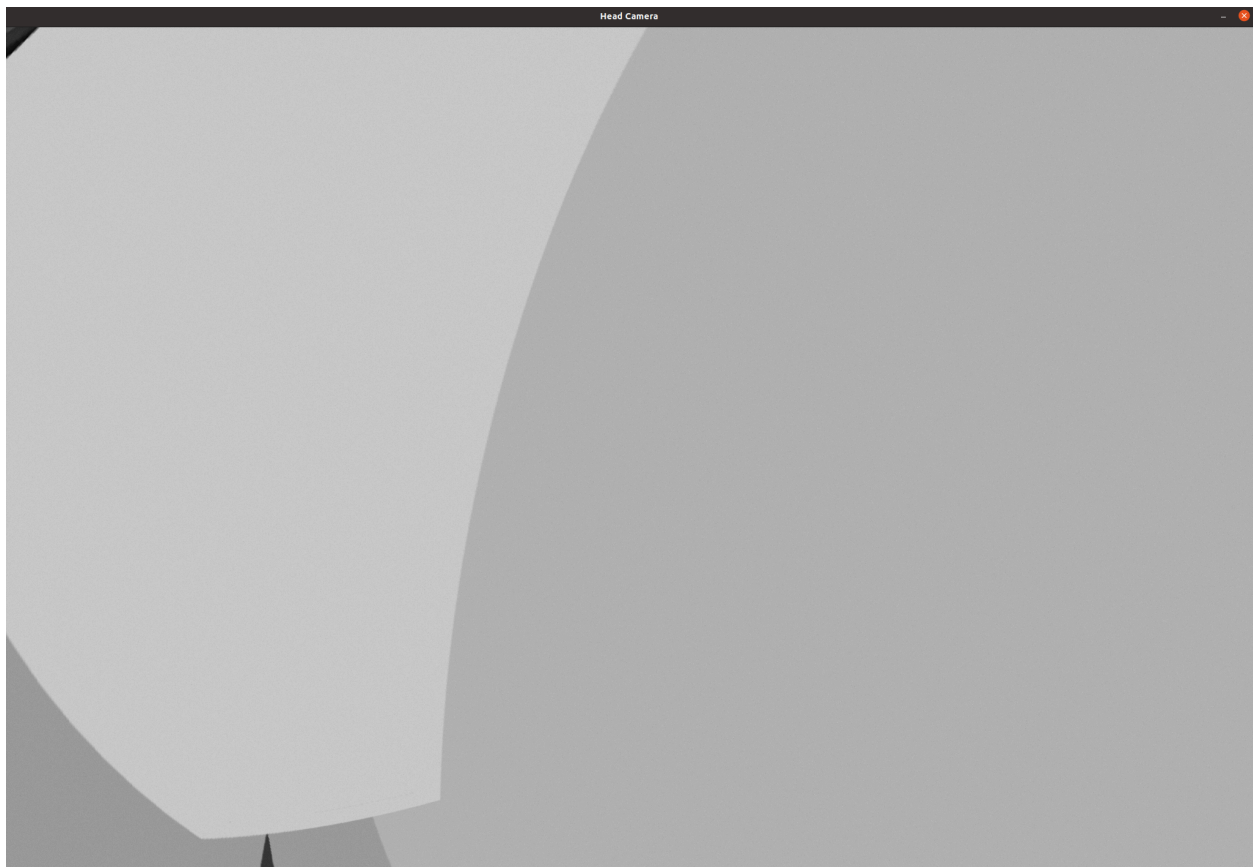
rospy.init_node("camera_viewer")

rospy.Subscriber("/head_cam/image_raw",

```
Image,  
callback,  
queue_size=1)
```

```
rospy.spin()
```

```
labuser@lab-ubuntu-01:~/catkin_ws/src/privacy_environment_map/scripts$ source  
/opt/ros/noetic/setup.bash  
labuser@lab-ubuntu-01:~/catkin_ws/src/privacy_environment_map/scripts$ source  
~/catkin_ws/devel/setup.bash  
labuser@lab-ubuntu-01:~/catkin_ws/src/privacy_environment_map/scripts$ rosrn  
privacy_environment_map camera_view.py
```



```
*****
```

```
labuser@lab-ubuntu-01:~/catkin_ws$ rospack find laser_geometry
/opt/ros/noetic/share/laser_geometry
labuser@lab-ubuntu-01:~/catkin_ws$
labuser@lab-ubuntu-01:~/catkin_ws$ rostopic type /scan_unified
sensor_msgs/LaserScan
labuser@lab-ubuntu-01:~/catkin_ws$
labuser@lab-ubuntu-01:~/catkin_ws$ rospack find laser_geometry
/opt/ros/noetic/share/laser_geometry
labuser@lab-ubuntu-01:~/catkin_ws$
labuser@lab-ubuntu-01:~/catkin_ws$ rostopic list | grep points
```

```
/sensoring_cam3d_back/depth/points
/sensoring_cam3d_back/depth_registered/points
/sensoring_cam3d_back_upright/depth_registered/points
/sensoring_cam3d_front/depth/points
/sensoring_cam3d_front_upright/depth/points
/torso_cam3d_down/depth/points
/torso_cam3d_left/depth/points
/torso_cam3d_left/depth_registered/points
/torso_cam3d_left_upright/depth_registered/points
/torso_cam3d_right/depth/points
/torso_cam3d_right/depth_registered/points
/torso_cam3d_right_upright/depth_registered/points
labuser@lab-ubuntu-01:~/catkin_ws$
labuser@lab-ubuntu-01:~/catkin_ws$
```

```
labuser@lab-ubuntu-01:~/catkin_ws$ rostopic type /sensoring_cam3d_front/depth/points
sensor_msgs/PointCloud2
labuser@lab-ubuntu-01:~/catkin_ws$
```

```
*****+
```

```
#!/usr/bin/env python3
```

```
import rospy
import laser_geometry.laser_geometry as lg
```

```
from sensor_msgs.msg import LaserScan
from sensor_msgs.msg import PointCloud2
```

```
projector = lg.LaserProjection()
```

```
pub = None

def callback(scan):

    cloud = projector.projectLaser(scan)

    pub.publish(cloud)

rospy.init_node("scan_to_cloud")

pub = rospy.Publisher(
    "/scan_cloud",
    PointCloud2,
    queue_size=1
)

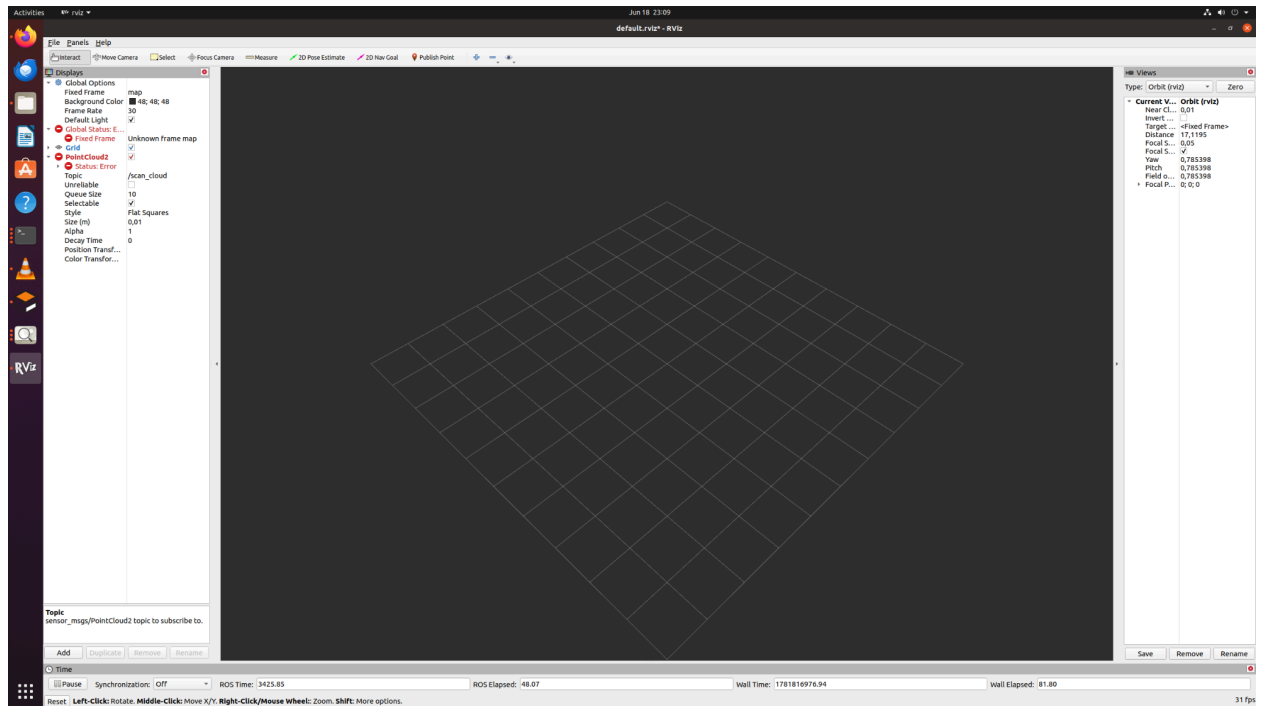
rospy.Subscriber(
    "/scan_unified",
    LaserScan,
    callback
)

rospy.spin()
```

```
labuser@lab-ubuntu-01:~/catkin_ws/src/privacy_environment_map/scripts$ source
~/catkin_ws/devel/setup.bash
```

```
labuser@lab-ubuntu-01:~/catkin_ws/src/privacy_environment_map/scripts$ rosrn
privacy_environment_map scan_to_cloud.py
```

```
labuser@lab-ubuntu-01:~$ rviz
```



```
labuser@lab-ubuntu-01:~/catkin_ws$ rostopic echo -n 1 /scan_cloud/header
```

```
seq: 759
```

```
stamp:
```

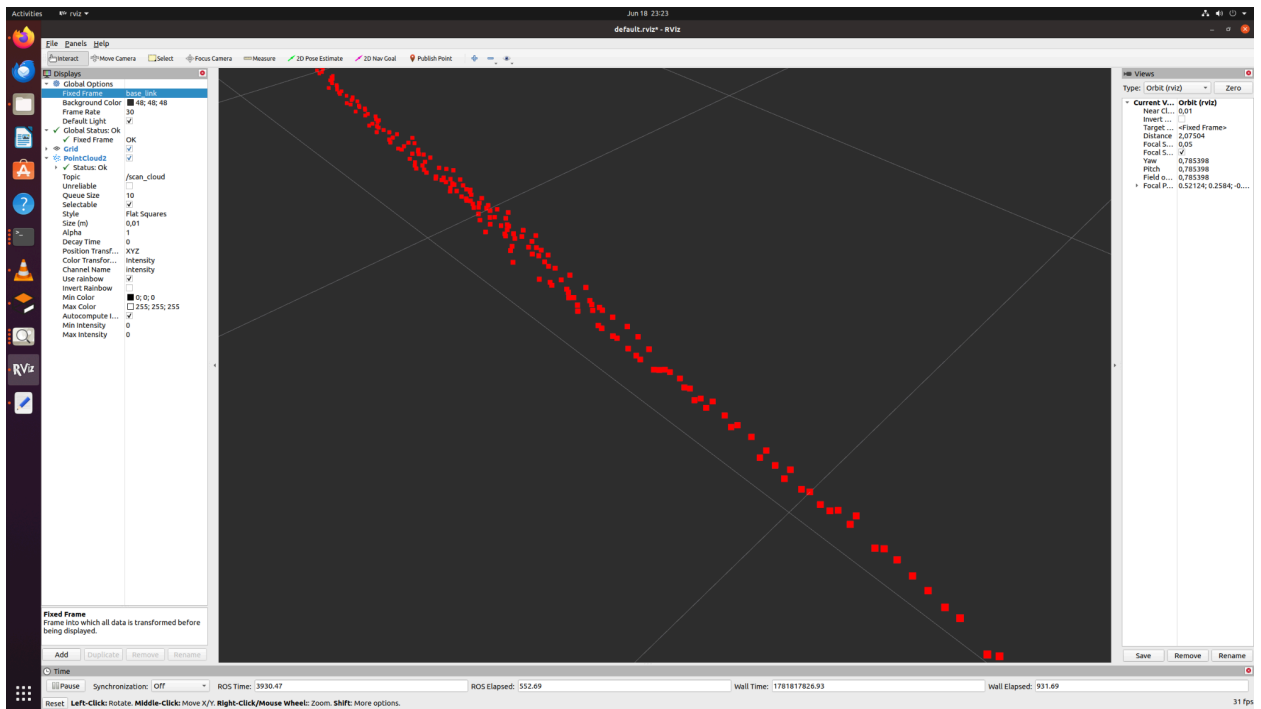
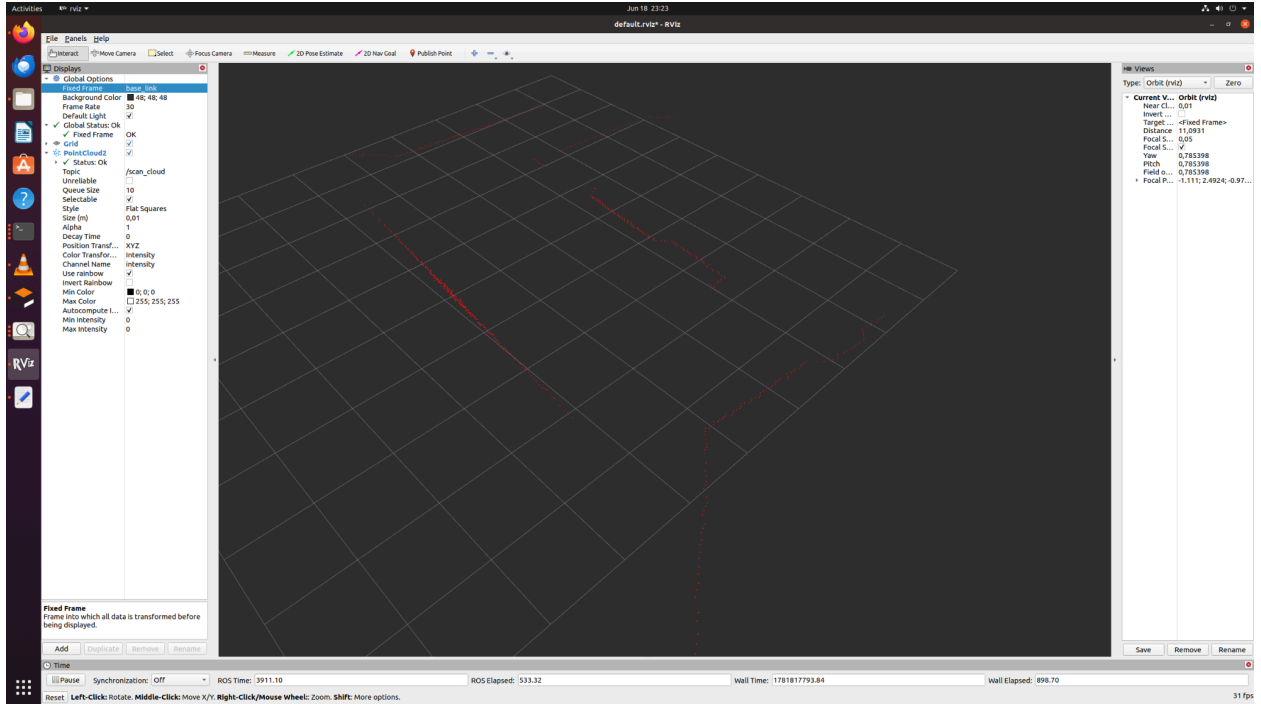
```
  secs: 3461
```

```
  nsecs: 509000000
```

```
frame_id: "base_link"
```

```
---
```

```
labuser@lab-ubuntu-01:~/catkin_ws$
```



```
rostopic echo -n 1 /sensorring_cam3d_front/depth/points
```

```
127, 0, 0, 192, 127, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 192, 127, 0, 0, 192,
127, 0, 0, 192, 127, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 192, 127, 0, 0, 192,
127, 0, 0, 192, 127, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 192, 127, 0, 0, 192,
127, 0, 0, 192, 127, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 192, 127, 0, 0, 192,
127, 0, 0, 192, 127, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 192, 127, 0, 0, 192,
127, 0, 0, 192, 127, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 192, 127, 0, 0, 192,
127, 0, 0, 192, 127, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
```

is_dense: False

labuser@lab-ubuntu-01:~/catkin_ws\$

